

Essentials of Innovation & Entrepreneurship (EIE) - II

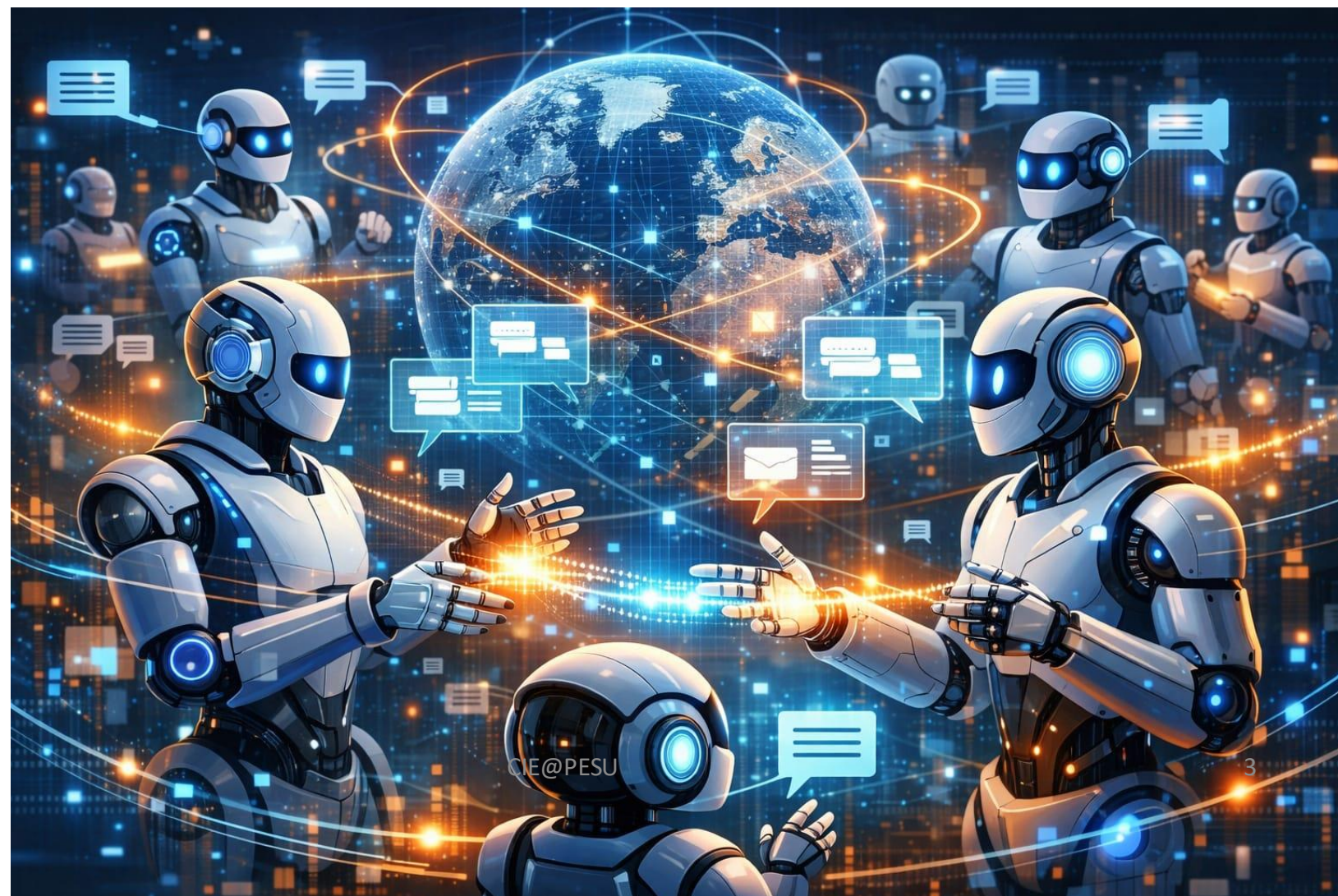
Module (M3): Customer Development (CD) Model: Testable/Measurable Hypothesis

EIE2-Icebreaker:

I&E Roundup this Week (Feb 11)



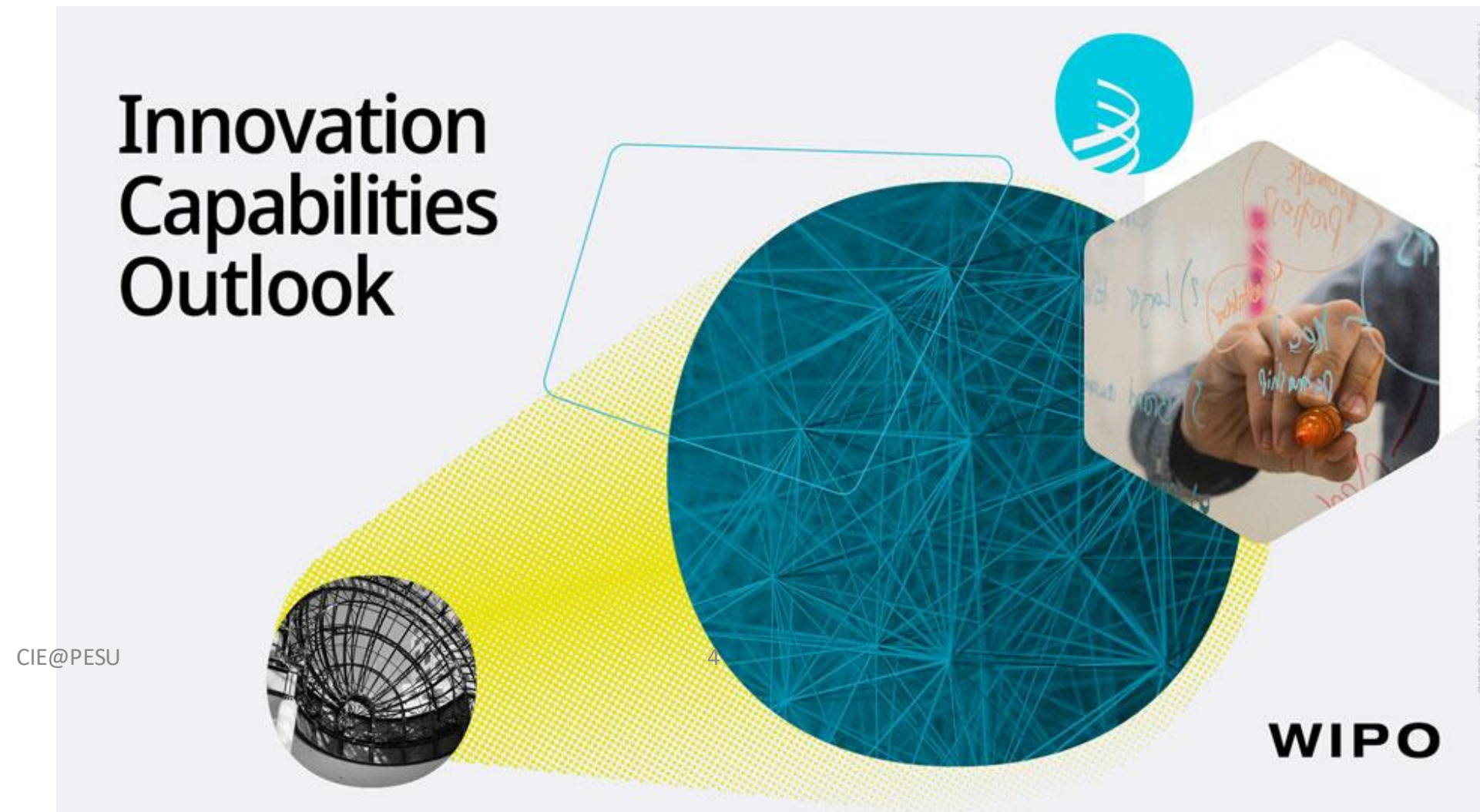
Moltbook: A First!



Moltbook is an AI-first platform where autonomous AI agents post, comment, argue, agree and evolve conversations on their own.

The site lets humans only observe as bots interact freely without human control.
(created by entrepreneur Matt Schlicht)

Can Innovation Change a Country's Success (Destiny)?

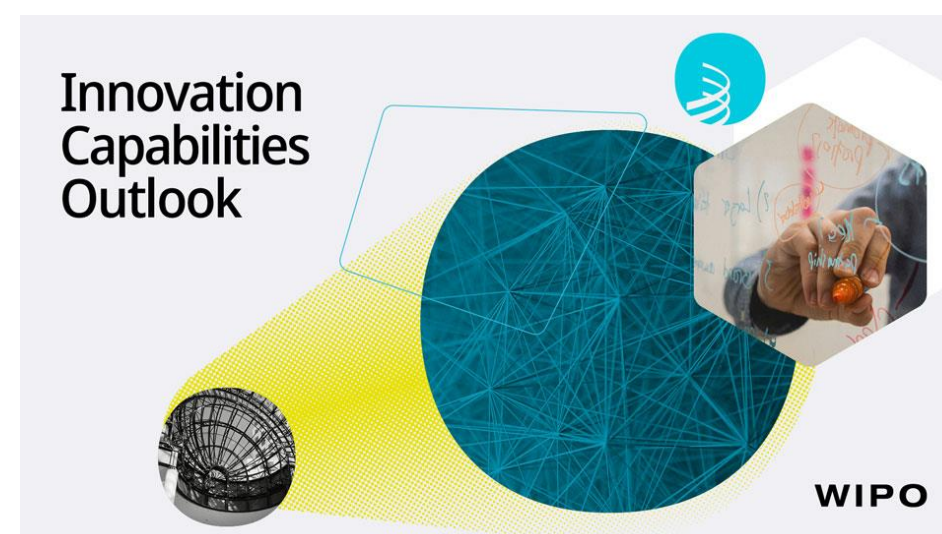


*Innovation Capabilities Outlook 2026:
New WIPO* Report Shows Countries' Innovation-Related Strengths*

* WIPO – World Intellectual Property Organization

<https://www.wipo.int/pressroom/en/stories/innov-capab-outlook2026.html>

ICO 2026: Summary



1. The world can use the existing innovation capabilities more efficiently:

Countries could generate up to 26% more technologies, 15% more entrepreneurial innovations and 12% more scientific discoveries every year by better leveraging the innovation capabilities they already possess.

2. Most economies have not yet developed complex technological capabilities:

Technological capabilities in patents represent the rarest and most demanding innovation capabilities to develop, but they are also the most rewarding. Yet less than 5% of the countries manage to develop them.

3. Developing more complex innovation capabilities is the key to success:

Every yr, only 1 in 3 countries upgrades its capabilities towards rarer/more rewarding fields of innovation.

What takeaways can you extract from the above for ***yourself***?

India Budget (FY 2026-27)

What does this mean for Innovation & Entrepreneurship?

I. **Startup Capital**

- I. *Startup & MSME funds (Rs 10,000 Crores)*
- II. *Deep-tech focus*
- III. *Easier access to early funding*

II. **Technology & R&D**

- I. *AI, Semiconductors, Biotech*
- II. *Innovation-led growth*
- III. *Future-ready careers*

III. **Ease of Doing Business**

- I. *Simpler tax & compliance*
- II. *Faster startup formation*
- III. *Lower entry barriers*

IV. **Infrastructure for Scale**

- I. *Digital + physical infra*
- II. *Manufacturing & logistics*
- III. *Global competitiveness*

V. **Inclusive Entrepreneurship**

- I. *Women & youth focus*
- II. *Community enterprises*
- III. *Entrepreneurship for all*

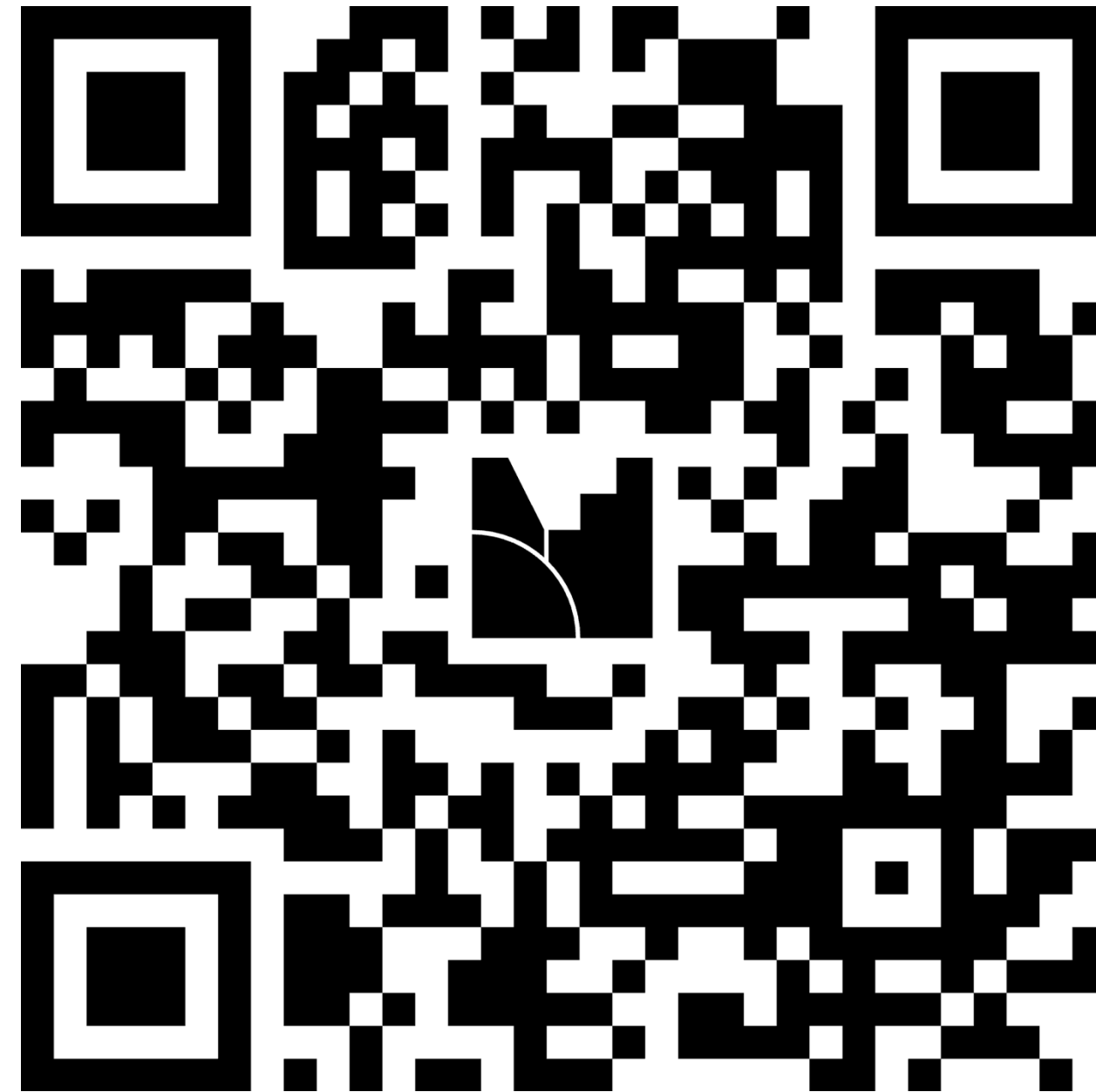
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Innovation is no longer optional; it's national priority—and your opportunity

M2 Recap – Quiz

MentiMeter

Use the code: **2973 8724**



M2: Customer Development Model Recap

➤ **Customer Development precedes Product Development**

Test problems before building solutions.

➤ **Get Outside the Building (GOOB)**

No real facts exist inside assumptions — talk to customers.

➤ **Hypothesis → Experiments → Learning → Pivot/Proceed**

Startups search for a business model, not execute one.

➤ **Validate Problem–Solution Fit before Scaling**

Sales and early adopters prove viability.

➤ **Iteration beats Perfection**

Fast feedback cycles reduce risk and waste.

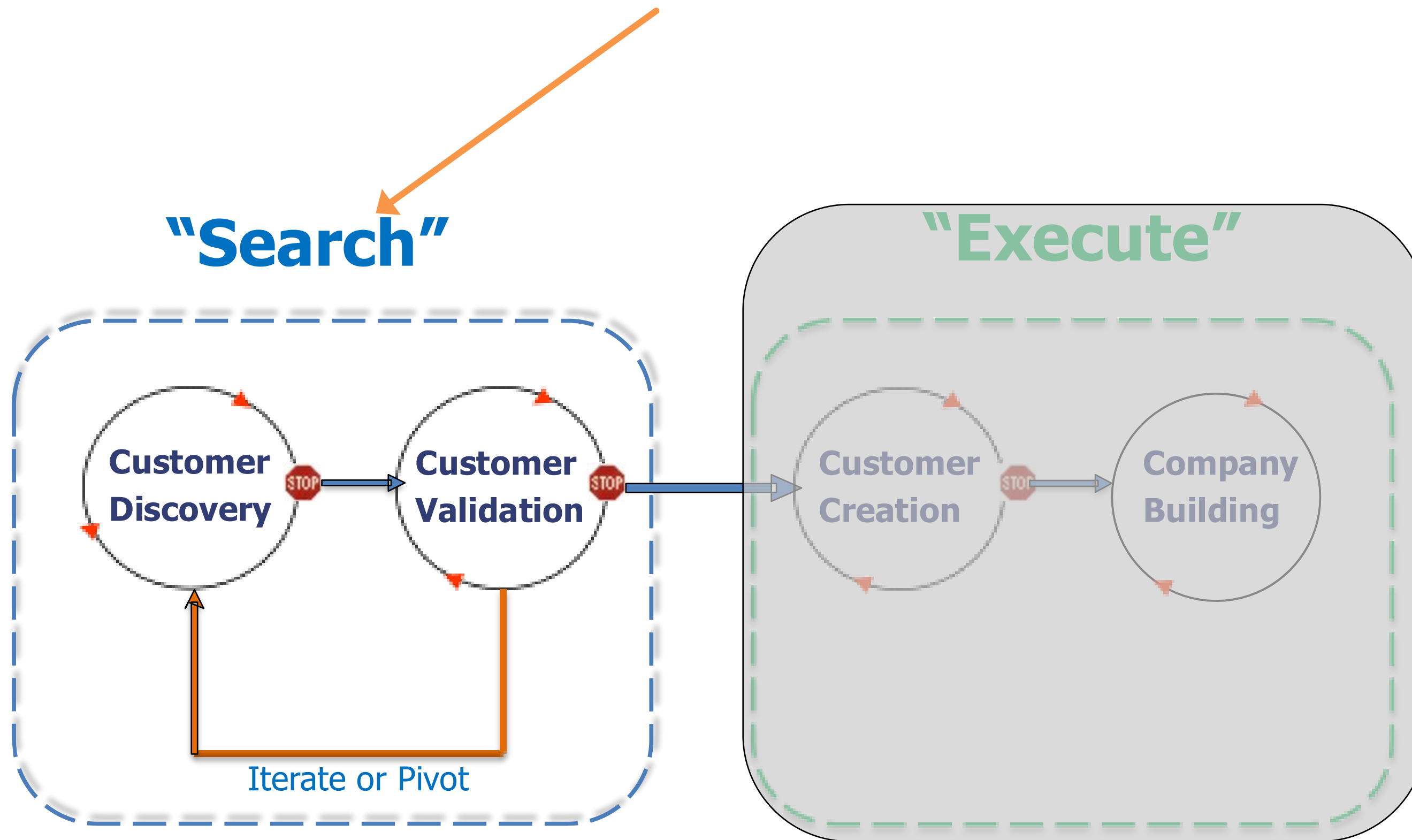
➤ **Teams thrive in Chaos and Uncertainty**

Shared values, resilience, and trust drive execution.

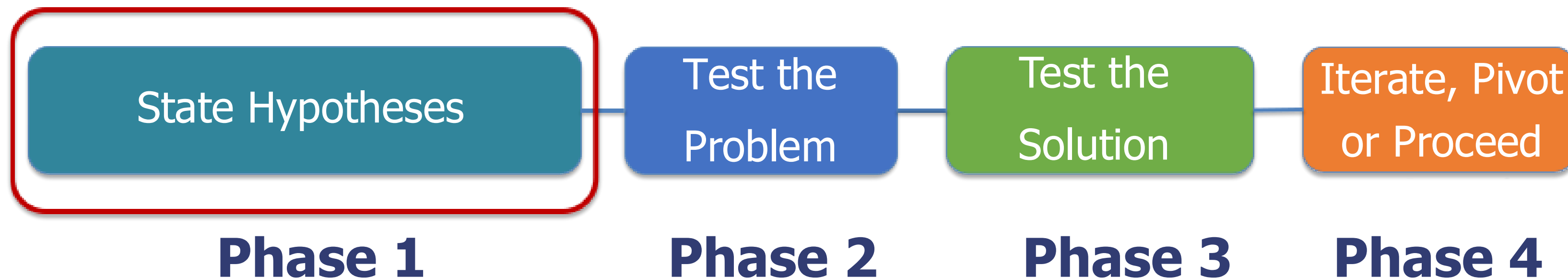
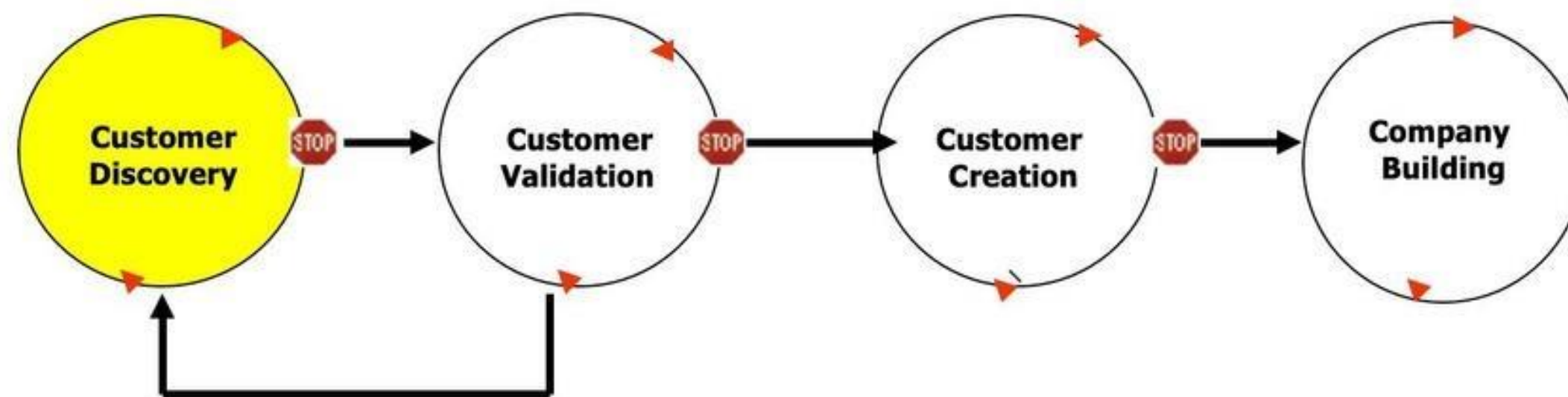
Essentials of Innovation & Entrepreneurship (EIE) - II

Customer Development (CD) Model: Testable & Measurable Hypotheses

Focus Area for EIE2 & CIE-Ignite

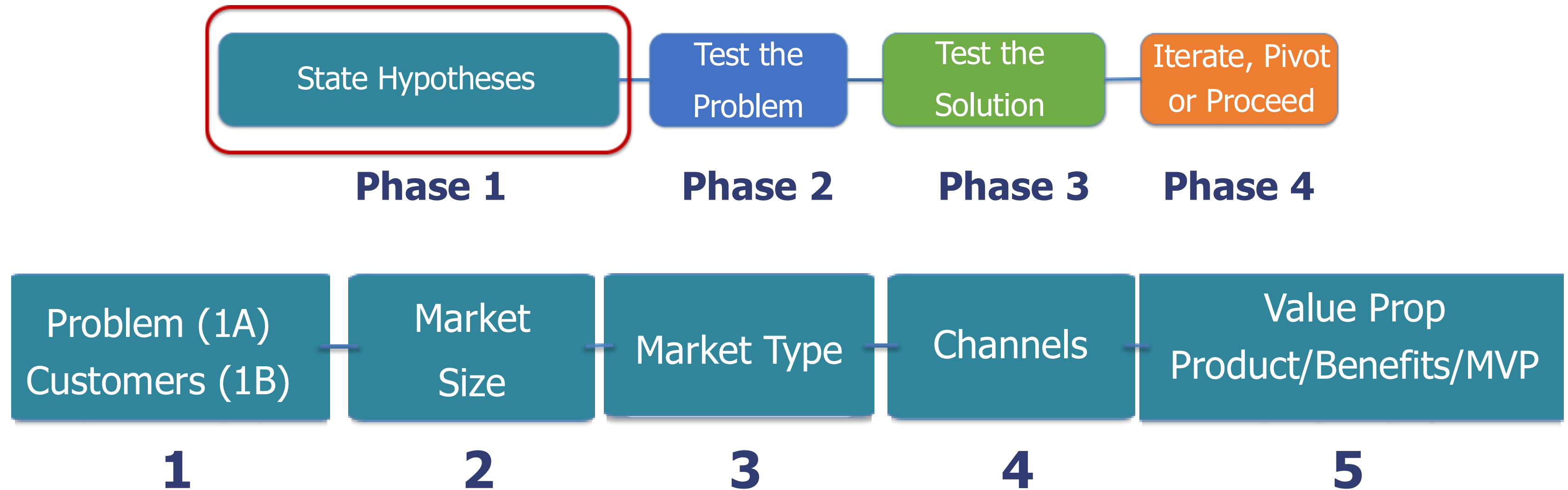


Customer Discovery Overview



Customer Discovery Overview

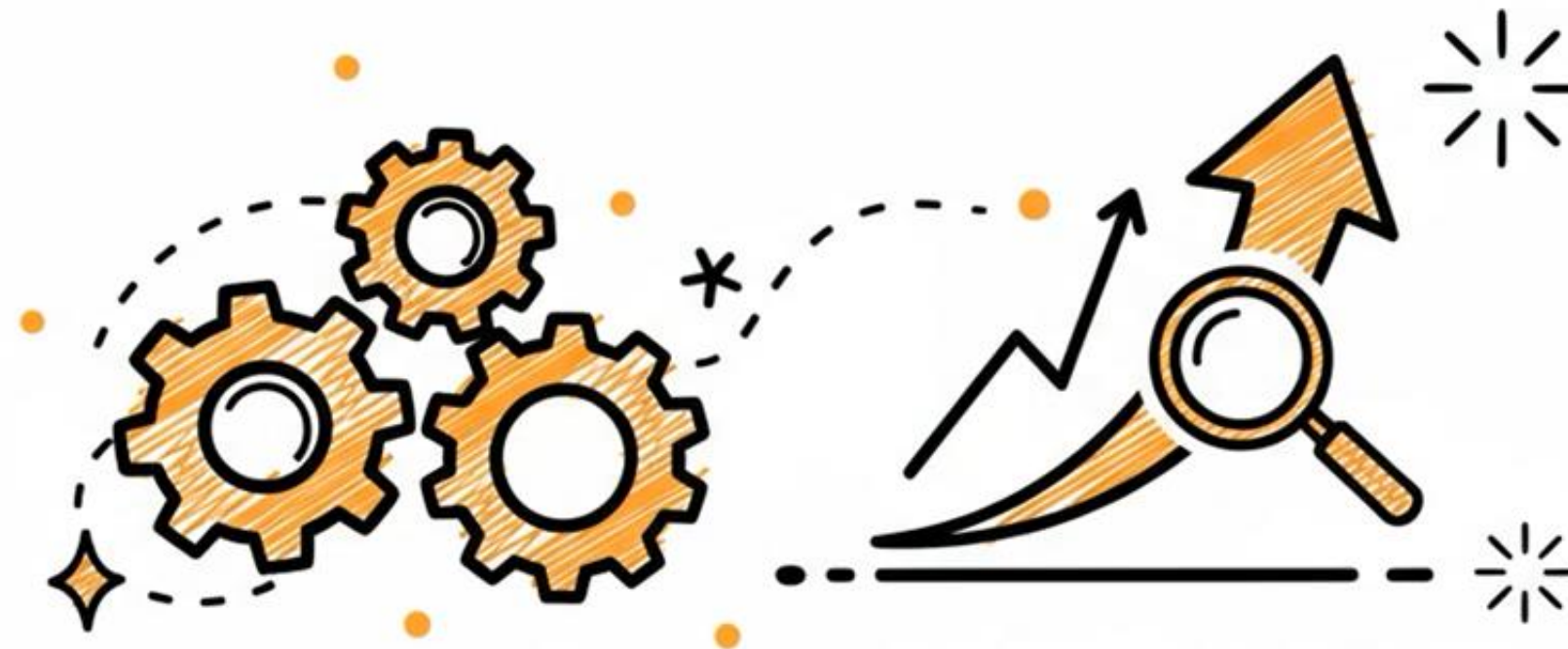
Phase 1 : Hypothesis



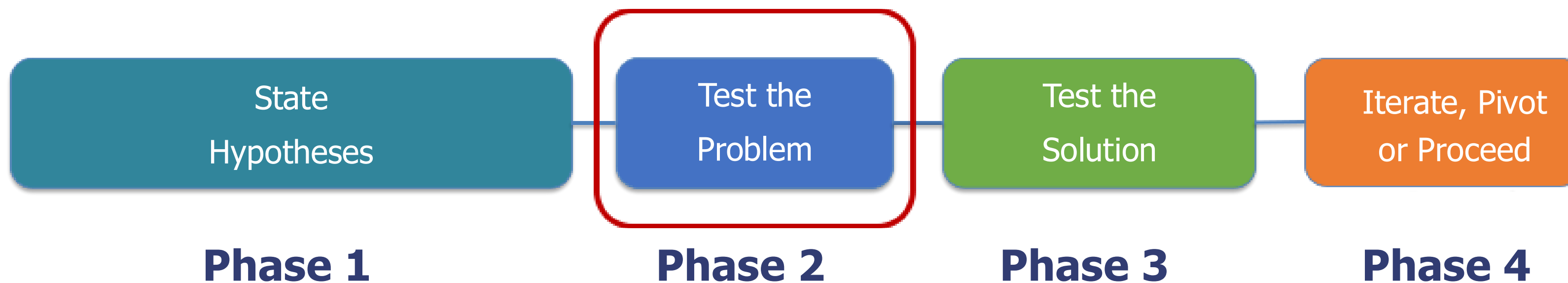
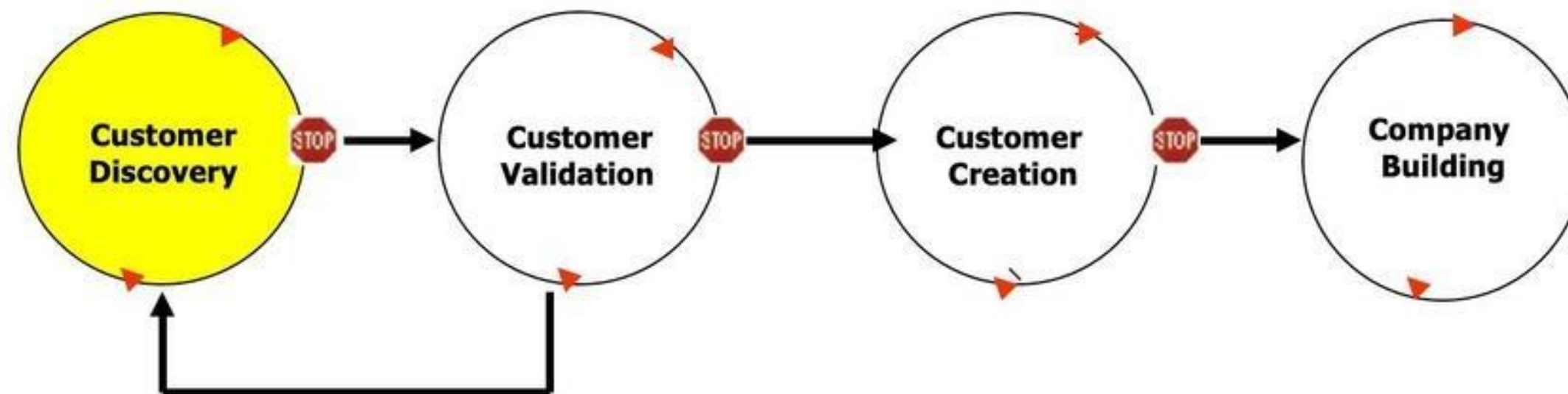
**Capture your Hypothesis (Assumptions) on a
Business Model Canvas (BMC)**

Recall – What is BMC

The Business Model Canvas



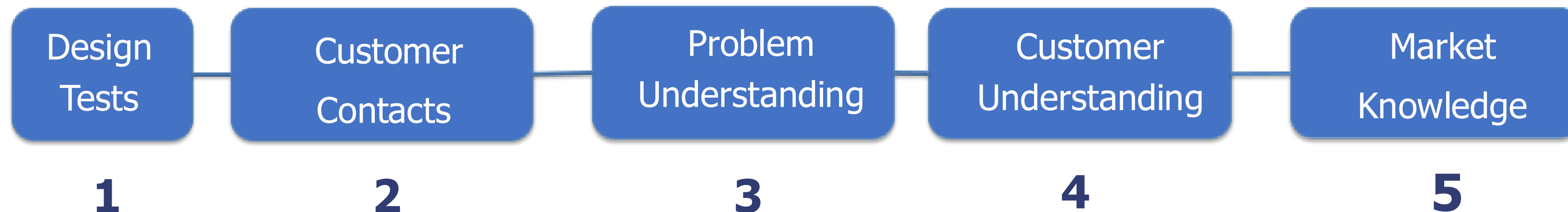
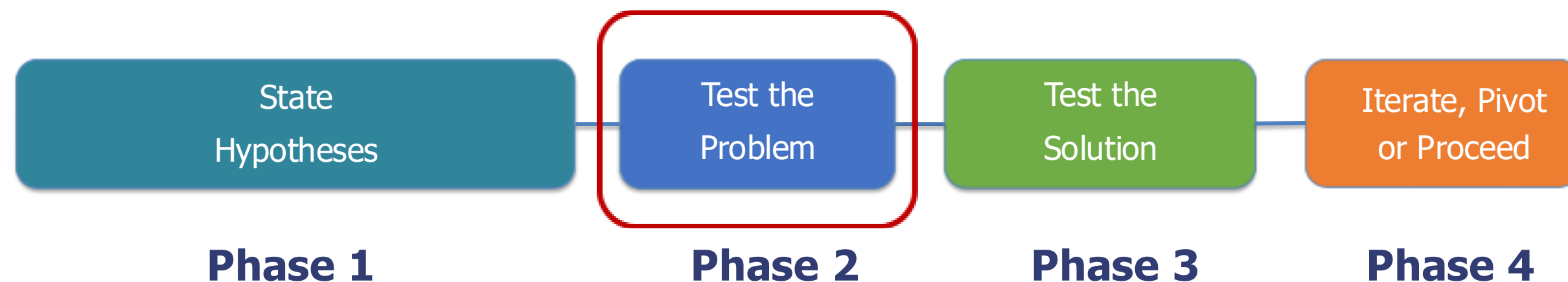
Customer Discovery Overview



Notice that testing the Problem comes before testing the Solution

Customer Discovery Overview

Phase 2 : Test the Problem



Revise your BMC based on Testing your Problem

Recall - Hypothesis

- A **tentative, testable statement or assumption** made about a phenomenon, relationship, or outcome, which can be **validated or falsified through observation or experimentation**.
- It is formulated **before data is collected**, based on existing knowledge, reasoning, or assumptions, and serves to **guide systematic inquiry**.

Examples:

- a. Sounds Familiar:** “Students who supplement regular course material with 5–7 minute concept videos and audio will perform better in quizzes, ISAs and ESAs than students who rely only on live lectures and standard course material.”: **Predicts a relationship between an intervention and an outcome**
- b.** If the college canteen introduces UPI-only fast lanes during peak hours, average waiting time for students will reduce by at least 30%.”: **Testable with a measurable outcome**
- c.** “Urban Indian working professionals aged 22–35 prefer ordering groceries via mobile apps rather than visiting local kirana stores at least three times a week.”: **Customer behaviour validated through data**
- d.** “In T20 cricket, teams that score more than 50 runs in the powerplay have a higher probability of winning the match.”: **Data driven assumption validated by match statistics**

How to Write a Good/Effective Hypothesis

Characteristics of a Good Hypothesis: “QRST”

Quantifiable

Minimal **Resources** (& time required to validate the hypothesis)

Specific

Testable (Pass or Fail)

Method: “X Y Z Hypothesis” – A least X% of Y (target market) will do Z

- Example: 20% of the luxury car market will buy at least 1 EV (electric) car
- Example: 20% of the luxury car market will buy at least 1 EV (electric) car priced at Rs 50 Lakh or more
- Example: 20% of the luxury car market will buy at least 1 EV (electric) car priced at Rs 50 Lakh or more in next 3 years

Note: More specific is better, but sometimes you can start with a simple test and add details as you go along

QRST: High-Impact Takeaways for Customer Development

PESU CIE - Hypthesis Framing Model

Q "No numbers = no hypothesis."



R "If it's expensive to test, it's the wrong wrong hypothesis—right now."



S "Vague hypotheses always look successful."



T "If you can't fail, it's can't teach."



B Belief is not evidence.



L In the Search phase, learning speed being right.



QRST – Your Mental Model

Before you test any idea, ask:

- Q** Can I measure it?
- R** Can I test it quickly and cheaply?
- S** Do we all interpret it the same way?
- T** Will the result clearly say Pass or Fail?

If the answer is 'no' to even one—rewrite the hypothesis.

QRST – Summary

QRST — Takeaways



A good hypothesis makes failure cheap.



If you need a product to test it,
you waited too long.



Specificity is kindness—to your future self.



Customer Development is not about being
right, it's about finding out fast.

Framing: What's our Hypothesis?

Template – What's our Hypothesis (*before* the test)

1	Hypothesis (H₁)	We believe that
		<i>Critical:</i> Low Med High
2	Test(s)	To verify that
		<i>Test Cost:</i> Low Med High <i>Data Reliability</i> Low Med High
3	Metric(s)	We will measure
		<i>Time Required:</i> Low Med High
4	Criteria	We are right if

For sample template filled out, see:

CIE-L2-Template -Hypotheses Capture and Testing

Framing: What's our Hypothesis?



Hypothesis Template

Answer these 4 questions before any test.

1

#1 What do we believe?

Our assumption about the customer or problem

2

#2 How will we test it quickly?

Tests we can run quickly and cheaply



3

#3 What number will we track?

Metrics we will measure to check the test

4

#4 What result means we're right?

Criteria to decide Pass or Fail



For sample template filled out, see:

CIE-L2-Template: Hypotheses Capture and Testing

Hypothesis Framing – A Karaoke Club in Bangalore



CASE STUDY: THE BOLLYWOOD KARAOKE COLLECTIVE - VALIDATING A PASSION-BASED BUSINESS (Bangalore, India)

1. HYPOTHESIS (What we believe)



Criticality: High

"We believe that there is a sufficient market of Indian Language Film Songs Enthusiasts in Bangalore willing to pay a premium (INR 2,000) for a curated Karaoke or Live Music singing experience, allowing for 8–10 sustainable events per year."

2. TESTING ROADMAP (How we verify)



Test 1: Smoke Test
(Medium Reliability)
Landing page + INR 15k Instagram ads to capture interest.



Test 2: Deposit Test
(High Reliability)
Asking "interested" leads to pay a ₹500 refundable deposit.



Test 3: Pilot (MVP)
(Very High Reliability)
Running one small-scale event (25 singers) to test ops and "vibe."

3. KEY METRICS (What we measure)



Demand & Conversion

- CPA: Cost Per Acquisition
- Deposit Conversion Rate



Financials & Sustainability

- Contribution Margin
- Audience-to-Singer Ratio



Retention

- Repeat Intent: % returning

4. SUCCESS CRITERIA (The "Go/No-Go" Signal)

Validated If: ✓

- ☒ **Direct Validation** 🎯: ≥ 20 singers pay ₹500 deposit within 14 days.
- ☐ **Financial Health** 💰: Revenue ≥ Total Costs (Breakeven).
- ☐ **Acquisition Viability** 💵: Cost to acquire singer ≤ ₹400.
- ☐ **Retention Signal** ❤️: ≥ 40% intent to return for next month.

Sample Template (Filled Out) for Capturing Hypotheses

Example: UBER

One of Uber's PROBLEM hypothesis (say, P1) is that existing solutions (i.e. before Uber came to market, e.g. taxi, chauffeur-driven cars, rental cars, etc) for car-based transportation in metros/cities) had several 'pain' points (costly, time-inefficient, cumbersome, etc)

(Note there may be more problems and can be captured as P2, P3, etc).

Other hypotheses related to Solution (S1, S2, etc), Value Prop (VP1, VP2, etc), Customer segments (CS1, CS2, etc), Price (Pr1, Pr2, etc) should be done along similar lines.

1	Hypothesis (H_P1)	We (Uber Team) believe that <i>Existing road transportation solutions based on cars in metropolitan cities is <u>costly</u> and <u>time-consuming</u>*</i>. <u>Critical</u>: Low Med High (marked High as this is the fundamental assumption on which Uber business is built)
2	Test(s)	To verify that <i>T1. <u>Cost</u> is a barrier that prevents would-be customers from using existing solutions</i> <i>T2. <u>Cost</u> is a barrier that prevents customers from using existing solutions sparingly (i.e. they may take fewer rides)</i> <i>T3. Customers spend way more <u>time</u> getting taxi/rental-car/etc than they wish for intra-city transportation</i> <u>Test Cost</u>: Low Med High <u>Data Reliability</u> Low Med High (Note: I have arbitrarily selected High and Med above; actual value may vary)
3	Metric(s)	We will measure <i>M1. Average <u>cost</u> per KM spent by an existing customer and what % lower would be delightful to the customer</i> <i>M2. <u>Cost</u> per km that would allow new customers to take Uber</i> <i>M3. <u>Time</u> taken to find a taxi during morning / evening peak hrs</i> <u>Time Required</u>: Low Med High
4	Criteria	We are right if <i>1. our solution (Uber) can be at least 50%** less than existing solutions, on cost per km basis</i> <i>2. we can sign up 10% more people than who use taxis today if we offer cost per KM which is 50% less than current solution</i> <i>3. Uber takes 80% less time than now to get a ride for customer</i>

* Note: For sake for this example, we have combined 2 hypotheses (cost & Time). Do not combine, separate out hypotheses.

Testing: Hypothesis Validation & Insights

Template – Learnings/Insights *after* testing the hypothesis

1	Hypothesis	We believed that	<u>Critical</u> : Low Med High
2	Observation	We observed that	<u>Data Reliability</u> : Low Med High
3	Learnings and Insights	From that we learned	<u>Action Required</u> : Low Med High
4	Decisions and Actions	Therefore we will	

Sample Template for Learnings from Testing Hypotheses

Template – Learnings from Testing Hypotheses

(what decisions you can make)

Template – Learnings After Testing Your Hypotheses (Assumptions)

Example: Uber

See Page 1 for the hypotheses we wanted to test:

Existing road transportation solutions based on cars in metropolitan cities is costly and time-consuming.

To test the above hypotheses, we (Uber team) went out and sought current users of taxi and potential/aspiring customers (not using taxi because it is costly &/or time-inefficient). After we ran the experiments (also called test), we can summarize the findings as below and reach some decisions/conclusions.

Note: It is not necessary that we have to prove the hypotheses to be correct! The experiment should be run in an unbiased manner. You can take appropriate decision based on results of the test/experiment.

1	Hypothesis (H_ P1)	We believed that <i>Existing solutions inside metropolitan cities (for passenger vehicles) transportation is <u>costly</u> and <u>time-consuming</u>.</i> <i>Critical: Low Med High</i>
2	Observation	We observed that <i>1. Average cost per KM is Rs 25 for existing solutions (compared to our Rs 10/KM)</i> <i>2. New users would use Uber if average cost per KM is Rs 12</i> <i>3. Time taken to call/get a taxi during morning & evening peak hours was 25 minutes and 40 minutes respectively</i> <i>Data Reliability Low Med High</i>
3	Learnings and Insights	From that we learned <i>1. Uber is cost competitive for existing users (Rs 10/KM vs. Rs 25/KM)</i> <i>2. New users would very likely use Uber (willing to pay Rs 12/KM while our cost is Rs 10/KM)</i> <i>3. That we can cut the time to find taxi by >60%</i> <i>Action Required: Low Med High</i>
4	Decisions and Actions	Therefore <i>1. We will continue with the app-based taxi calling service Uber and target <u>existing</u> users of taxi</i> <i>2. We will continue with app-based taxi calling service Uber and target <u>new</u> users of taxi</i> <i>3. We have a compelling differentiation on time-saving for customer and use this as a key value-proposition</i>

Hypothesis Insights – A New Singing Club in Bangalore

LEARNINGS & INSIGHTS: THE BOLLYWOOD KARAOKE COLLECTIVE (After Pilot Events 1 & 2)

1. HYPOTHESIS

(We believed that)

Critical: High

There is a sufficient market of Indian Language Film Songs Enthusiasts in Bangalore willing to pay a premium (INR 2,000) for a curated Karaoke or Live Music singing experience, allowing for 8–10 sustainable events per year.



2. OBSERVATION

(We observed that)

Data Reliability: High

✓ **Demand:** 45 singers paid ₹500 deposit (Events 1 & 2); 42 converted to full ₹2,000 payment.



📁 **Financials:** 3-star hotel venue costs 15% higher than projected; Singer Acquisition Cost (CPA) was lower at ₹310. ↓

🔄 **Retention:** 65% of singers from Event #1 immediately requested to book for Event #2.



3. LEARNINGS & INSIGHTS

(From that we learned)

Action Required: Med

📺 **The "Hook":** Professional video recording is the primary value driver; social shares drove 40% of organic inquiries for Event #2.

🪑 **The "Gap":** Audience tier is lagging; difficult to sell non-singer tickets at high price without a headliner.

🏨 **The Cost:** 4-star hotels look premium, but F&B minimum guarantees are a financial risk without 100% attendance.



4. DECISIONS & ACTIONS

(Therefore we will)

➡ **Scale:** Commit to a 6-month calendar of events (₹2,000 price point validated).

➡ **Optimize:** Shift to boutique performance studio/social space to reduce fixed F&B costs while maintaining "premium vibe".

➡ **New Feature:** Introduce a "Bring a Friend" discounted audience pass to solve the audience-to-singer ratio issue.



Visualizing the move from assumption to validated learning.

How do you Test your Hypotheses (Steps)

- **Capture** your hypotheses (assumptions/opinions)
- **Design** the Experiments to test assumptions/opinions
- **Test** your hypotheses by conducting your key hypotheses
 - Run the Experiments ("Get out of the Building", Customer interviews, *not* surveys)
 - Gather Data/Results from the Experiments (hypothesis is validated or invalidated)
 - What insights did you gather? How does this affect your problem/ solution/ BMC/ etc?
 - ***Tip:*** *For your experiments, recommend having some artifact (Pretotype/Prototype/MVP)*
 - ***Artefact can be:*** *Paper, Digital, Native (in order of ascending fidelity: low -> high)*

Tips/Tricks – Some useful Practical Tips in Validating Your Hypothesis

- Find 'real' customer(s)
 - 'Real': who you (potentially) plan to sell your solution to
- While your hypothesis may target global/country/city, test it 'near' you
 - "Think global, test local" first (you can always expand this out later)
- How to test your hypothesis?
 - Verbal is good, but "prototype" is better! (People pay attention if you show something)
 - Prototype is none of the following: *expensive, takes time to build, or complex!*
- Develop customer interviewing skills (see CD Toolkit)
 - See "Tips and Tricks", videos and other resources listed in subsequent slides

Tips/Tricks: Before 'Getting Out of the Building'

- Insights are not 'inside the building', therefore, 'Get Out of the Building'
- What are Insights?
 - Honest, data driven, careful analysis of validation of your hypotheses
- Prepare for 'Get-out-of-the-Building' (*GOOB*)
 - See: Tips/Tricks: Before Leaving the Building slide
- Test your hypotheses with actual Customers
 - As a team, plan on 'Who / What / When' for 'Get Out of the Building'
 - Meet at least 35 customers (per team per week, ~5 per team member)

Video

Reasons Not to Be an Entrepreneur

(Stanford eCorner, 2:35m)



Resources



Subsequent slides list resources (videos/articles/etc) related to
Customer Discovery / Development

Tips/Tricks: Before 'Getting Out of the Building'

Before Leaving the Building: Helpful videos (by Steve Blank):

(<https://venturewell.org/i-corps/llpvideos/customer-discovery/before-leaving-the-building/>)

- Discovery is for Founders (1:30) CD
- Pre-Planning Contacts (4:34) CD
- Customer Interview Dry Runs (0:49) CD
- Pass/Fail Experiments (1:32) CD
- **Customer Dev: Insight not just Data!** (2:12, Kauffman Foundation)

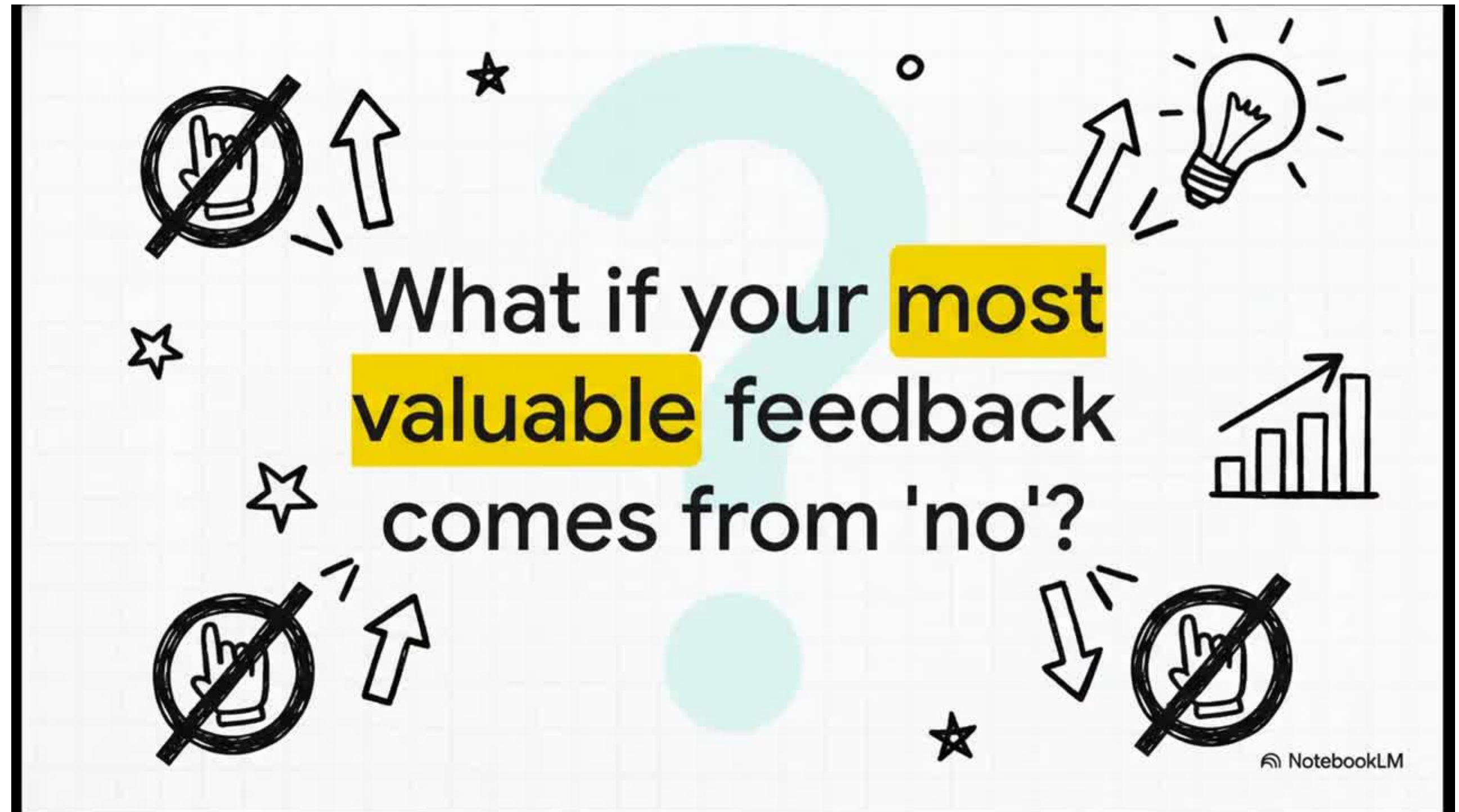
Tips/Tricks: Before 'Getting Out of the Building'

- **12 Tips for Early Customer Development Interviews** (Giff Constable): [Link](#)
- **Interviews vs. Experiments** (Giff Constable): [Link](#)

"Interviews give you the greatest insights. Experiments give you the greatest proof"

- Results of hypothesis validation fall into 4 buckets:
- See Progress Board template: **I**nvalidated, **V**alidated, **W**IP*, **U**nclear results

Tips/Tricks: Before 'Getting Out of the Building'



Tips/Tricks: 12 Tips for Early Customer Development Interviews

12 Tips for Early Customer Development Interviews



Know your Goals & Questions ahead of time



**One person at a time
(not a group)**



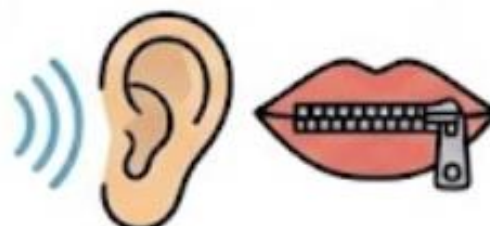
**Be open to hearing things
you DON'T want to hear**



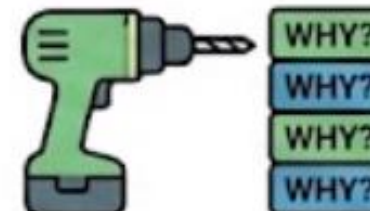
**Disarm "Politeness
Training" (tell the person
to be brutally honest)**



**Ask open-ended
questions**



Listen, Do NOT Talk



**Drill down on details –
"5 Whys" method?**



**Separate behavior and
feedback in a discussion**



**Focus on actual
behavior**



**Recap what you heard
and/or misrepresent to
confirm!**



**Ask for Introductions –
Get leads to talk to
another 1-3 people!**



**Write up your notes as
quickly as possible**

Tips/Tricks: 12 Tips for Early Customer Development Interviews

- 1. Know your goals & questions ahead of time**
- 2. One person at a time (not a group)**
- 3. Be open to hearing things you DON'T want to hear**
- 4. Disarm "Politeness Training" (tell the person to be brutally honest)**
- 5. Ask open ended questions**
- 6. Listen, don't talk**
- 7. Drill down on details –"5 Whys" method?**
- 8. Separate behavior and feedback in a discussion**
- 9. Focus on actual behavior**
- 10. Recap what you heard and/or misrepresent to confirm!**
- 11. Ask for introductions –Get leads to talk to another 1-3 people!**
- 12. Write up your notes as quickly as possible**

Resources - More Videos on Customer Development

- 1.*Engaging the Customer (CD05-06) **Right & Wrong way!**
2. Assuming you know (CD09-10) **Right and Wrong way!**
3. Customers Lie (CD11) **Example!**
- 4.* Extracting Insight from Data (CD12)
5. Attention to Outliers (CD13) **Right and Wrong way!**
6. The User, Buyer & Saboteur (CD14-15)
7. Pre-Planning Pt1 Pt2 Pt3 (CD30, 31, 32)
8. Interviews Pt1, Pt2 (CD33-34) **Right and Wrong way!**
- 9.* Conducting a Customer Interview (CD46)
10. Letting the Customer interview flow (CD47)
11. Looking for Insights (CD50)
12. * The Distracted Customer (CD2021) **Right/Wrong way!**
13. * Customer Empathy (CD23-24) **Right/Wrong way!**
14. * Asking the Right Question (CD25-26) **Right/Wrong way!**
15. Finding Patterns (CD49)
16. * Communicating your discoveries (CD52)

<https://vimeo.com/groups/204136/videos>

Resources

Subsequent slides list **step-by-step** overview and tip-and-tricks for each Phase of **Customer Discovery**



Customer Discovery Overview

- Hypothesis

State your Hypothesis (for your startup opportunity)

- **Problem** (why?)
- Faced by target **Customers**, Customer Segments (whom?)
- Your **Solution? Value proposition** you will offer (what?)
- Your assumptions on Market (size, type), Team, etc
- Based on above, create BMC (BMC#1) or LC (Lean Canvas #1)

State Hypotheses
Create Business Model
Canvas

Phase 1

Customer Discovery Overview

- Tips & Tricks

What If this gets Overwhelming?

- **Prioritize, Prioritize, Prioritize!**
 - **Customer** Jobs
 - **Customer** Pains, Customer Gains
 - **Customer** segments
 - **Solution** (Features/ Benefits you offer)
 - **Market** (type, size, channel)
- **Prioritize Speed over Accuracy**
- **Divide & Conquer (team/ delegation)**

State Hypotheses
Create Business Model
Canvas

Phase 1

Customer Discovery Overview

- Test the Problem

Test the Problem

- **Design** the **tests** (to validate the problem hypothesis)
- Run the tests' –use data / insights to (in)validate hypothesis
 - **Interview** potential **customers**
- Update the BMC (BMC#2)

Test the Problem

Phase 2

Customer Discovery Overview

- Tips & Tricks

What If this gets Overwhelming?

- Customer Contact / Interview
 - Go through the tips/tricks, videos to prepare
 - Have a goal for the discussion
 - Be relaxed, let the 'customer' do most of the talking
- Preparing hypotheses
 - Is it testable?
 - (pass/fail –this tells if you are on the right path or not)
 - Is it measurable?
 - (if the right path, what would be meaningful?)

Test the Problem

Phase 2

Customer Discovery Overview

- Test the Solution

Problem - Solution fit ?

Test the Solution

- Create solution representative (preto/prototype, etc)
- Test this with target customers
- Update your solution hypothesis (& pretotypes, etc)
- Update the BMC (BMC#3)

Test the Solution

Phase 3

Customer Discovery Overview

- Tips & Tricks

What If this gets Overwhelming?

- Test the most fundamental assumption/hypothesis
- Build a pretotype (not even a prototype)
- Even a simple pretotype is better than no pretotype!
- Don't be afraid to show your pretotype to Customer
most people will see that you are really committed!

Test the Solution

Phase 3

Customer Discovery Overview

– Pivot or Proceed?

Pivot or Proceed?

- Having a better understanding of the opportunity, would you continue with this idea?
- If so, why? What will be your next steps?
- What help do you need?
- If not, why? What will be your next steps?

Pivot or proceed

Phase 4

Customer Discovery Overview

- Tips & Tricks

What if it gets Overwhelming ?

- Do you have enough insights? (Phase I, 2, 3)
- Everyone in the team engaged during Ph 1, 2, 3?
- All the team members aligned on startup goals?
- Can you make the tough decision on changes to keepgoing?
- (if needed, identify any help needed)

Pivot or proceed

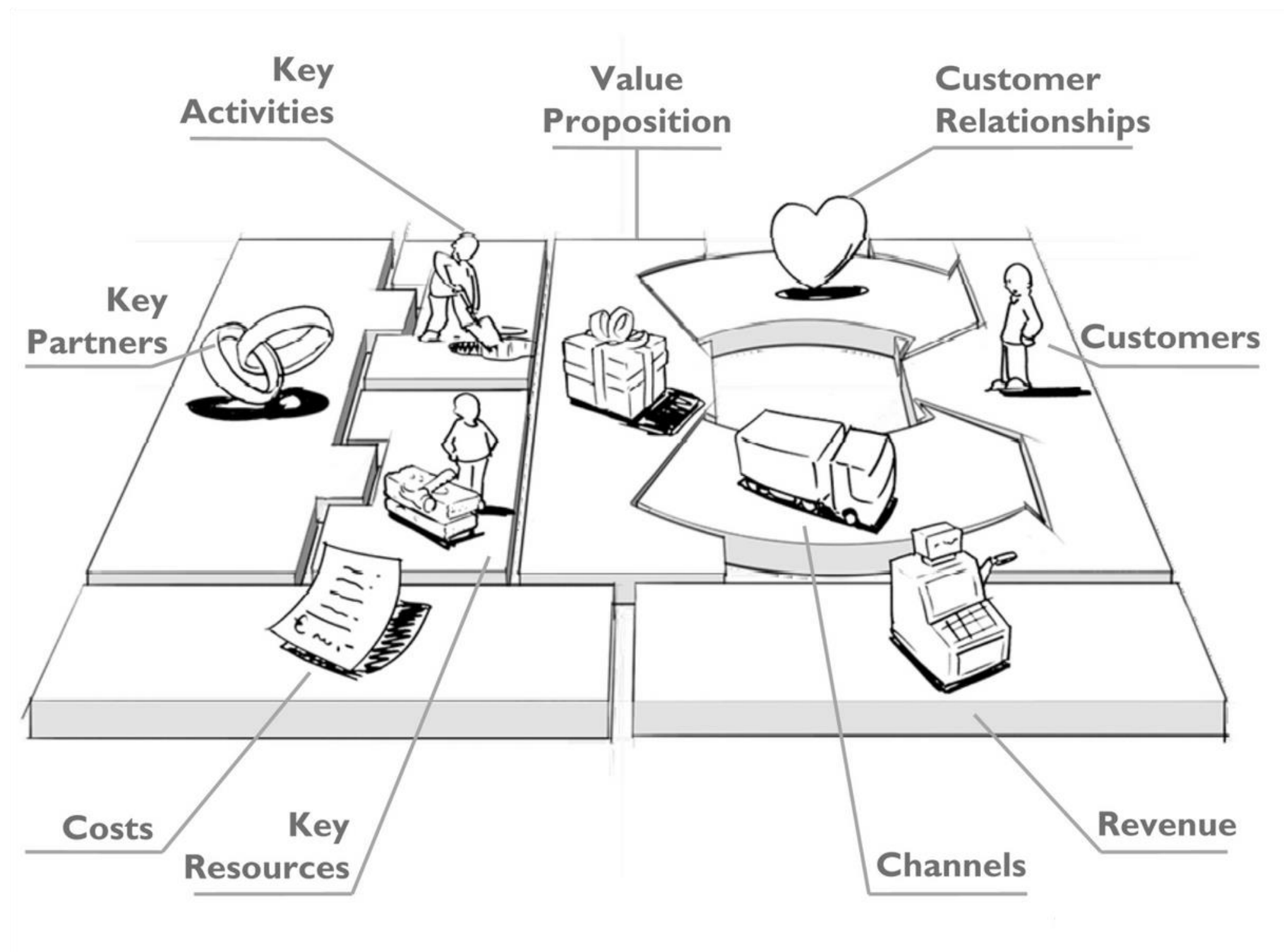
Phase 4

Capturing your Hypotheses in a BMC

- For your team opportunity, capture your hypotheses (see template)
 - Customer (customer segments, channel, relationship, etc)
 - Value Proposition (proposed benefits to customer from your solution)
 - Market Size
 - Price and Cost

Create an initial BMC (don't worry about all the details or accuracy)

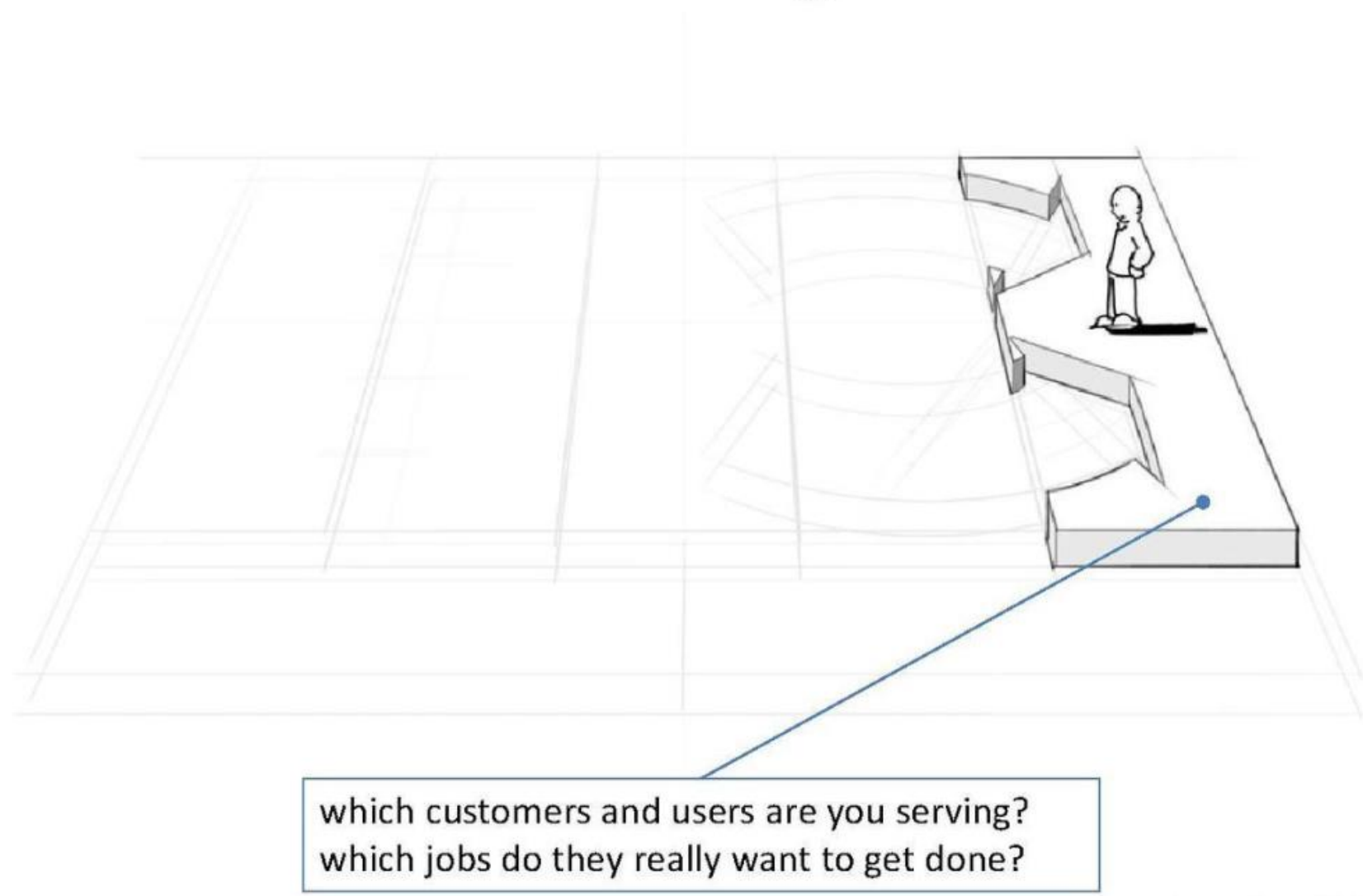
Business Model Canvas (BMC)



Create / Update the BMC for this opportunity

Business Model Canvas (BMC)

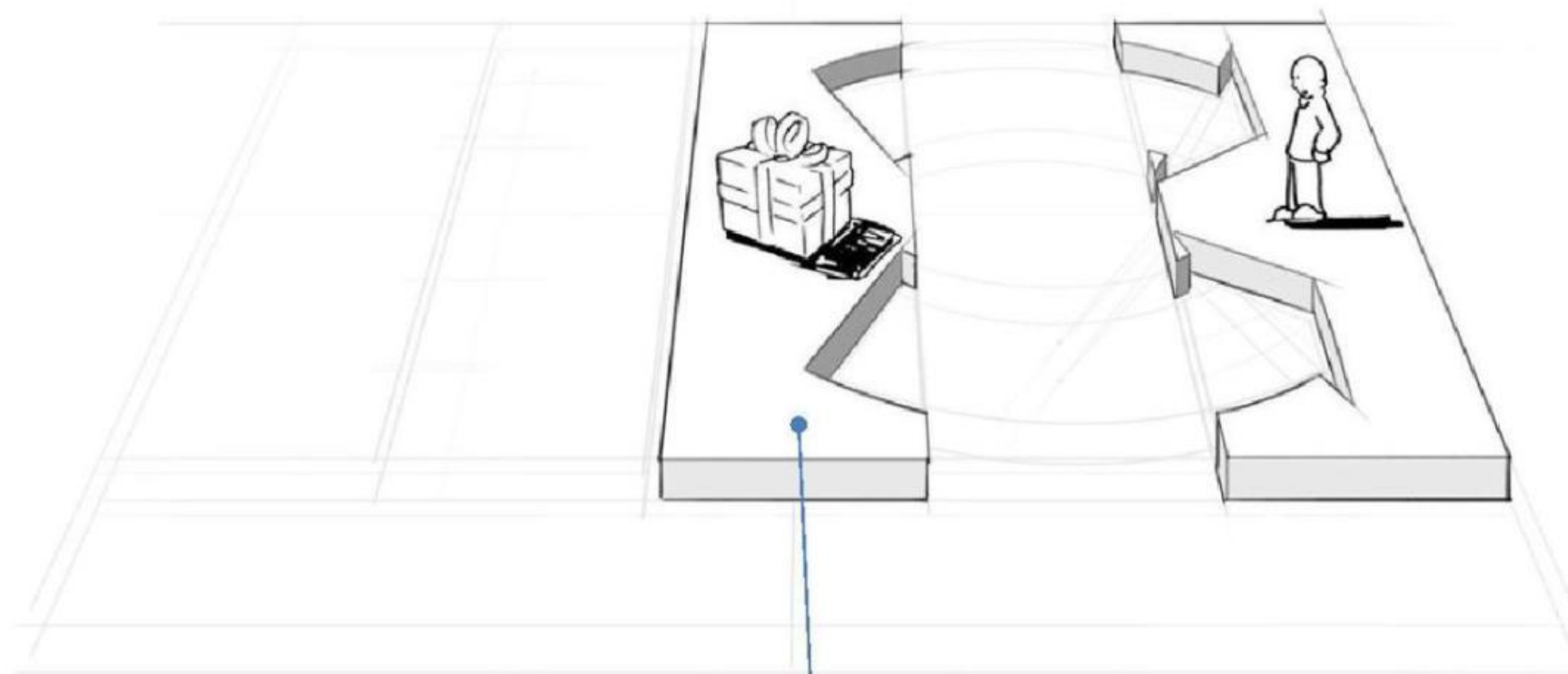
Customer Segments



- Mass market
- Niche market
- Segmented market
- Diversified market

Business Model Canvas (BMC)

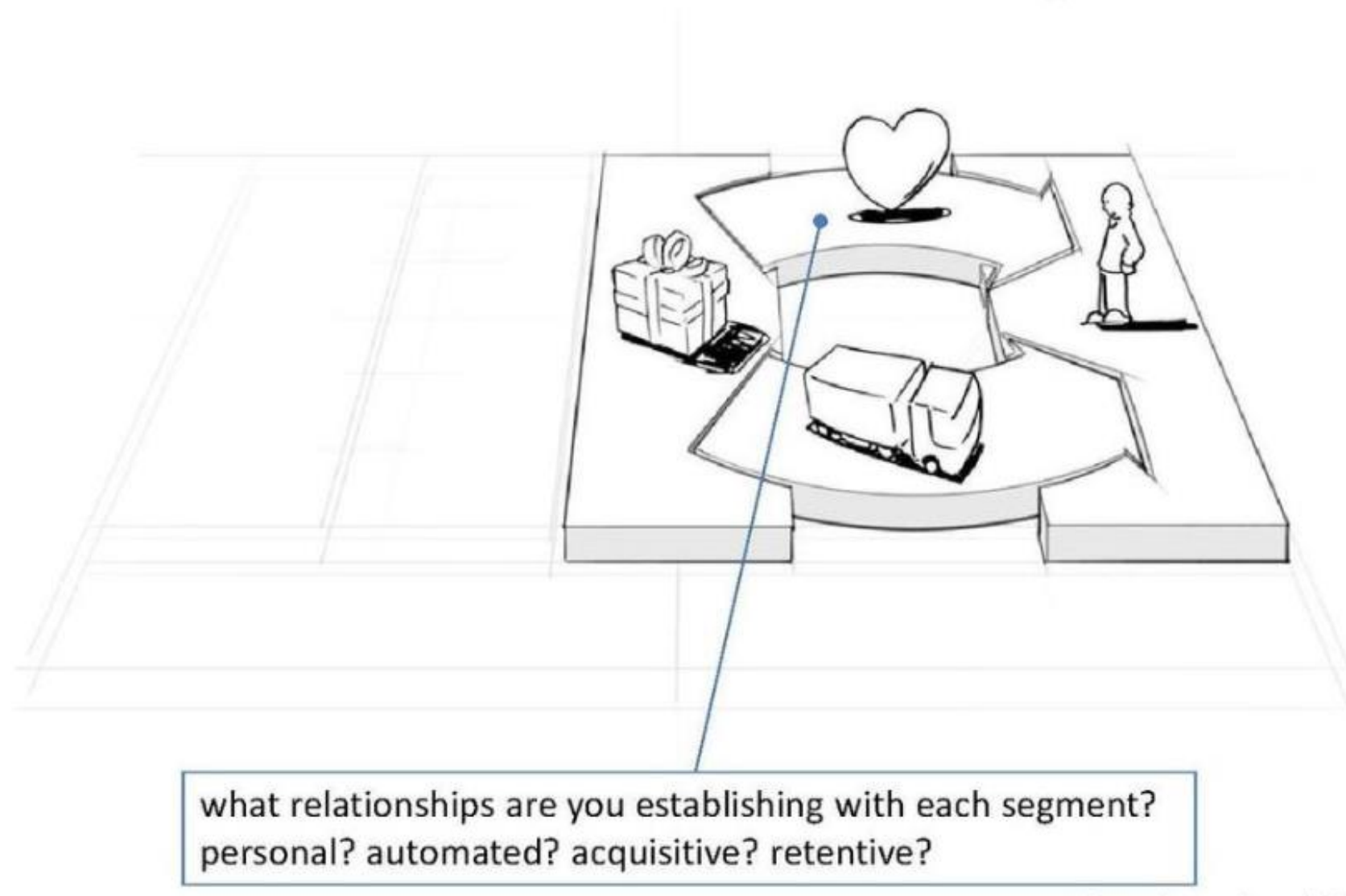
Value Proposition



what are you offering them? what is that getting done for them? do they care?

Business Model Canvas (BMC)

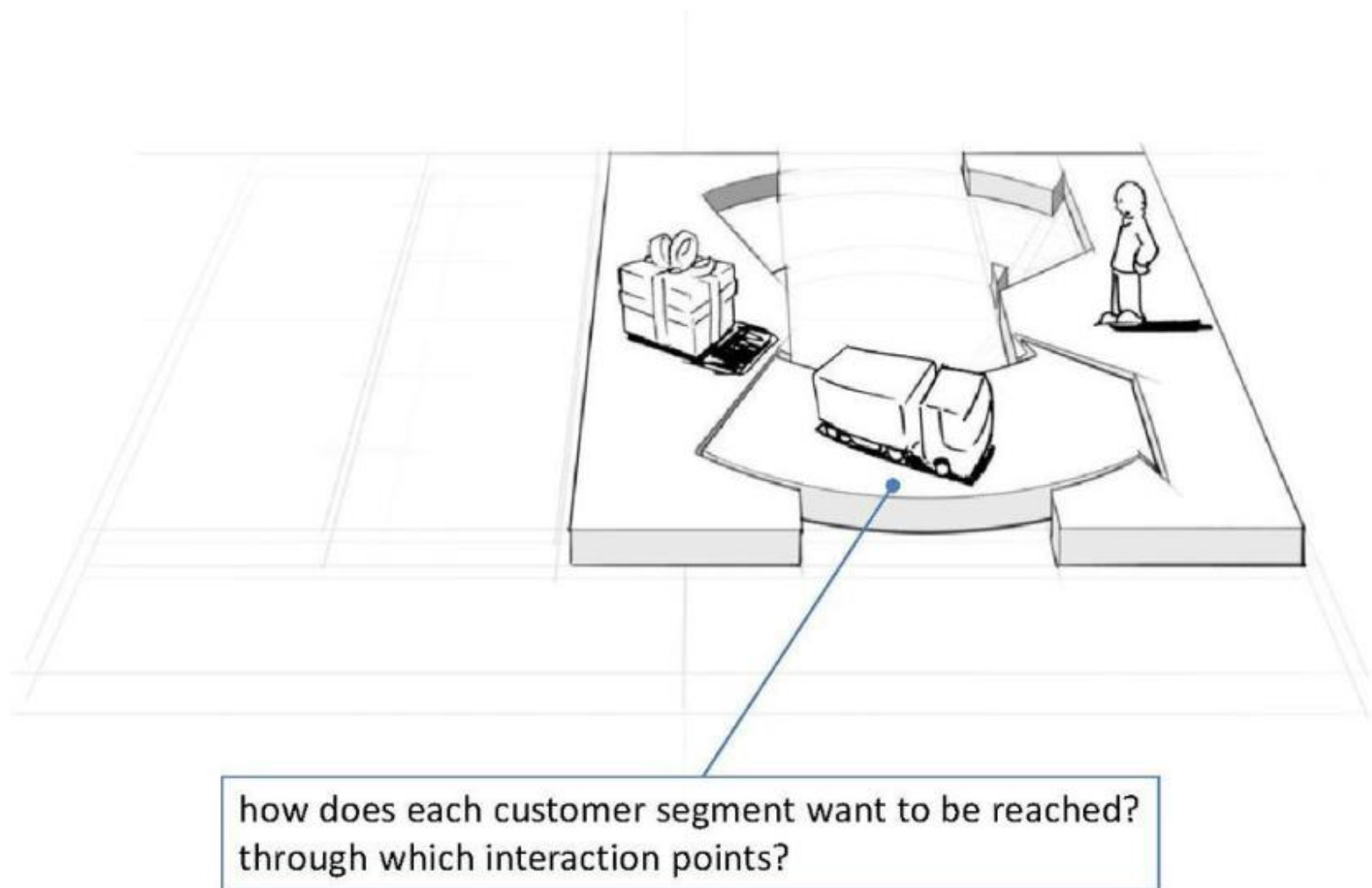
Customer Relationships



- Personal Assistance
- Dedicated Personal Asst
- Self-Service
- Automated Services
- Communities Co-Creation

Business Model Canvas (BMC)

Channels



Channels for:

Awareness



Evaluation



Purchase



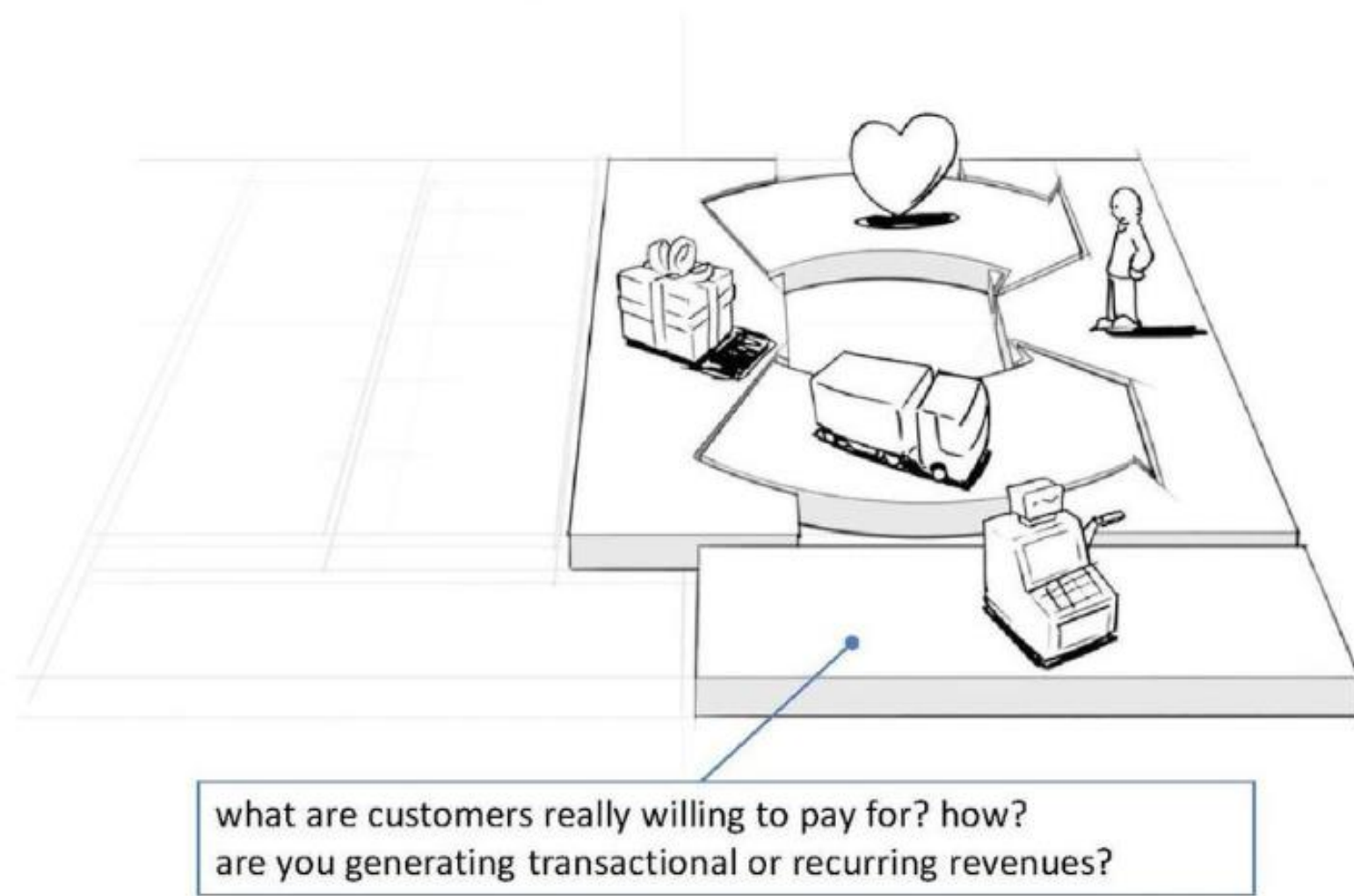
Delivery



After Sales

Business Model Canvas (BMC)

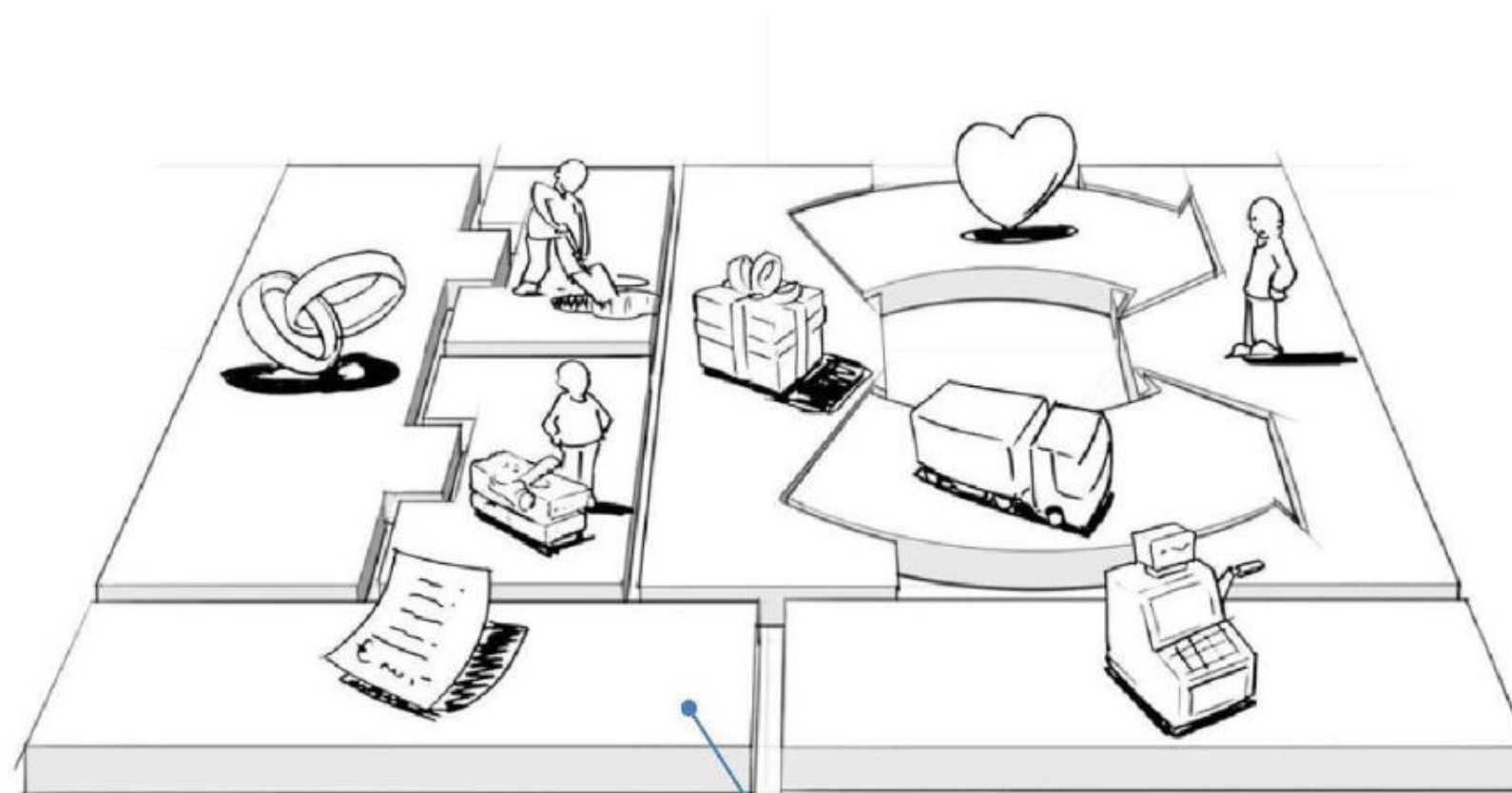
Revenue Streams



- Product Sale
- Usage fee
- Subscription fee
- Lending /Renting
- Licensing
- Types: Transaction, Recurring

Business Model Canvas (BMC)

Cost Structure

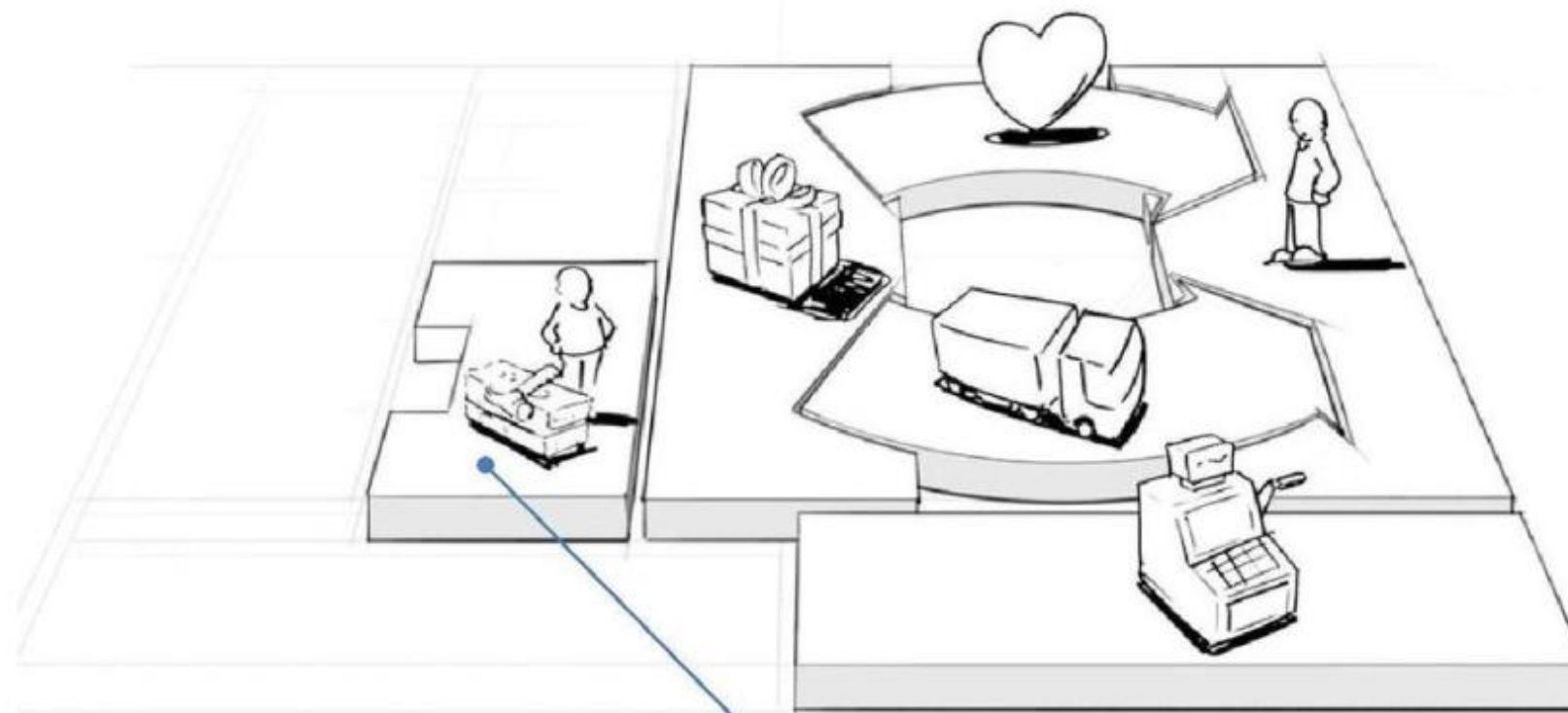


what is the resulting cost structure?
which key elements drive your costs?

- Cost-driven
- Value-driven
- Fixed-costs
- Variable-costs
- Economies of scale
- Economies of scope

Business Model Canvas (BMC)

Key Resources

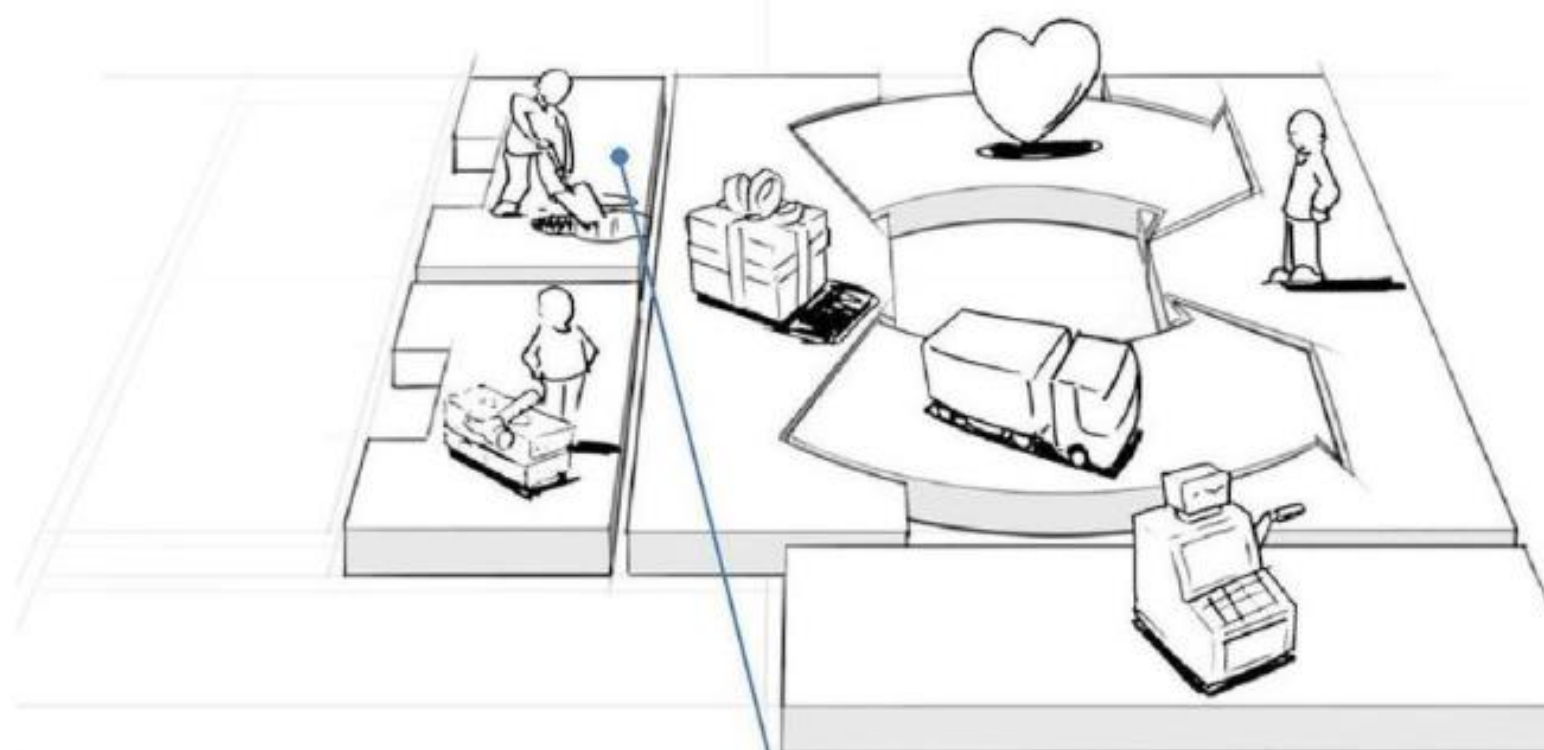


which resources underpin your business model?
which assets are essential?

- Physical
- Intellectual
- Human
- Financial

Business Model Canvas (BMC)

Key Activities

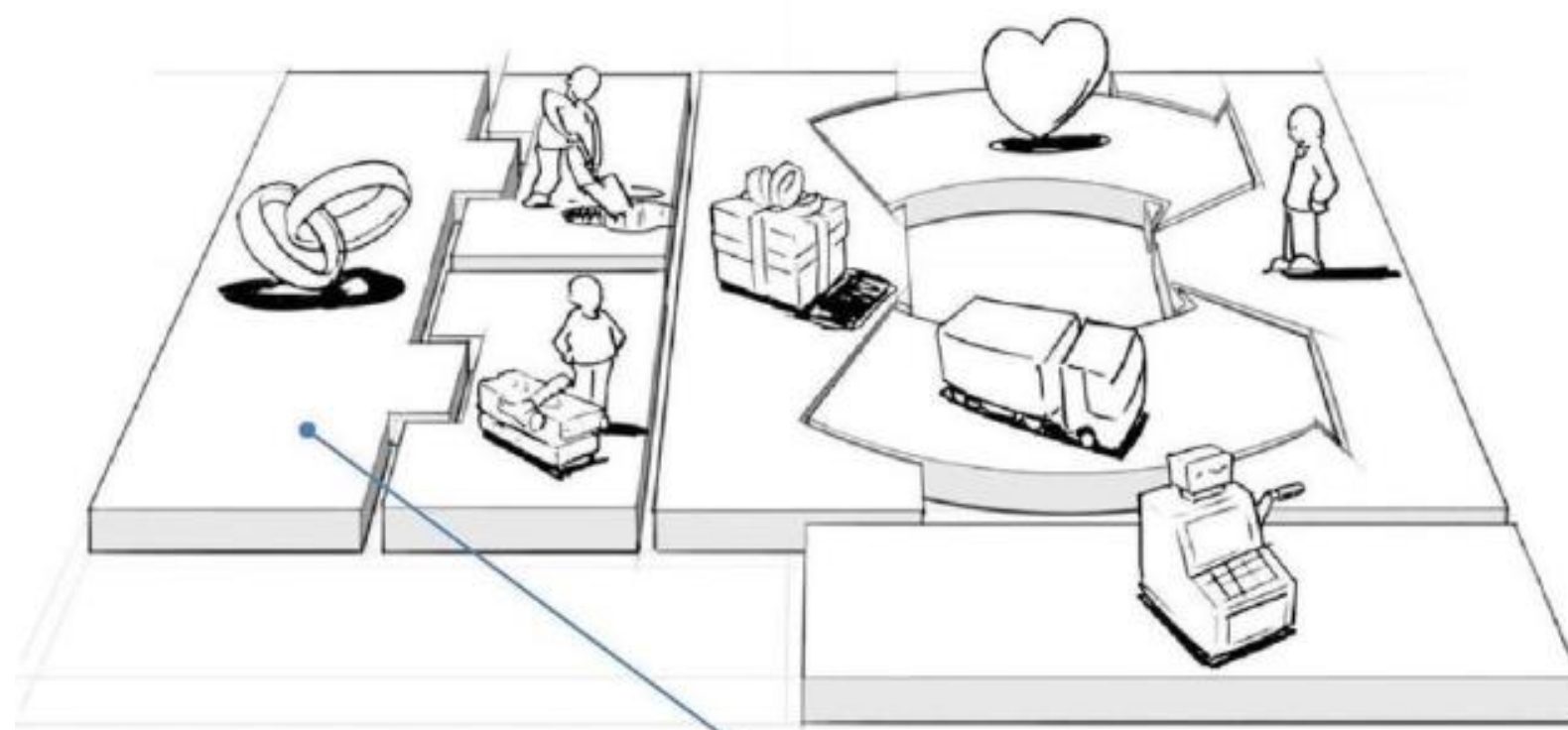


which activities do you need to perform well in your business model? what is crucial?

- Production Services Platform/Network

Business Model Canvas (BMC)

Key Partners

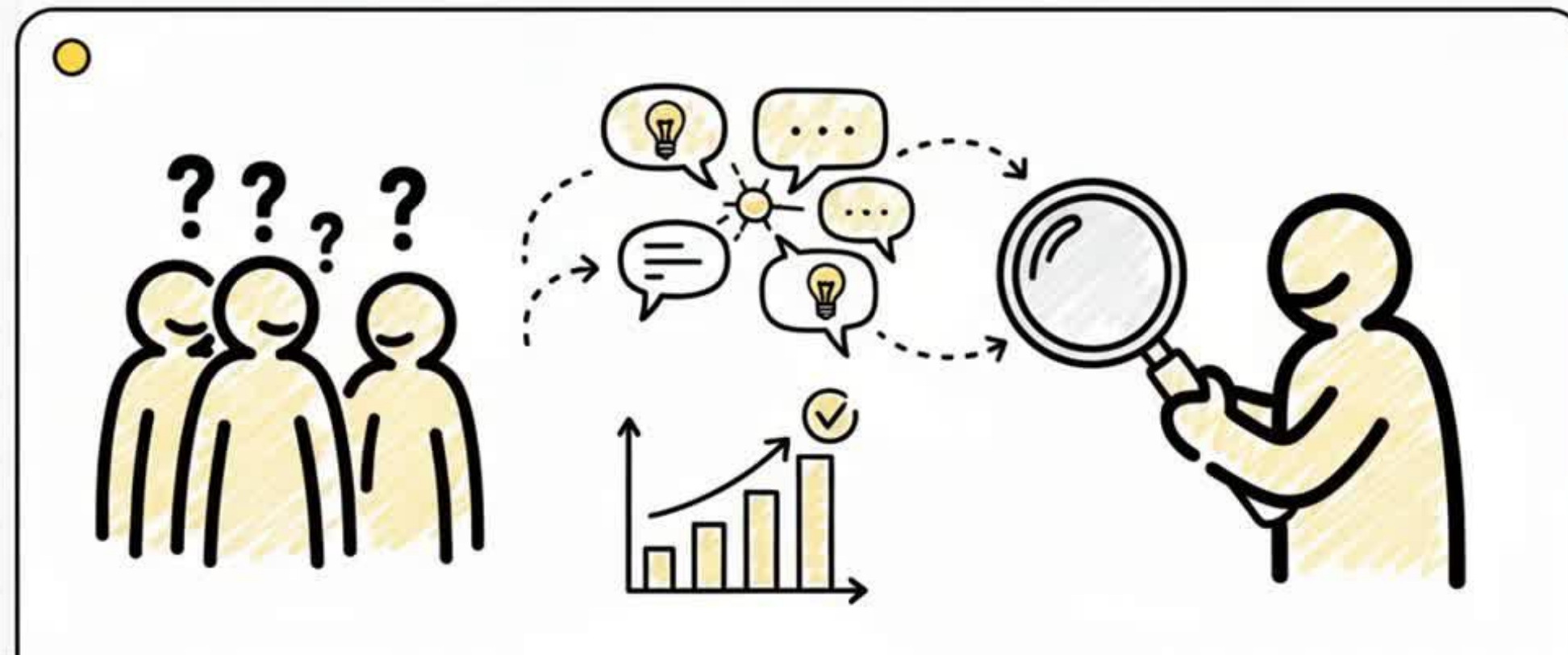


which partners and suppliers leverage your model?
who do you need to rely on?

- Economy of scale
- Specific resources/ capabilities
- Reduction of risk & uncertainty
- Strategic Co-operation
- Joint Venture
- Buyer-Supplier

Summary

Customer Discovery



Summary

- **“QRST” Hypothesis:** *Quantifiable, Resource-efficient, Specific, Testable.*
- **Structured Method:** Define hypothesis clearly.
- **Experiments Design:** Test assumptions smartly.
- **Real Customer Interaction:** Seek 'real' feedback.
- **Data-Driven Insights:** Gather results, analyze.
- **Decisive Steps:** Pivot or proceed with clarity.

Summary

- **Outside Perspective:** Facts lie beyond your walls.
- **Problem-Solution Testing:** Hypotheses validation.
- **Product-Market Alignment:** Early adopter validation.
- **Sales as Proof:** Create after validating sales.
- **Transition Wisely:** Evolve from startup to company.
- **Adaptive Growth:** Embrace chaos for innovation.

Customer Development Manifesto

Steve Blank's Customer Development Manifesto: 17 Principles for Startups



1. Get Outside the Building



2. Pair with Agile Development



3. Failure is Part of Search



4. Don't Fear Failure



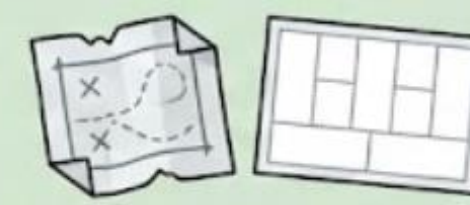
5. Pivot on Insight



6. Test Your Hypotheses



7. Get Early Buy-in



8. No Plan Survives Contact



9. Not All Startups Are Alike



10. Startup Metrics Differ



11. Agree on Market Type



12. Speed and Tempo



13. It's All About Passion



14. Job Titles are Different



15. Preserve Cash While Searching



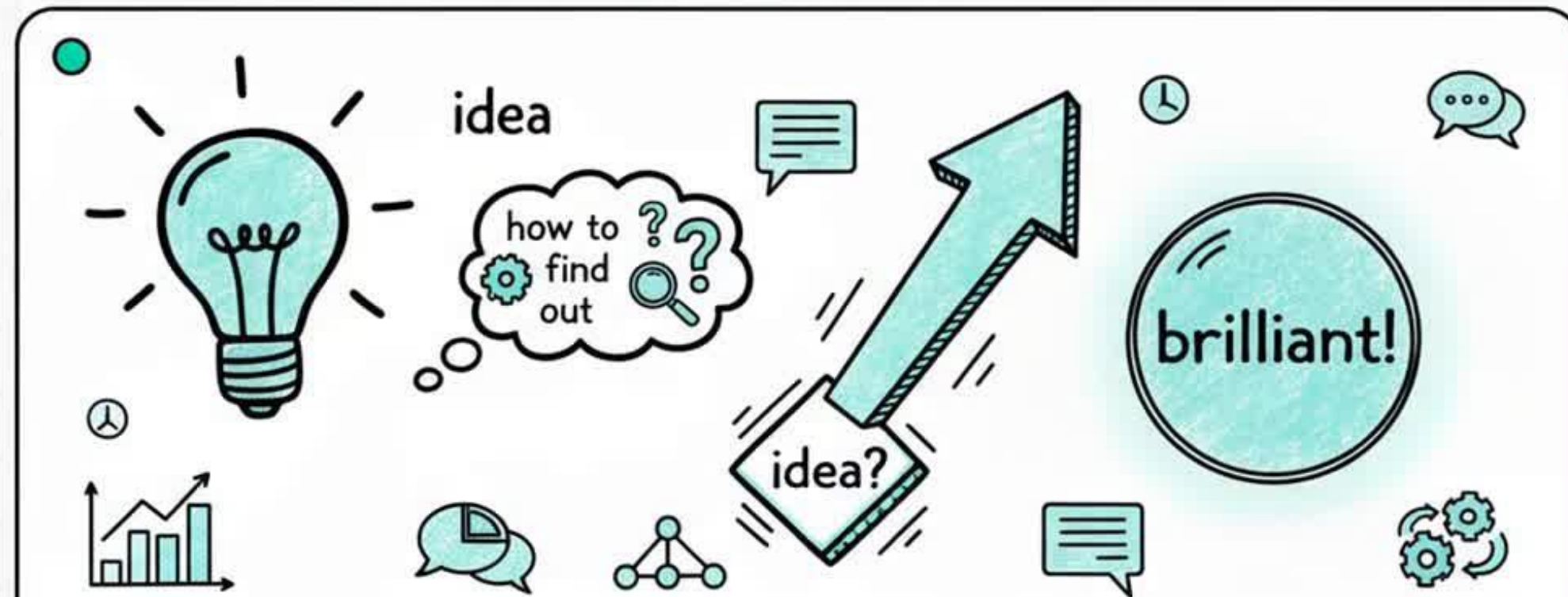
16. Communicate & Share



17. Comfort with Chaos

Customer Development Manifesto

Customer Discovery



Customer Development Manifesto

[\(Link\)](#)

1. There Are No Facts Inside Your Building, So **Get Outside**
2. Combine **Customer** Development with **Agile** Development
3. **Failure** is an Integral Part of the Search for the Business Model
4. Iterations and Pivots are Driven by **Insight (not just data)**
5. Validate Your Hypotheses with **Experiments**
6. No Business Plan Survives **First Contact with Customers**
7. **Communicate** and **Share** Learning (within the team)
8. Agree on **Market Type** - It Changes Everything
9. **Fast** (speed and tempo) and **Fearless Decision-Making**
10. Startups Demand Comfort with **Chaos** and **Uncertainty**

Customer Development Manifesto



11. If it's not About **Passion**, You're Dead the Day You Opened your Doors

12. Startup **Titles/Functions** Are Very Different from a regular firm/company

13. **Preserve** Cash While Searching. After It's Found, **Spend**

14. Success Begins with Buy-In from **Co-Founders** and **Investors**

15. If You're **Afraid to Fail** You're **Destined to Do So**



'9 Deadly Startup Sins' by Steve Blank

9 DEADLY STARTUP SINS



PRODUCT & MARKET MISSTEPS

**Prematurely scaling
based on a presumption
of success.**
Scaling before validating your
business model burns cash quickly.



**Assuming you
know what the
customer wants.**
Startups fail when they build
something nobody needs.



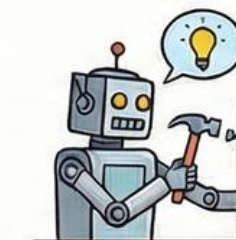
**The "I know
what features
to build" flaw.**
Focus on solving customer
problems, not just adding
features.



**Executing a
sales and
marketing
plan too early.**
Find product-market fit
before spending heavily
on customer acquisition.



FLAWED PLANNING & EXECUTION



**Emphasizing
execution over
learning and
iteration.**
Startups must test
hypotheses and pivot
based on feedback.

**Focusing only on
the launch date.**
The goal is to learn and adopt,
not just to ship a product.



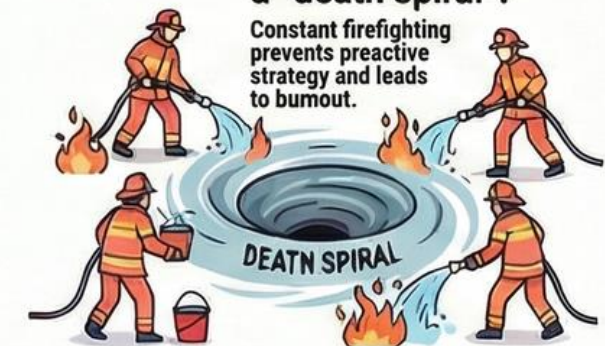
**Writing a rigid
business plan
that disallows
trial and error.**
Your initial plan is a set
of guesses, not a fixed
roadmap.



**Confusing traditional
job titles with a
startup's needs.**
Early teams need versatile
members who focus on tasks,
not titles.



**Management by
crisis, leading to
a "death spiral".**
Constant firefighting
prevents proactive
strategy and leads
to burnout.



'9 Deadly Startup Sins' by Steve Blank

[Link](#)

Assuming “I know what the customer wants”

The “I know what features to build” flaw

Focus on launch date

Emphasis on execution instead of hypotheses, testing, learning and iteration

Traditional business plans presume no trial and no errors

Confusing traditional job titles with what a startup needs to accomplish

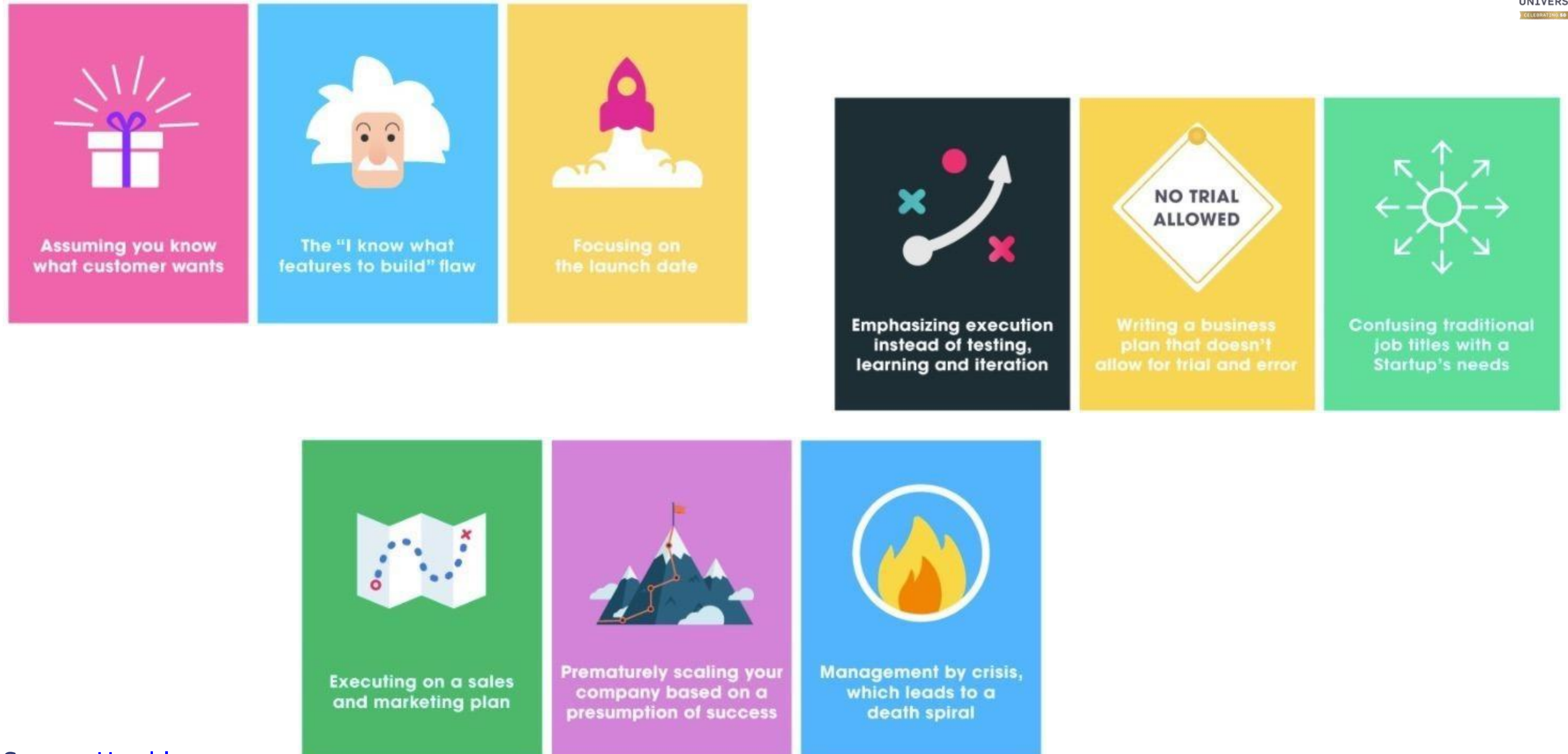
Sales and Marketing execute to a plan

Presumption of success leads to premature scaling

Management by Crisis leads to “Death Spiral”

System Preferences

'9 Deadly Startup Sins'





For Tomorrow's Innovators

CIE Ignite Season 02: A curriculum-integrated innovation program delivering to 4,500+ engineering and non-engineering students at your most motivated moment.

From classroom theory to real-world impact!

Partnership Announcement



Innovation Partners



Research Partners

Phase 1 - Department Hackathons

Have you participated?



Phase 2 – Sprint with Mentor Support

15 – Alumni Mentors

05 – Industry Mentors

The CIE Ignite Journey

A three-phase innovation pipeline that transforms classroom learning into validated, market-ready solutions



Phase 1: Department Hackathons

January 2026

600+ teams compete across 21 departments. Top 20% advance with validated concepts and initial prototypes.

Phase 3: Grand Finale

April 2026

15 finalists pitch refined solutions to investors, partners, and faculty. Winners announced with prizes and opportunities.

1

2

3

4

Phase 2: Mentor Sprint

February 2026

100+ selected teams receive structured guidance from industry & alumni experts through two intensive mentorship sessions.

Beyond: Sustained Innovation

April 2026+

Top projects can continue as capstone work, enter pre-incubation programs, or launch as startups with ongoing support.

Innovation starts here—and it doesn't stop.



cie.pes.edu

tarunrama@pes.edu

Expert Talk: Dr. V. Jaykumar

Chairperson, BBA-Hospitality and Event Management



Expert Talk: Dr. V. Jaykumar



Designation:

Chairperson, BBA-Hospitality and Event Management

Key Roles:

Executive Committee Member, IFCA (Indian Federation of Culinary Association)

Nominated Member, Karnataka State Higher Education Council for UG Courses in Hotel and General Management

Achievements and Recognitions:

Awarded Educator Champion by Economic Hero (Singapore-based auditing firm)

Invited Panel Member for QS Summit, Malaysia (2022)

Member of FSSAI, Skill Council of India, and Karnataka Tourism

Industry Contributions:

- Consultant and Auditor for Leading Hotel and Event Management Organizations
- Treasurer, Skal BLR Chapter
- Member of CII, CII-Yuva, and Karnataka Tourism



Essentials of Innovation & Entrepreneurship (EIE) - II

Module (M4): Customer Development (CD) Model: Customer Interviews

M3 Recap: Customer Discovery Hypothesis

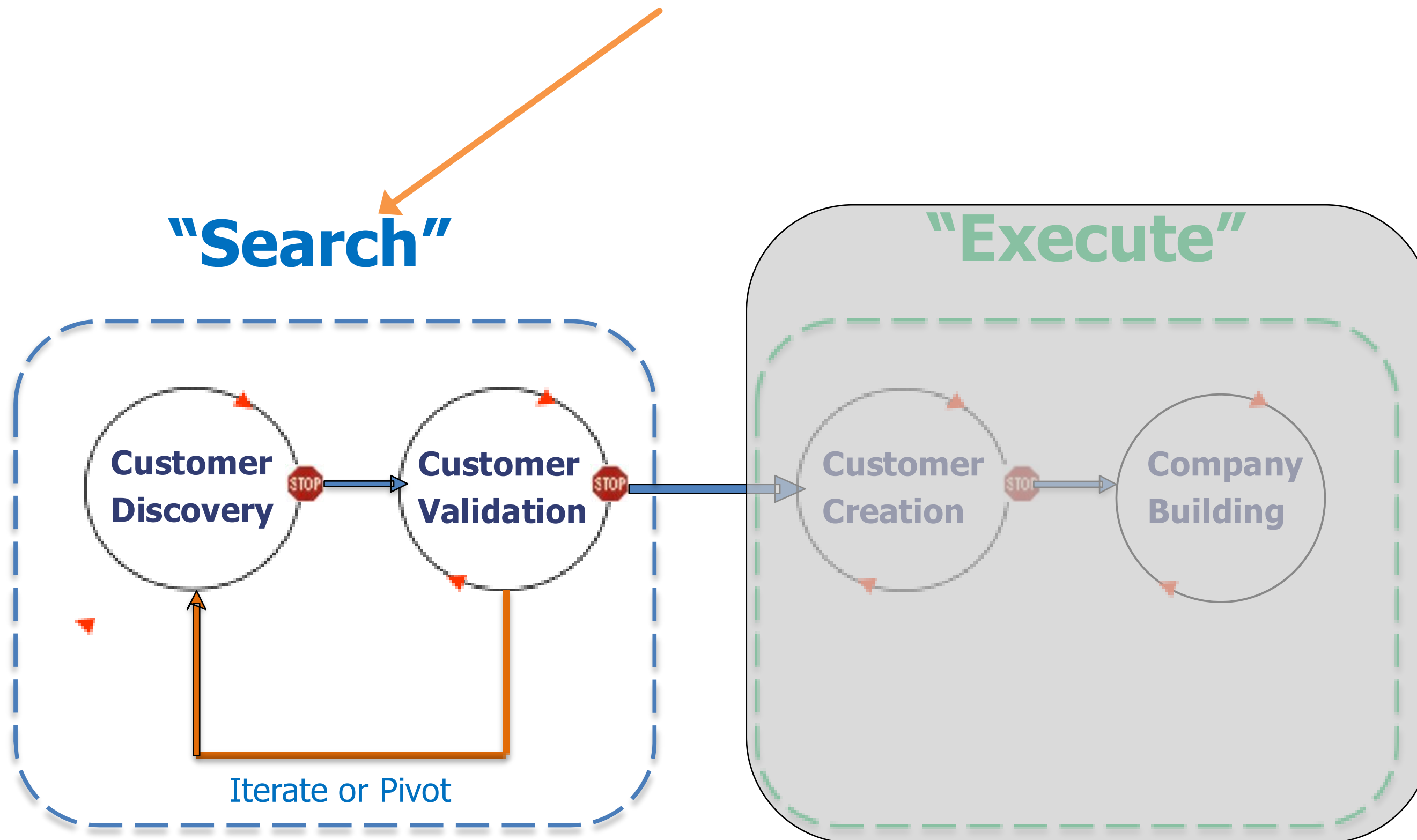
- **Hypothesis:** A **tentative, testable statement or assumption** made about a phenomenon, relationship, or outcome, which can be **validated or falsified through observation or experimentation**.
- **Process:** *State Hypothesis → Test the Problem → Test the Solution → Iterate/Pivot/Proceed.*
- **Effective Hypothesis** – QRST aka **Q**uantifiable, **M**inimal **R**esources, **S**pecific and **T**estable.
- Get Out of the Building aka **GOOB** for better customer reach.
- Use a **BMC** to capture your Hypothesis.
 - Remember BMC from EIE1 and Dr. Atre's Talk (RR Campus)
- **Customer Development Manifesto (CDM)** – by Steve Blank

Essentials of Innovation & Entrepreneurship (EIE) II:

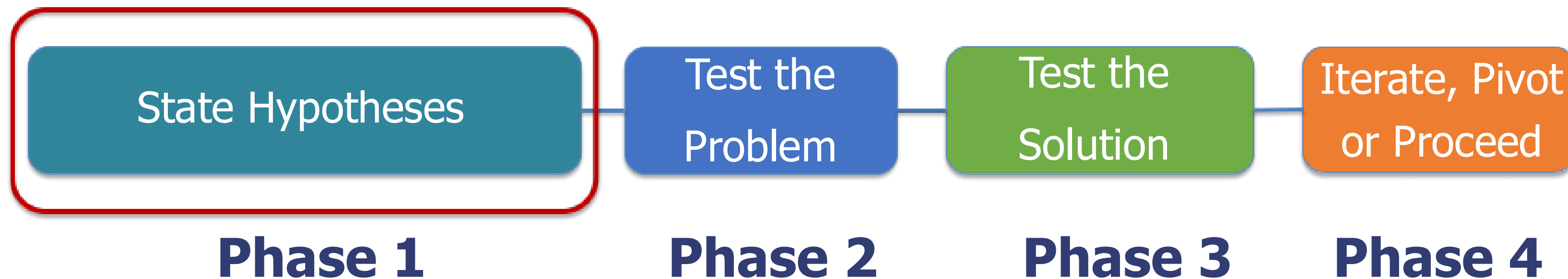
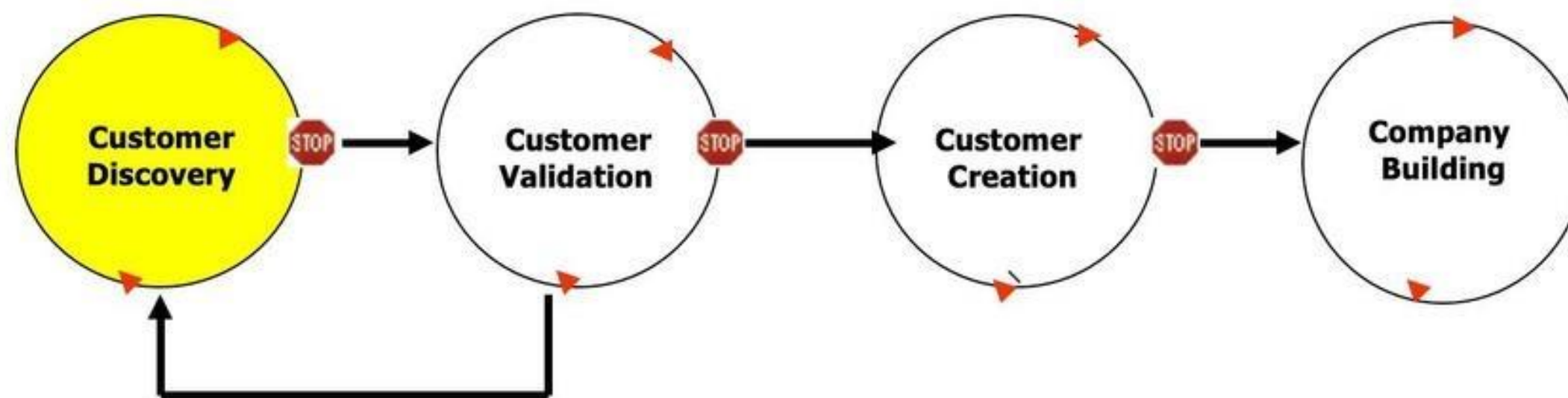
Customer Development (CD) Model:

Customer Interviews

Focus Area for EIE2 & CIE-Ignite

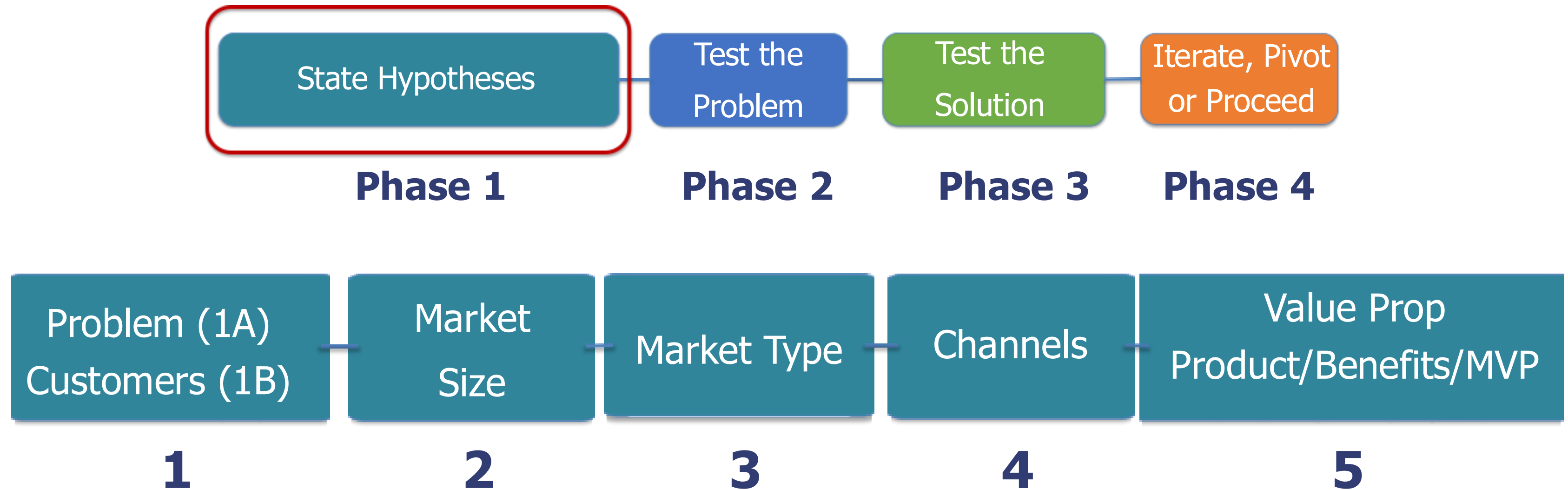


Customer Discovery Overview



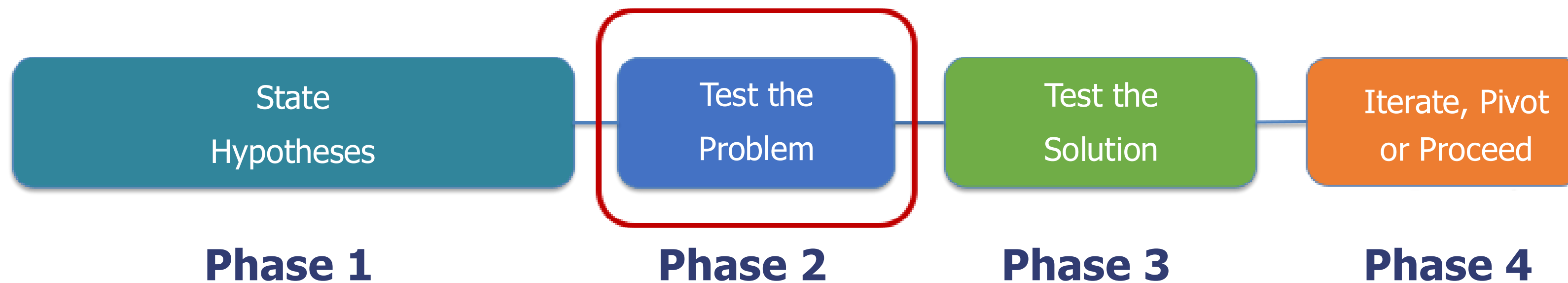
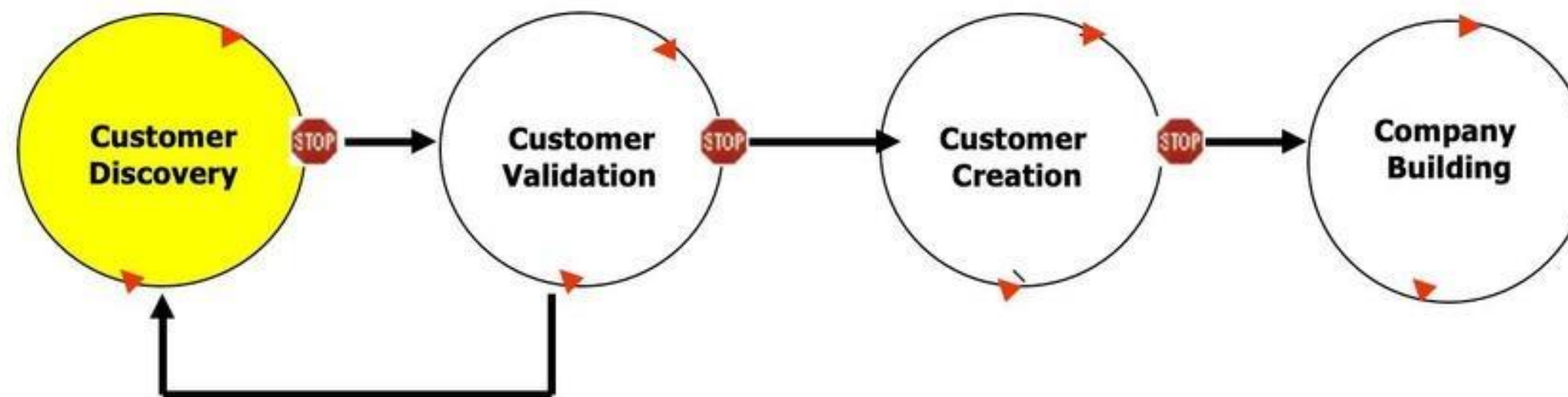
Customer Discovery Overview

Phase 1 : Hypothesis



**Capture your Hypothesis (Assumptions) on a
Business Model Canvas (BMC)**

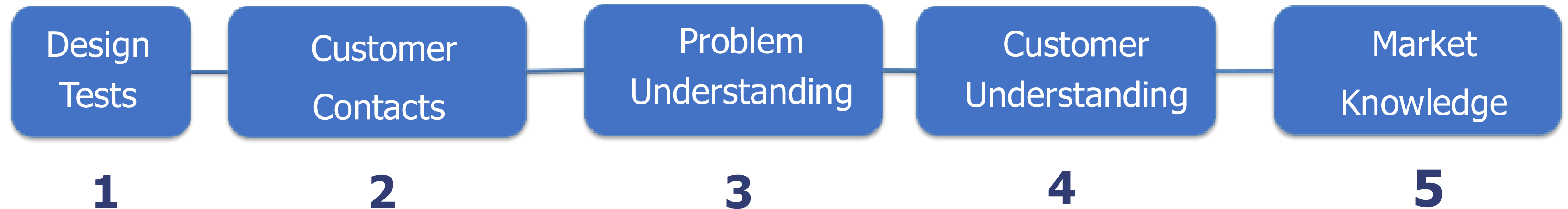
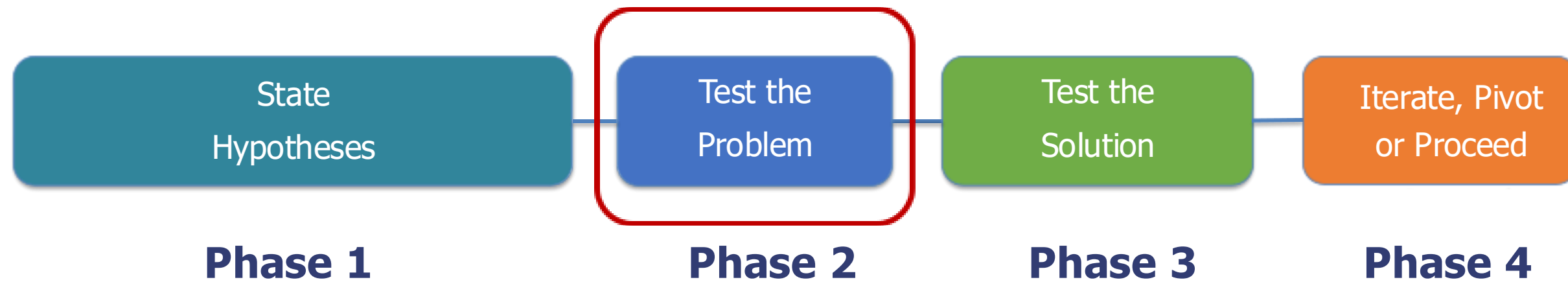
Customer Discovery Overview



Notice that testing the Problem comes before testing the Solution

Customer Discovery Overview

Phase 2 : Test the Problem



Revise your BMC based on Testing your Problem

Recall - Hypothesis

- A **tentative, testable statement or assumption** made about a phenomenon, relationship, or outcome, which can be **validated or falsified through observation or experimentation**.
- It is formulated **before data is collected**, based on existing knowledge, reasoning, or assumptions, and serves to **guide systematic inquiry**.

Examples:

- a. Sounds Familiar:** “Students who supplement regular course material with 5–7 minute concept videos and audio will perform better in quizzes, ISAs and ESAs than students who rely only on live lectures and standard course material.”: **Predicts a relationship between an intervention and an outcome**
- b.** If the college canteen introduces UPI-only fast lanes during peak hours, average waiting time for students will reduce by at least 30%.”: **Testable with a measurable outcome**
- c.** “Urban Indian working professionals aged 22–35 prefer ordering groceries via mobile apps rather than visiting local kirana stores at least three times a week.”: **Customer behaviour validated through data**
- d.** “In T20 cricket, teams that score more than 50 runs in the powerplay have a higher probability of winning the match.”: **Data driven assumption validated by match statistics**

How to Write a Good/Effective Hypothesis

Characteristics of a Good Hypothesis: “QRST”

Quantifiable

Minimal **Resources** (& time required to validate the hypothesis)

Specific

Testable (Pass or Fail)

Method: “X Y Z Hypothesis” – A least X% of Y (target market) will do Z

- Example: 20% of the luxury car market will buy at least 1 EV (electric) car
- Example: 20% of the luxury car market will buy at least 1 EV (electric) car priced at Rs 50 Lakh or more
- Example: 20% of the luxury car market will buy at least 1 EV (electric) car priced at Rs 50 Lakh or more in next 3 years

Note: More specific is better, but sometimes you can start with a simple test and add details as you go along

Customer Discovery: Experiments vs. Interviews

Customer Discovery: Experiments

Experiments are structured, measurable tests designed to observe real customer behaviour in response to a value proposition under controlled conditions.

They test behaviours — not opinions.

Experiments answer:

“Will they click?”

“Will they register?”

“Will they show up?”

“Will they pay?”

In Lean / Customer Development Context

Experiments are primarily used in:

Customer Validation

Transition from qualitative insight → behavioural proof

They test hypotheses such as:

“At least 20 people will sign up within 48 hours.”

“10% of landing page visitors will register.”

“People will prepay for a trial singing event.”

Customer Discovery: Experiments

What Experiments Produce

They produce:

- Behavioural data
- Conversion rates
- Evidence of demand
- Signal strength

They move you from:

"They said they like it" to "They actually acted."

Customer Discovery: Interviews

Customer Interviews are structured, hypothesis-driven conversations with potential customers conducted to discover, validate, or invalidate assumptions about their problems, behaviours, motivations and willingness to act.

They are:

- Not sales conversations
- Not pitching sessions
- Not opinion polls
- Not feature feedback sessions
- They are **evidence-gathering instruments** for problem–customer fit.

In Lean / Customer Development Context

In Steve Blank's model, interviews are primarily used in:

Customer Discovery → "Do they have this problem?"

Early Customer Validation → "Will they act on it?"

They test hypotheses such as:

"Working professionals in Bangalore feel socially isolated."

"Amateur singers want a safe, non-judgmental stage."

"People are willing to pay ₹499–₹999 for curated singing events."

Customer Discovery: Interviews

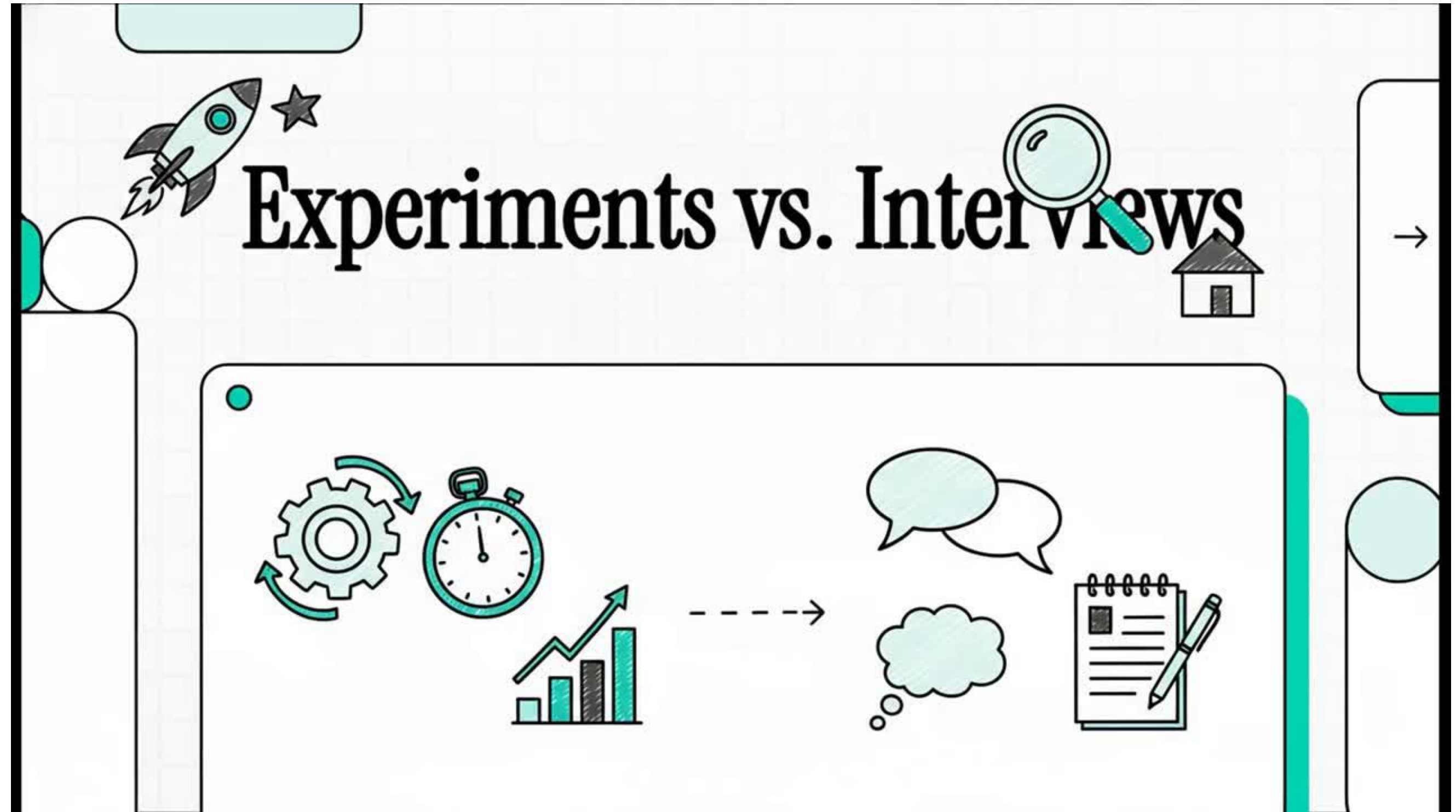
What Interviews Actually Produce

- ✓ Qualitative insights
- ✓ Language customers use
- ✓ Emotional intensity signals
- ✓ Behavioural evidence ("Tell me what you did last time...")

They do not produce:

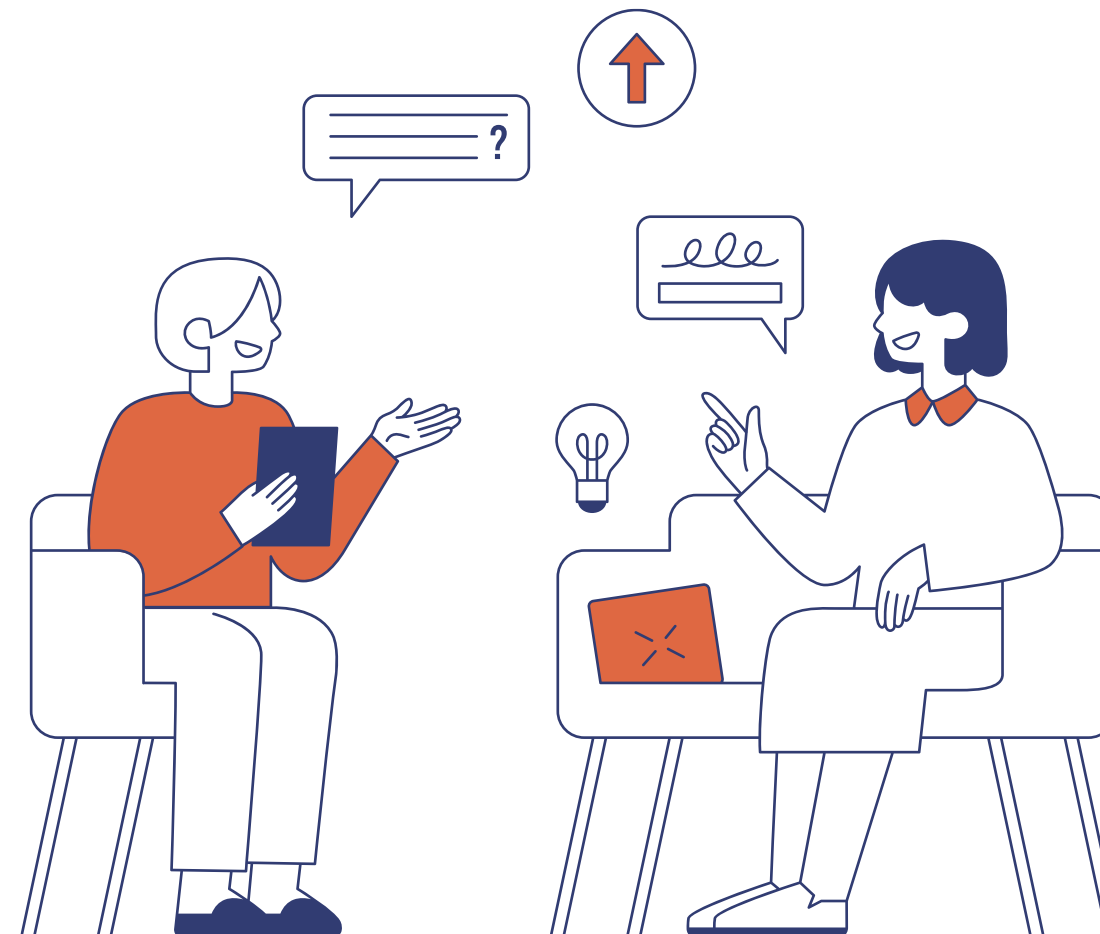
- ☐ Market validation
- ☐ Revenue proof
- ☐ Scalable evidence

Customer Discovery: Experiments vs. Interviews



Customer Discovery: Experiments vs. Interviews

Experiments give you the greatest proof,
Interviews give you the greatest
insights!



Customer Discovery: Experiments vs. Interviews

Experiments give you evidence whether something will happen.

Interviews help you understand why.

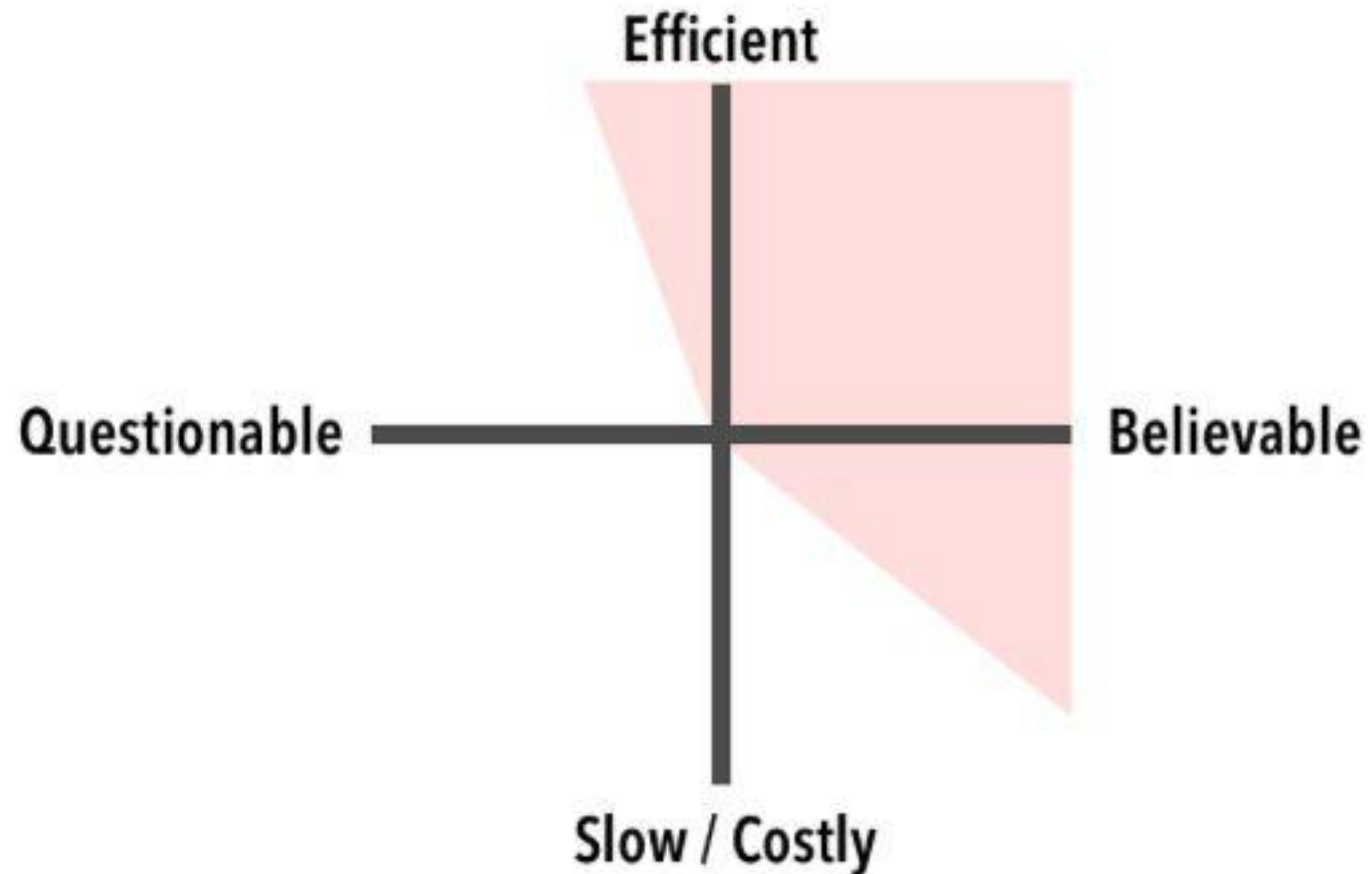


Customer Discovery: Experiments vs. Interviews

Dimension	Interviews	Experiments
Nature	Qualitative	Quantitative / Behavioral
Stage	Discovery	Validation
Evidence	Words	Actions
Risk Level	Low cost	Slightly higher cost
Output	Insights	Measurable signals

Customer Discovery:

How to Choose Experiments & Interviews



12 Tips for Early Customer Development Interviews

12 Tips for Early Customer Development Interviews



Know your Goals &
Questions ahead of time



One person at a time
(not a group)



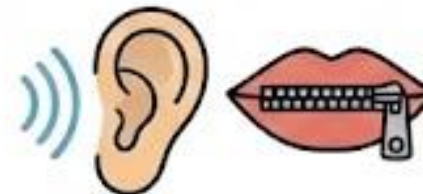
Be open to hearing things
you DON'T want to hear



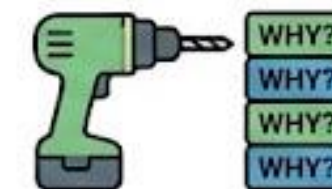
Disarm "Politeness
Training" (tell the person
to be brutally honest)



Ask open-ended
questions



Listen, Do NOT Talk



Drill down on details –
"5 Whys" method?



Separate behavior and
feedback in a discussion



Focus on actual
behavior



Recap what you heard
and/or misrepresent to
confirm!



Ask for Introductions –
Get leads to talk to
another 1-3 people!



Write up your notes as
quickly as possible

12 Tips for Early Customer Development Interviews

- Know your **Goals & Questions** ahead of time
- **One person** at a time (not a group)
- Be **open** to hearing things you **DON'T** want to hear
- **Disarm "Politeness Training"** (tell the person to be brutally honest)
- Ask **open-ended questions**
- Listen, Do NOT Talk

12 Tips for Early Customer Development Interviews

- Drill down on details — "**5 Whys**" method?
- Separate **behavior** and **feedback** in a discussion
- Focus on **actual behavior**
- **Recap** what you heard and/or misrepresent to confirm!
- Ask for **Introductions** — Get leads to talk to another 1-3 people!
- Write up your **notes** as **quickly** as possible

Tips for Early Customer Development Interviews



Expect Criticism



Set Goals



Solo Interview



Dig Deeper



Document Quickly



Disarm Politeness



Listen More



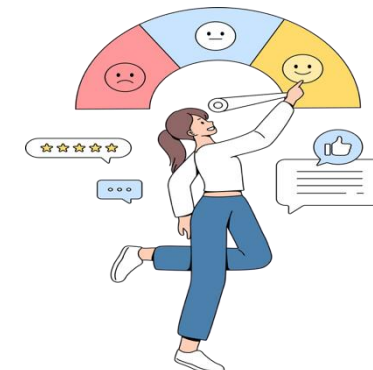
Observe Behavior



Request Referrals



Confirm Understanding



Separate Feedback



Ask Open-ended

CD - Experiments

Testing Your Big Idea



Examples of Experiments (1/2)

1. Landing Page: A simple web page/website conveying your value proposition and allows visitors to express their interest through a call to action.

2. Advertising: Paying to display your value proposition to a relevant audience through an ad to see if people respond or convert.

3. Promotional Material: A type of advertising test where you create online or offline promotional materials to gauge reactions or generate demand.

4. Pre-selling (including crowdfunding): Attempting to secure orders before the product is built.

5. Paper Testing:

Applying primarily to software and information (data, analysis, media etc) products, paper tests are where you mock-up an example of an application user interface or report and put it in front of a potential customer.

Examples of Experiments (2/3)

6. Product Prototype: A working version of your product/experience built for learning and fast iteration rather than for robustness or scale.

7. Dummy Product (Wizard of Oz): Where the customer thinks they are interfacing with a real product (or feature), but where your team provides the service manually, hidden behind the scenes (hence the name).

7. Concierge MVP: Manually/openly act as the product you eventually want to build (unlike 'Wizard of Oz' where people are behind the scenes).

8. Pilots: Put an early version of your product in the hands of your customers, scale down the size of the implementation, and assign a finite time period to the project.

9. Usability: Check to see if someone can effectively use a product without getting stuck or blocked.

Examples of Experiments

9 Smart Experiments to Validate Your Business Idea

Before committing resources, use these experimental methods to test assumptions, gather real-world data, and de-risk your concept.

GAUGING MARKET INTEREST & DEMAND



Landing Page

A simple web page conveying your value proposition that allows visitors to express interest through a call to action.



Advertising

Paying to display your value proposition to a relevant audience through an ad to see if people respond or convert.



Promotional Material

An advertising test where you create online or offline materials to gauge reactions and generate demand.



Pre-selling (including Crowdfunding)

Attempting to secure orders from customers before the final product has been built.

TESTING THE PRODUCT & USER EXPERIENCE



Paper Testing

Mock up a user interface or report and show it to a potential customer, primarily for software and information products.



Product Prototype

A working version of your product built for learning and fast iteration, not for robustness or scale.



Dummy Product (Wizard of Oz) vs. Concierge

Both involve manual service delivery. In 'Wizard of Oz,' the team is hidden, and customers think they're using a real product. In 'Concierge,' the team openly acts as the product.



Pilots

Put an early version of your product in the hands of customers, scaling down the implementation and setting a finite time period.



Usability Testing

A direct check to see if someone can effectively use a product without getting stuck or blocked.

Examples of Experiments

← → ↺ roognis-landing.pages.dev



R  gnis

[About](#) [Customers](#) [Newsletter](#) [Instagram](#) [Community \(Coming Soon\)](#)



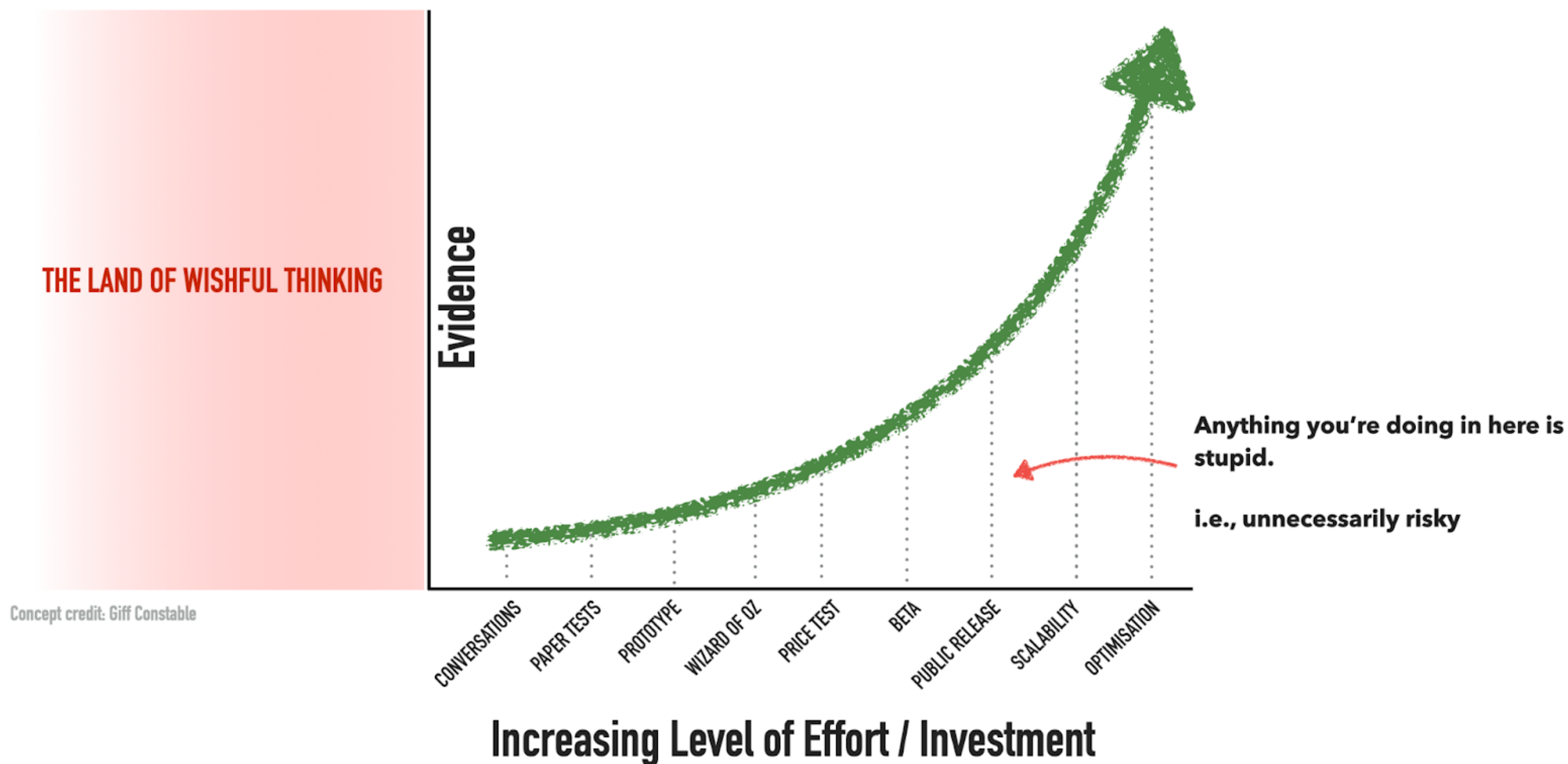
Coming Soon!

Empower Every Student with AI-Powered Learning

Roognis is an AI academic infrastructure designed for NGOs and educational institutions—delivering 24x7 personalized tutoring, reducing volunteer burnout, and scaling impact without diluting mission.

[Community \(Coming Soon\)](#)

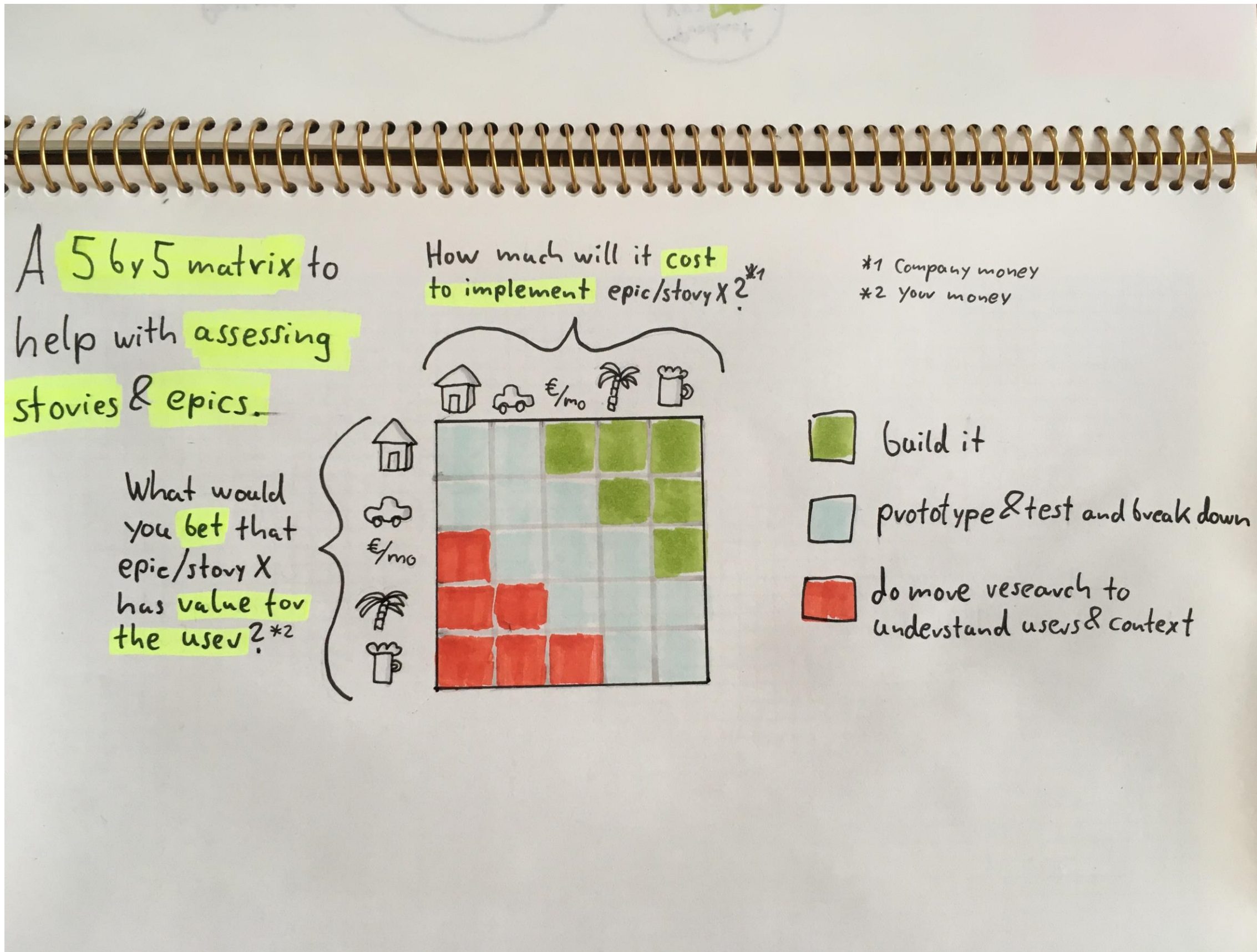
THE TRUTH CURVE



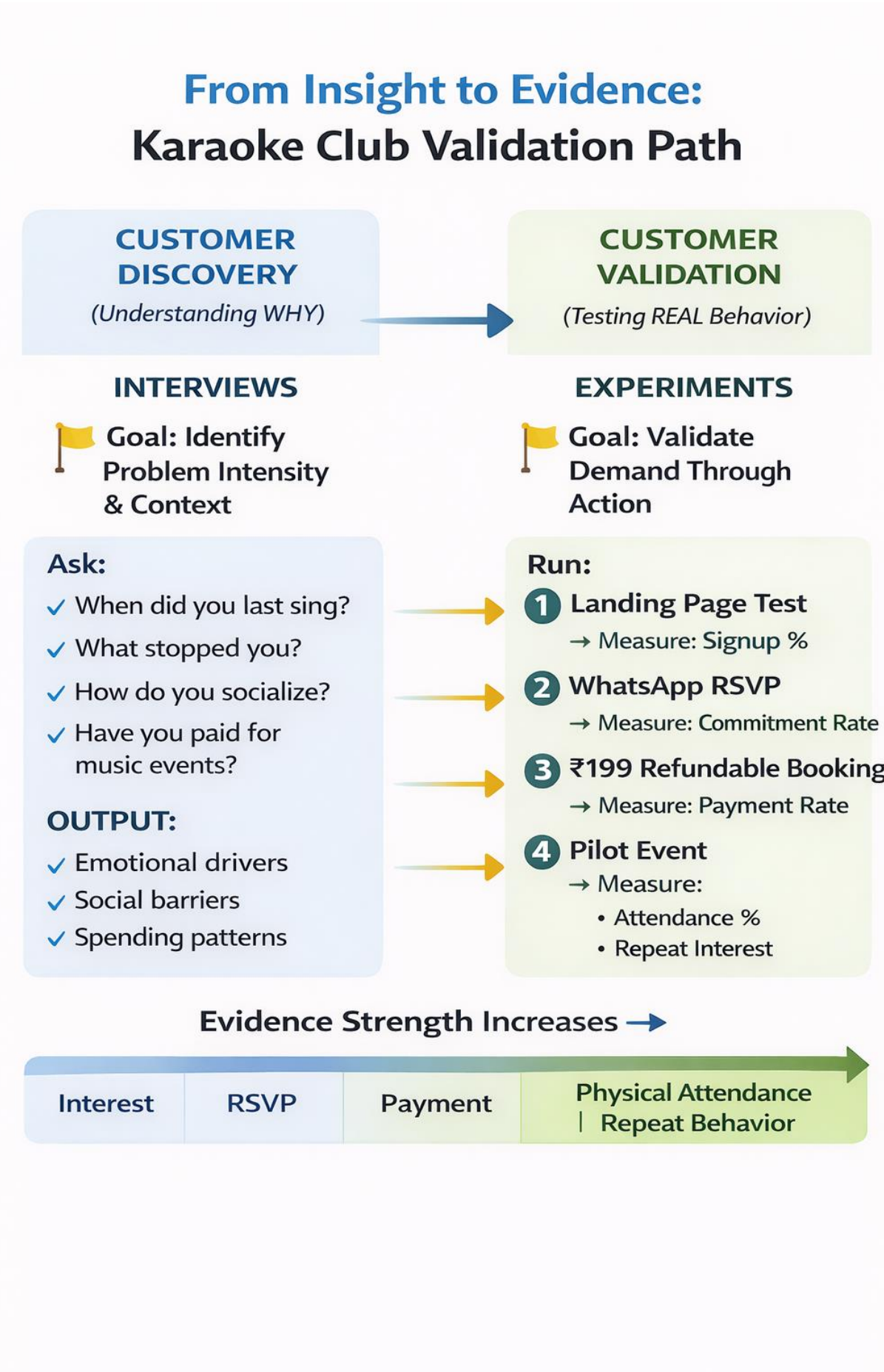
Concept credit: Giff Constable

GOTHELF.CO /

Interesting Visual



Example of the Karaoke Club



Experiments vs. Interviews

Aspect	Experiments	Interviews
Purpose/Approach	Test the hypotheses	Gather Insights
Data (type) Collected	Quantitative (conversion rate, click-through rate, A/B test results, etc)	Qualitative (emotional reason/response, decision-making reasoning, user stories, etc)
Bias & Interpretation	Less prone to bias	More prone to bias (e.g. confirmation-bias)
Speed & Scale	High (can be automated, scaled in many cases)	Low (requires more time/effort, typically done individually)
Depth (of understanding)	What people do (not why they do it)	Why customers behave the way they do

Important to Know When to Run Experiments vs. Conducting Interviews

Customer Development Data ([link](#))

(Steve Blank, Kauffman Foundation, 2:12 m)



Conducting Interviews ([link](#))

(Steve Blank, Kauffman Foundation, 5:36 m)

Resources



Subsequent slides list resources (videos/articles/etc) related to
Customer Discovery / Development

Tips/Tricks: Before 'Getting Out of the Building'

Before Leaving the Building: Helpful videos (by Steve Blank):

(<https://venturewell.org/i-corps/llpvideos/customer-discovery/before-leaving-the-building/>)

- Discovery is for Founders (1:30) CD
- Pre-Planning Contacts (4:34) CD
- Customer Interview Dry Runs (0:49) CD
- Pass/Fail Experiments (1:32) CD
- **Customer Dev: Insight not just Data!** (2:12, Kauffman Foundation)

Tips/Tricks: Before 'Getting Out of the Building'

- **12 Tips for Early Customer Development Interviews** (Giff Constable): [Link](#)
- **Interviews vs. Experiments** (Giff Constable): [Link](#)

"Interviews give you the greatest insights. Experiments give you the greatest proof"

- Results of hypothesis validation fall into 4 buckets:
- See Progress Board template: **I**nvalidated, **V**alidated, **W**IP*, **U**nclear results

Resources - More Videos on Customer Development

- 1.*Engaging the Customer (CD05-06) **Right & Wrong way!**
2. Assuming you know (CD09-10) **Right and Wrong way!**
3. Customers Lie (CD11) **Example!**
- 4.* Extracting Insight from Data (CD12)
5. Attention to Outliers (CD13) **Right and Wrong way!**
6. The User, Buyer & Saboteur (CD14-15)
7. Pre-Planning Pt1 Pt2 Pt3 (CD30, 31, 32)
8. Interviews Pt1, Pt2 (CD33-34) **Right and Wrong way!**
- 9.* Conducting a Customer Interview (CD46)
10. Letting the Customer interview flow (CD47)
11. Looking for Insights (CD50)
12. * The Distracted Customer (CD2021) **Right/Wrong way!**
13. * Customer Empathy (CD23-24) **Right/Wrong way!**
14. * Asking the Right Question (CD25-26) **Right/Wrong way!**
15. Finding Patterns (CD49)
16. * Communicating your discoveries (CD52)

<https://vimeo.com/groups/204136/videos>

Resources

Subsequent slides list **step-by-step** overview and
tip-and-tricks
for each Phase of **Customer Discovery**

Customer Discovery Overview - Hypothesis

State your Hypothesis (for your startup opportunity)

- **Problem** (why?)
- Faced by target **Customers**, Customer Segments (whom?)
- Your **Solution? Value proposition** you will offer (what?)
- Your assumptions on Market (size, type), Team, etc
- Based on above, create BMC (BMC#1) or LC (Lean Canvas #1)

State Hypotheses
Create Business Model
Canvas

Phase 1

Customer Discovery Overview - Tips & Tricks

What If this gets Overwhelming?

- **Prioritize, Prioritize, Prioritize!**
 - **Customer** Jobs
 - **Customer** Pains, Customer Gains
 - **Customer** segments
 - **Solution** (Features/ Benefits you offer)
 - **Market** (type, size, channel)
- **Prioritize Speed over Accuracy**
- **Divide & Conquer (team/ delegation)**

State Hypotheses
Create Business Model
Canvas

Phase 1

Customer Discovery Overview - Test the Problem

Test the Problem

- **Design** the **tests** (to validate the problem hypothesis)
- Run the tests' –use data / insights to (in)validate hypothesis
 - **Interview** potential **customers**
- Update the BMC (BMC#2)

Test the Problem

Phase 2

Customer Discovery Overview - Tips & Tricks

What If this gets Overwhelming?

- Customer Contact / Interview
 - Go through the tips/tricks, videos to prepare
 - Have a goal for the discussion
 - Be relaxed, let the 'customer' do most of the talking
- Preparing hypotheses
 - Is it testable?
 - (pass/fail –this tells if you are on the right path or not)
 - Is it measurable?
 - (if the right path, what would be meaningful?)

Test the Problem

Phase 2

Customer Discovery Overview - Test the Solution

Problem - Solution fit ?

Test the Solution

- Create solution representative (preto/prototype, etc)
- Test this with target customers
- Update your solution hypothesis (& pretotypes, etc)
- Update the BMC (BMC#3)

Test the Solution

Phase 3

Customer Discovery Overview - Tips & Tricks

What If this gets Overwhelming?

- Test the most fundamental assumption/hypothesis
- Build a pretotype (not even a prototype)
- Even a simple pretotype is better than no pretotype!
- Don't be afraid to show your pretotype to Customer
most people will see that you are really committed!

Test the Solution

Phase 3

Customer Discovery Overview – Pivot or Proceed?

Pivot or Proceed?

- Having a better understanding of the opportunity, would you continue with this idea?
- If so, why? What will be your next steps?
- What help do you need?
- If not, why? What will be your next steps?

Pivot or proceed

Phase 4

Customer Discovery Overview - Tips & Tricks

What if it gets Overwhelming ?

- Do you have enough insights? (Phase I, 2, 3)
- Everyone in the team engaged during Ph 1, 2, 3?
- All the team members aligned on startup goals?
- Can you make the tough decision on changes to keepgoing?
- (if needed, identify any help needed)

Pivot or proceed

Phase 4

Capturing your Hypotheses in a BMC

- For your team opportunity, capture your hypotheses (see template)
 - Customer (customer segments, channel, relationship, etc)
 - Value Proposition (proposed benefits to customer from your solution)
 - Market Size
 - Price and Cost

Create an initial BMC (do not worry about all the details or accuracy)

Customer Development Manifesto

Steve Blank's Customer Development Manifesto: 17 Principles for Startups



1. Get Outside the Building



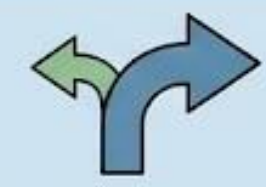
2. Pair with Agile Development



3. Failure is Part of Search



4. Don't Fear Failure



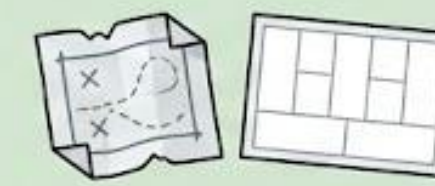
5. Pivot on Insight



6. Test Your Hypotheses



7. Get Early Buy-in



8. No Plan Survives Contact



9. Not All Startups Are Alike



10. Startup Metrics Differ



11. Agree on Market Type



12. Speed and Tempo



13. It's All About Passion



14. Job Titles are Different



15. Preserve Cash While Searching



16. Communicate & Share



17. Comfort with Chaos

Customer Development Manifesto

[\(Link\)](#)

1. There Are No Facts Inside Your Building, So **Get Outside**
2. Combine **Customer** Development with **Agile** Development
3. **Failure** is an Integral Part of the Search for the Business Model
4. Iterations and Pivots are Driven by **Insight (not just data)**
5. Validate Your Hypotheses with **Experiments**
6. No Business Plan Survives **First Contact with Customers**
7. **Communicate** and **Share** Learning (within the team)
8. Agree on **Market Type** - It Changes Everything
9. **Fast** (speed and tempo) and **Fearless Decision-Making**
10. Startups Demand Comfort with **Chaos** and **Uncertainty**

Customer Development Manifesto



11. If it's not About **Passion**, You're Dead the Day You Opened your Doors

12. Startup **Titles/Functions** Are Very Different from a regular firm/company

13. **Preserve** Cash While Searching. After It's Found, **Spend**

14. Success Begins with Buy-In from **Co-Founders** and **Investors**

15. If You're **Afraid to Fail** You're **Destined to Do So**



'9 Deadly Startup Sins' by Steve Blank

[Link](#)

Assuming “I know what the customer wants”

The “I know what features to build” flaw

Focus on launch date

Emphasis on execution instead of hypotheses, testing, learning and iteration

Traditional business plans presume no trial and no errors

Confusing traditional job titles with what a startup needs to accomplish

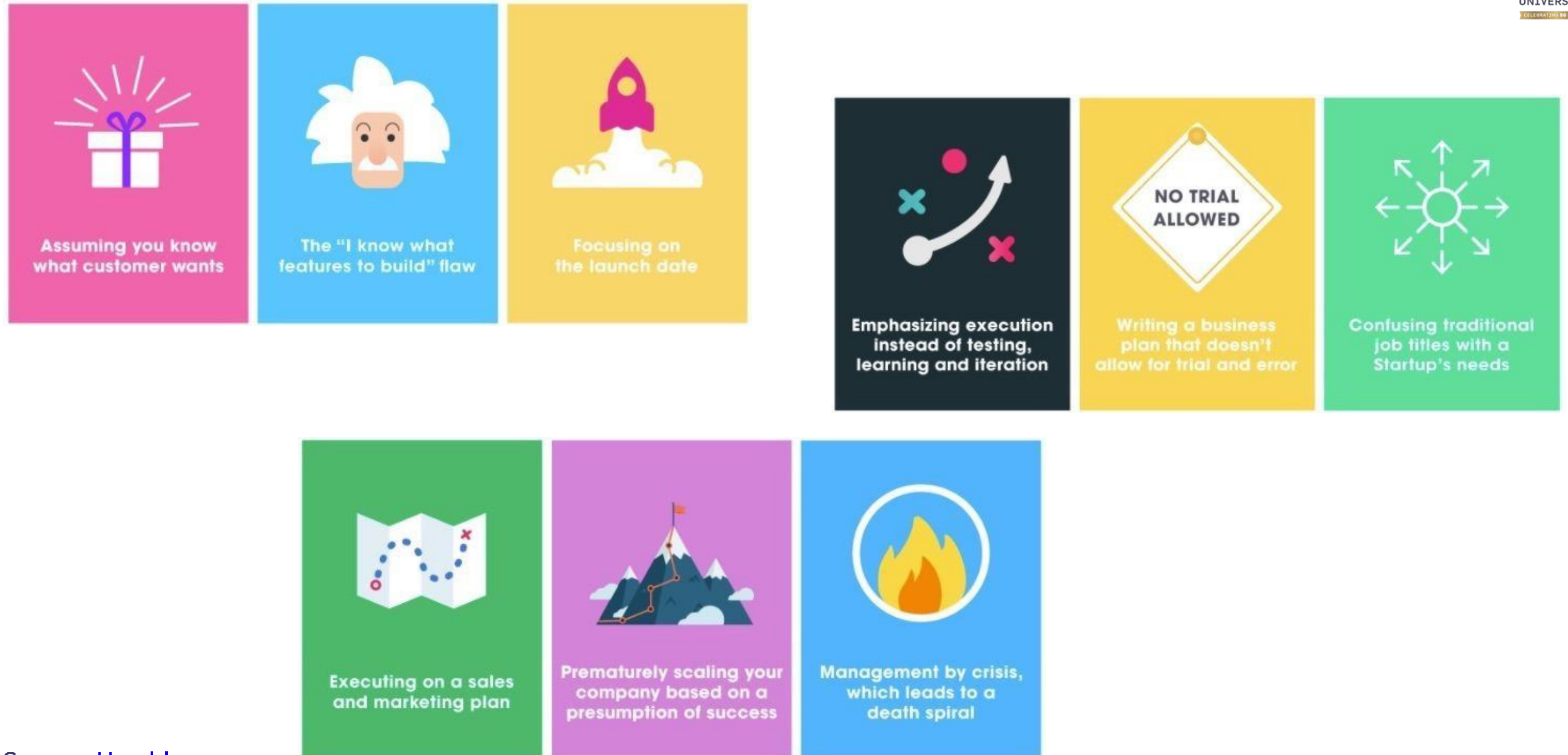
Sales and Marketing execute to a plan

Presumption of success leads to premature scaling

Management by Crisis leads to “Death Spiral”

System Preferences

'9 Deadly Startup Sins'



Examples



Dropbox

Customer Discovery: Drew Houston, the founder of Dropbox, initially created a video demo of the product and posted it on a technology forum to gauge interest and gather feedback. The overwhelmingly positive response from potential customers helped validate the idea and demonstrate the demand for an easy-to-use file storage and sharing solution.

Impact: Armed with valuable customer feedback, Dropbox was launched in 2007, offering cloud-based file storage and synchronisation. The company's customer-centric approach, which focused on addressing user needs and pain points, contributed to its rapid growth and widespread adoption. Today, Dropbox is a leading cloud storage provider with millions of users worldwide.

Examples



Zappos

Customer Discovery: Zappos, an online shoe and clothing retailer, has built its success on a foundation of exceptional customer service. In the early stages, founder Tony Hsieh recognized the importance of understanding customer needs and expectations. He personally spoke with customers to gain insights into their preferences and pain points.

Impact: Zappos prioritised customer satisfaction and went beyond selling products by focusing on creating a delightful shopping experience. This customer-centric approach paid off, and Zappos became known for its outstanding customer service and loyalty. In 2009, Amazon acquired Zappos for over \$1 billion, recognising the company's commitment to customer satisfaction.

Essentials of Innovation & Entrepreneurship (EIE) - II

Module (M5): Why Startups Fail, The Right It, Prototyping

EIE2-Icebreaker:

I&E Roundup this Week (Feb 18)



'Spatial Web' – Weaving a New Magic



CIE@PESU

A new Internet standard anticipates a broad range of devices and AI agents interacting and collaborating through the Spatial Web.

How will *you* be a part of this new 'playground'?

<https://spectrum.ieee.org/spatial-web-standard>

India AI Summit - 2026

Smattering of announcements in the last 2 days
(not a comprehensive list – check out <https://impact.indiaai.gov.in>)

The Three Sutras Shaping a Sustainable AI Future



People

AI must serve humanity in all its diversity, preserving dignity and ensuring inclusivity.



Planet

AI innovation must align with environmental stewardship and sustainability.



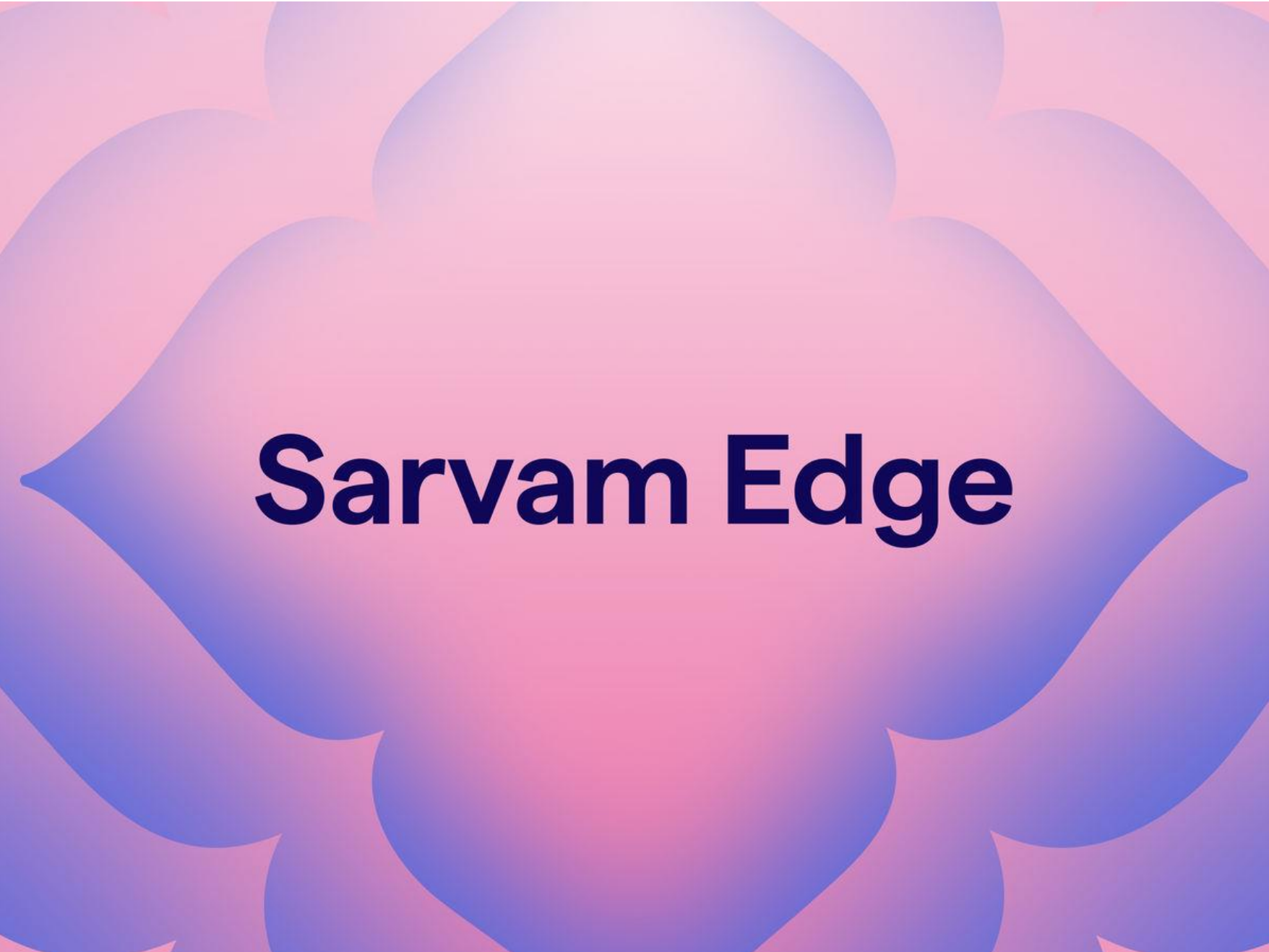
Progress

AI's benefits must be equitably shared, advancing global development and prosperity.

1. EkaScribe: A doctor in a busy rural clinic see 5 patients without touching a keyboard. The AI handles the prescriptions, history, and filing.
2. Ottobots: Made in India autonomous robots that can deliver medicines in hospitals, navigating elevators and corridors by themselves (no more family members rushing to pharmacies).
3. Sarvam Kaze: Sleek, Made-in-India AI spectacles that "see" what you see. They don't just record – they explain the world to you in your local language via bone conduction like a genius translator whispering in your ear.
4. Sarvam Edge: AI that runs on your phone with zero internet – translates 22 languages in real-time even while you're in a basement.
5. Mankomb's "Chewie": AI native kitchen appliance that literally eats your waste. It converts wet waste – bones, meat, and fluids into nutrient rich "RegenSoil" in hours using real-time AI sensors.

India AI Summit - 2026

Example announcement: Made in India, Made for India and the World



Sarvam Edge

- **Responses are instant**
(no round-trip to datacenter)
- **Your data never leaves your device**
(data/info stays local)
- **It works everywhere**
(rural/urban/no-connectivity is fine)
- **The economics are different**
(no per query cost / usage based pricing)

E-Summit (ECell @ PESU-RR)



The poster features a dark, crumpled paper background with yellow wavy lines. At the top left is the PES ECELL logo (a hand holding a wireframe head) and at the top right is the PES University logo. The central text reads 'E-SUMMIT 2026' in large white letters, with 'E-CELL PRESENTS' above it and 'Suit Up. Step In. Stand Out.' below it. To the right of the title is a calendar icon and the dates '21-22 February'. Below the title is a yellow-bordered box containing a trophy icon and the text 'PRIZE POOL WORTH ₹60,000+'. At the bottom, it says 'Compete. Collaborate. Build.' and 'Learn from mentors. Pitch like a founder. Think like a CEO.' On the right side, there is a large yellow QR code and the text 'REGISTER NOW'. Illustrations of a man with a megaphone and a man pointing are on the left and right respectively, with a rocket ship at the bottom right.

PES ECELL

E-CELL PRESENTS

E-SUMMIT 2026

Suit Up. Step In. Stand Out.

PRIZE POOL WORTH ₹60,000+

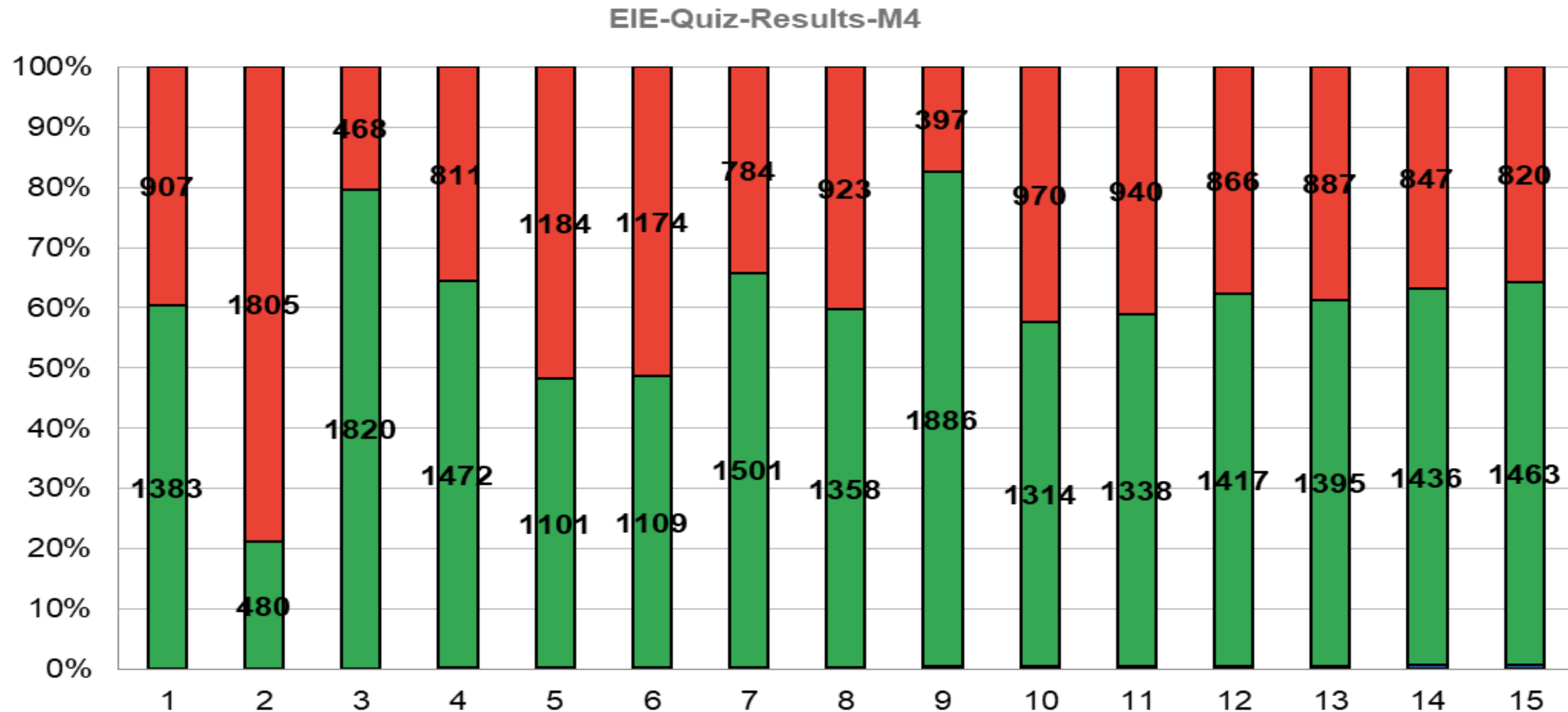
Compete. Collaborate. Build.
Learn from mentors. Pitch like a founder. Think like a CEO.

21-22 February

REGISTER NOW

PES UNIVERSITY

EIE M4 Session Quiz - Discussion / Recap



EIE M1 Session Quiz - Discussion / Recap

Qstn #	StudentScore (Correct Answer)	StudentScore (Wrong Answer)	Total Students	% Correct	Options-wise StudentScore			
					Option A	Option B	Option C	Option D
1	1383	907	2290	60%	289	416	1383	202
2	480	1805	2285	21%	908	291	480	606
3	1820	468	2288	80%	1820	197	134	137
4	1472	811	2283	64%	519	174	118	1472
5	1101	1184	2285	48%	683	1101	291	210
6	1109	1174	2283	49%	895	146	133	1109
7	1501	784	2285	66%	263	1501	400	121
8	1358	923	2281	60%	1358	655	174	94
9	1886	397	2283	83%	125	144	1886	128
10	1314	970	2284	58%	628	1314	205	137
11	1338	940	2278	59%	627	176	1338	137
12	1417	866	2283	62%	1417	354	359	153
13	1395	887	2282	61%	442	217	1395	228
14	1436	847	2283	63%	1436	163	145	539
15	1463	820	2283	64%	410	216	194	1463

EIE M1 Session Quiz - Discussion / Recap

2. According to the QRST framework, the "S" stands for:

- a) Scalable
- b) Strategic
- c) Specific - 480 students got it right!**
- d) Sustainable

5. A startup mocks up a user interface and shows it to potential customers. This is:

- a) Landing Page experiment
- b) Paper Testing - 1101 students got it right!**
- c) Wizard of Oz
- d) Concierge Testing

6. In a "Concierge" experiment, the distinguishing feature is:

- a) The team is hidden while manually providing the service
- b) The product is fully automated from the start
- c) No direct customer interaction takes place
- d) The team openly acts as the product manually - 1109 students got it right!**

M4 Recap – Placeholder for Quiz

Mentimeter.com

Menti Code: 6789 0331



M4: Interviews & Experiment Recap

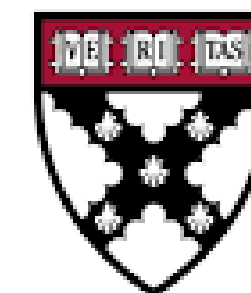
- **Customer Development = Search before Execution**
 - Validate customers and problems before building
- **Customer Discovery Flow**
 - Hypothesize → Test Problem → Test Solution → Decide
- **Problem First**
 - No validated problem = no product
- **Good Hypotheses (QRST)**
 - Quantifiable • Specific • Testable • Low-resource
- **Proof is **GREATER THAN** Opinions**
 - Interviews explain why
 - Experiments show what
 - Remember: Proof is in the Pudding
- **Interviews**
 - Listen, don't sell

Essentials of Innovation & Entrepreneurship (EIE) II:

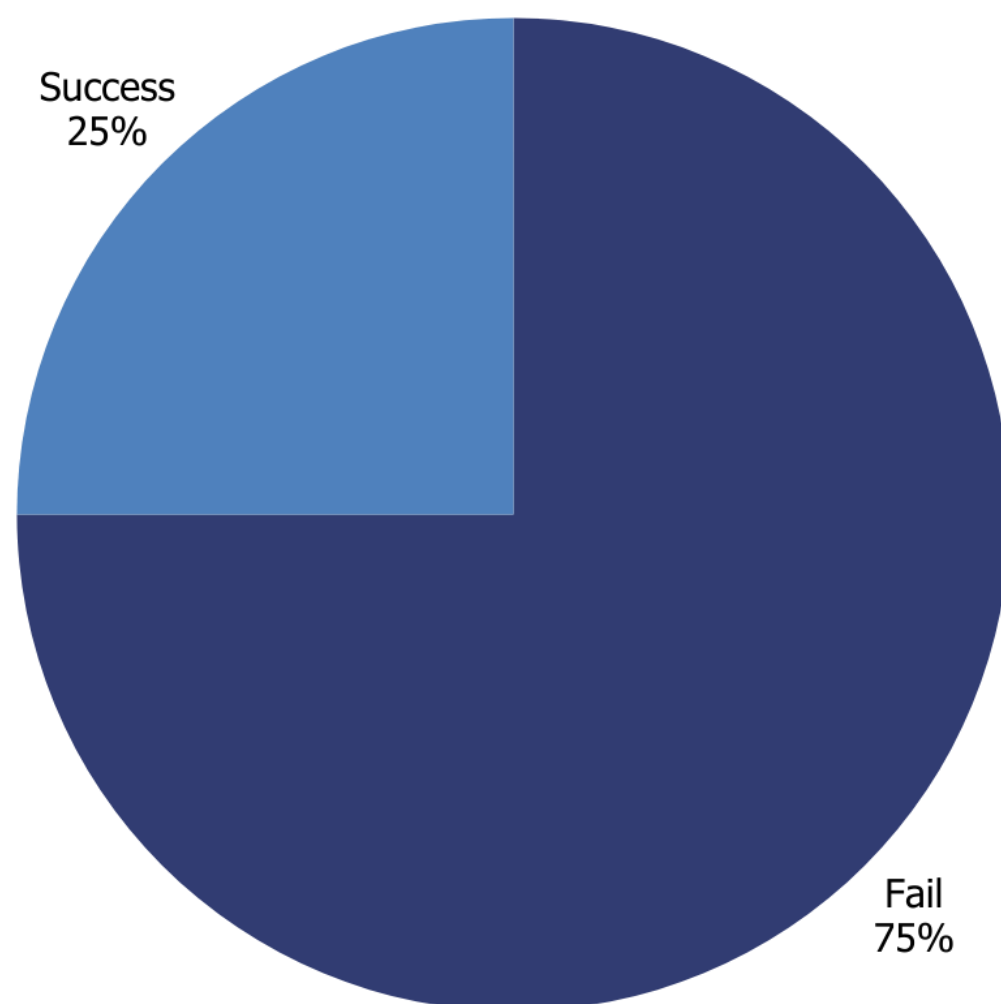
Module (M5):

‘Law of Market Failure’ and ‘Pretotype’ – What is it?

Venture backed Companies fail



**Harvard
Business
School**



**75% of Venture-Backed Startups
fail to return Investors' capital**

Faculty News | Fast Company | 10 DEC 2012

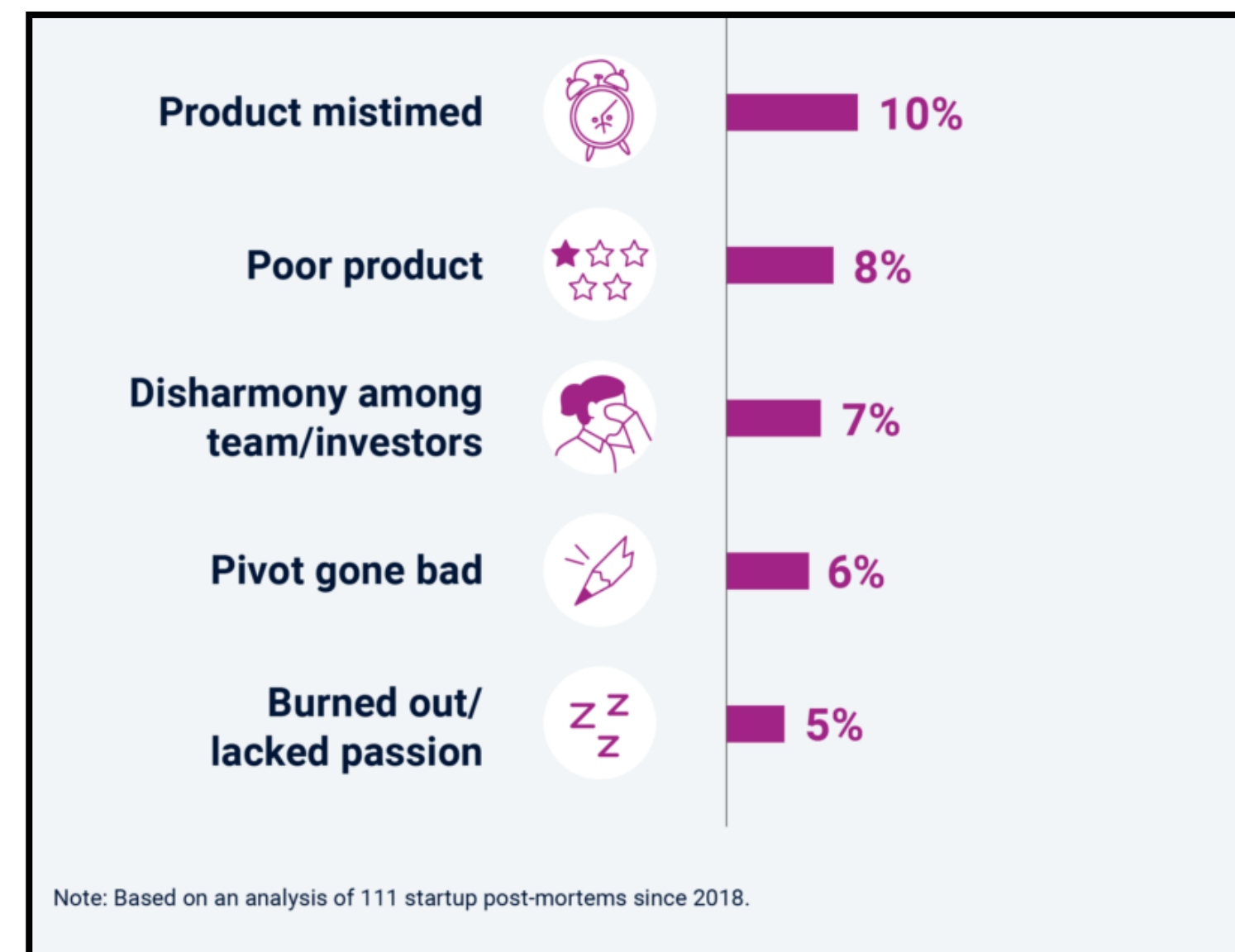
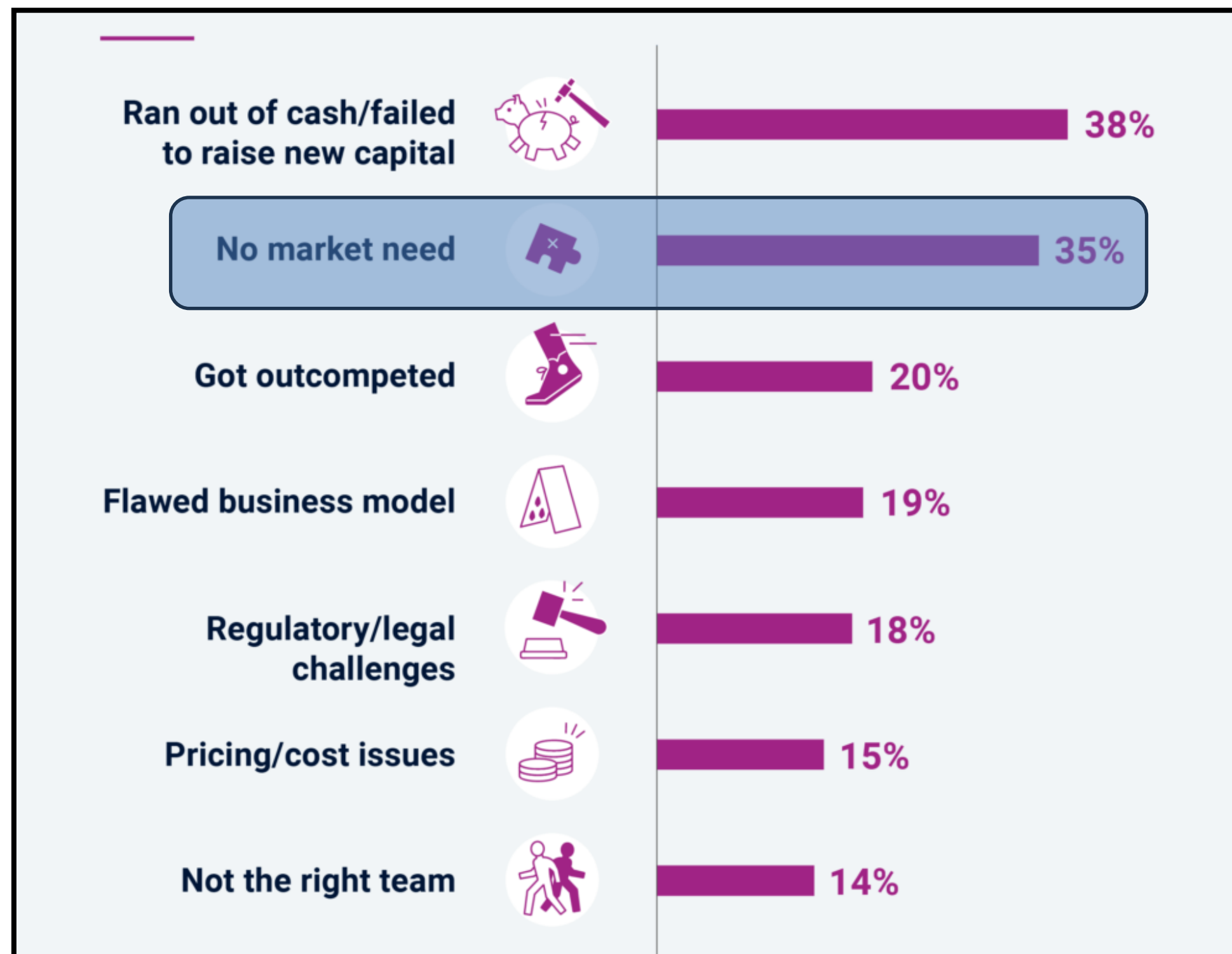
Why Most Venture-Backed Companies Fail

Re: Shikhar Ghosh

Ghosh's research indicates that as many as 75 percent of venture-backed companies never return cash to investors, with 30 to 40 percent of those liquidating assets where investors lose all of their money. His findings are based on research of more than 2,000 venture-backed companies that raised at least \$1 million from 2004 to 2010.

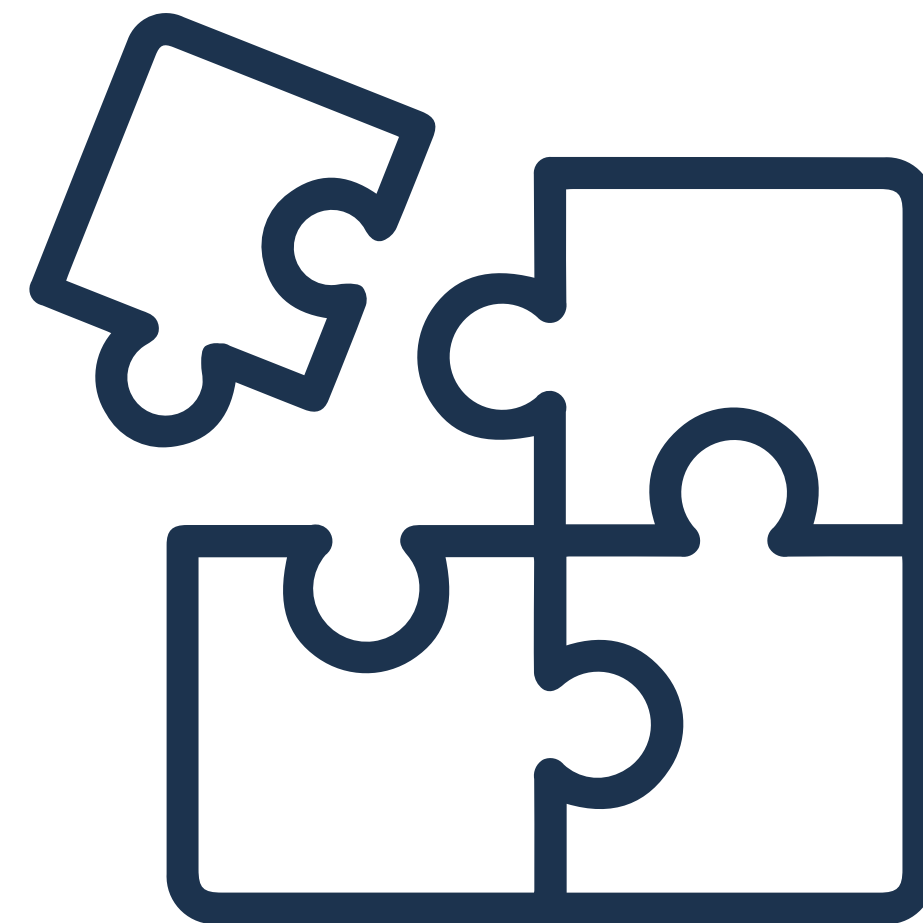
Why Startups fail?

Top 12 reasons for startups to



Why Startups Fail – Some Key Aspects

- “Startups fail when they are not solving a market problem”
- “We were not solving a large enough problem that we could universally serve with a **scalable** solution”
- “We had great technology, great data on shopping behavior, great reputation as a tough leader, great expertise, great advisors, etc, but what we did not have was technology or a business model that solved a pain point in a scalable way”



Why Products Fail?

2011

Harvard Professor Clayton Christensen said more than:

30,000

new **consumer products** are launched every year.

85%

of them fail.

2018

Nielsen Report:
"Setting the record straight on innovation failure"

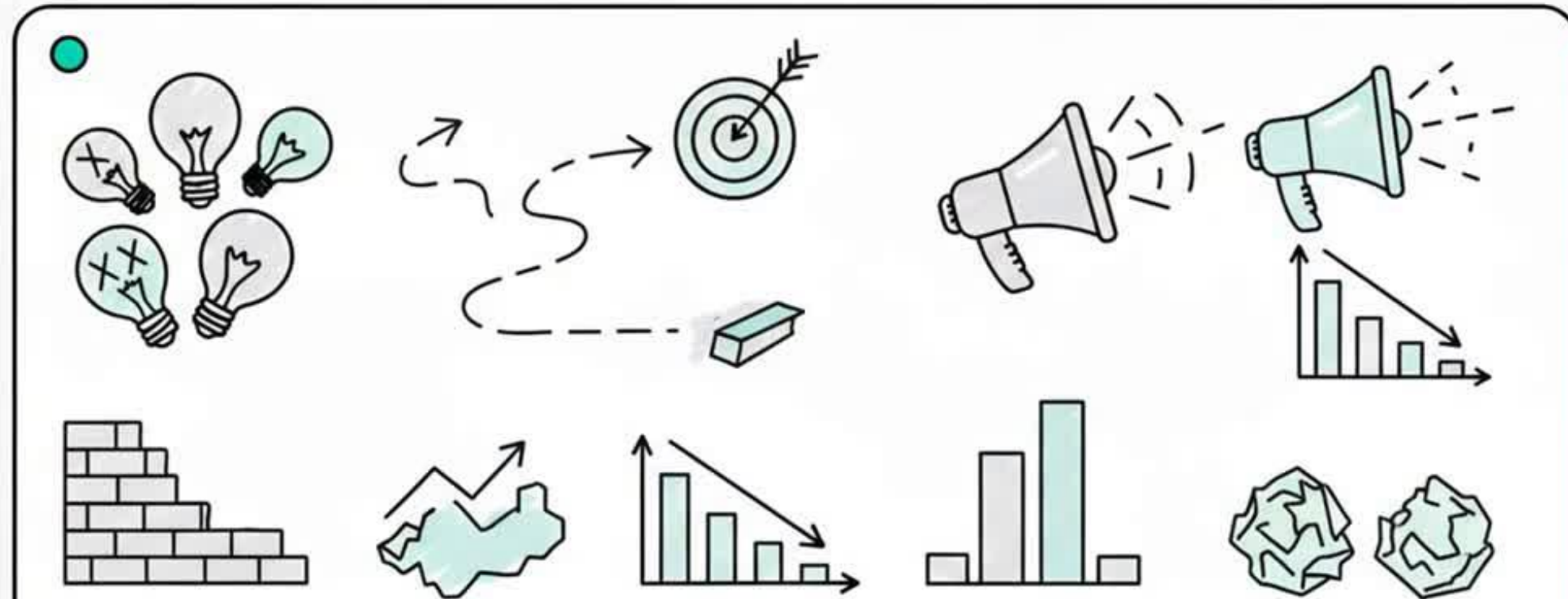
80%

of Products

FAIL

Why Products Fail – Deeper Dive

Why Most Products F.L.O.P.



F.L.O.P

**Failure in
Launch**

Sales/Marketing/distribution fails to reach intended market

Operations or

Fails user expectation
(Design/Functionality/Reliability)

Premise

People are simply NOT interested in your idea.
They may find it, try it or buy it, but they simply don't care.

Credit: Alberto Savoia

'Law of Market Failure'

“Most new products will fail in
the market,
even if competently executed”

~Alberto Savoia (ex-Google Director of
Engineering and Innovation)



<https://albertosavoia.medium.com/the-law-of-market-failure-1dba44b148f>,
<https://www.jamasoftware.com/blog/2014/09/02/law-market-failure/>

Beyond Market Risks ...



Technology Risk



Product Risk



Execution Risk



Market Risk

"Customers saying *yes* is necessary—but not sufficient."

The 'Right It'

How do you know early on that you are building the 'Right it'?*

*** The 'Right It' can be a product, service or an offering**



Google Wave and F.L.O.P

- Hypothesis/Definition in 2009: What Email would look like if we invented it today?
- Developed by 2 Engineers who built Google Maps – Lars and Jens Rasmussen
- This is how it was introduced: Here's how it works: In Google Wave you create a wave and add people to it. Everyone on your wave can use richly formatted text, photos, gadgets, and even feeds from other sources on the web. They can insert a reply or edit the wave directly. It's concurrent rich-text editing, where you see on your screen nearly instantly what your fellow collaborators are typing in your wave.
- Launch was successful, Many people waiting for invites of Google Wave (6K roll outs, 20K waiting list) – So Not a **Launch** issue.
- It had some initial hiccups, but they were addressed and users were able to use it – So perhaps not an **Operations** issue.
- It wasn't compelling enough for people to use it. – So it has to be the **Premise** that Google got **it** wrong.
- **Final Nail:** “We were equally jazzed about Google Wave internally, even though we weren't quite sure how users would respond to this radically different kind of communication. The use cases we've seen show the power of this technology: sharing images and other media in real time; improving spell-checking by understanding not just an individual word, but also the context of each word; and enabling third-party developers to build new tools like consumer gadgets for travel, or robots to check code. But despite these wins, and numerous loyal fans, Wave has not seen the user adoption we would have liked.”

The 'Right It' (Tesla Roadster)



**How Tesla tested if they were building the the 'Right It'
(likely the 1st EV sportscar)**

The 'Right It' (CIE Basecamp 2018)



How PES University tested the the 'Right It'
(which led to the CIE - Center for Innovation and Entrepreneurship)

Job #1 in Customer Discovery: Validating Key Assumptions

Do not Prototype it first
Pretotype first!

Pretotype: An artifact or technique that can be used to collect YODA (your-own-data assessment) very quickly and very inexpensively.

Pretotyping

Investment: hours,days

Main Question: Would we use it?

Deliverable: [Working] pretotype



Should We Build It?

Prototyping

Investment: days, weeks

Main Question: Can we build it?

Deliverable: Working prototype



Can We Build It?

Pretotype vs. Prototype

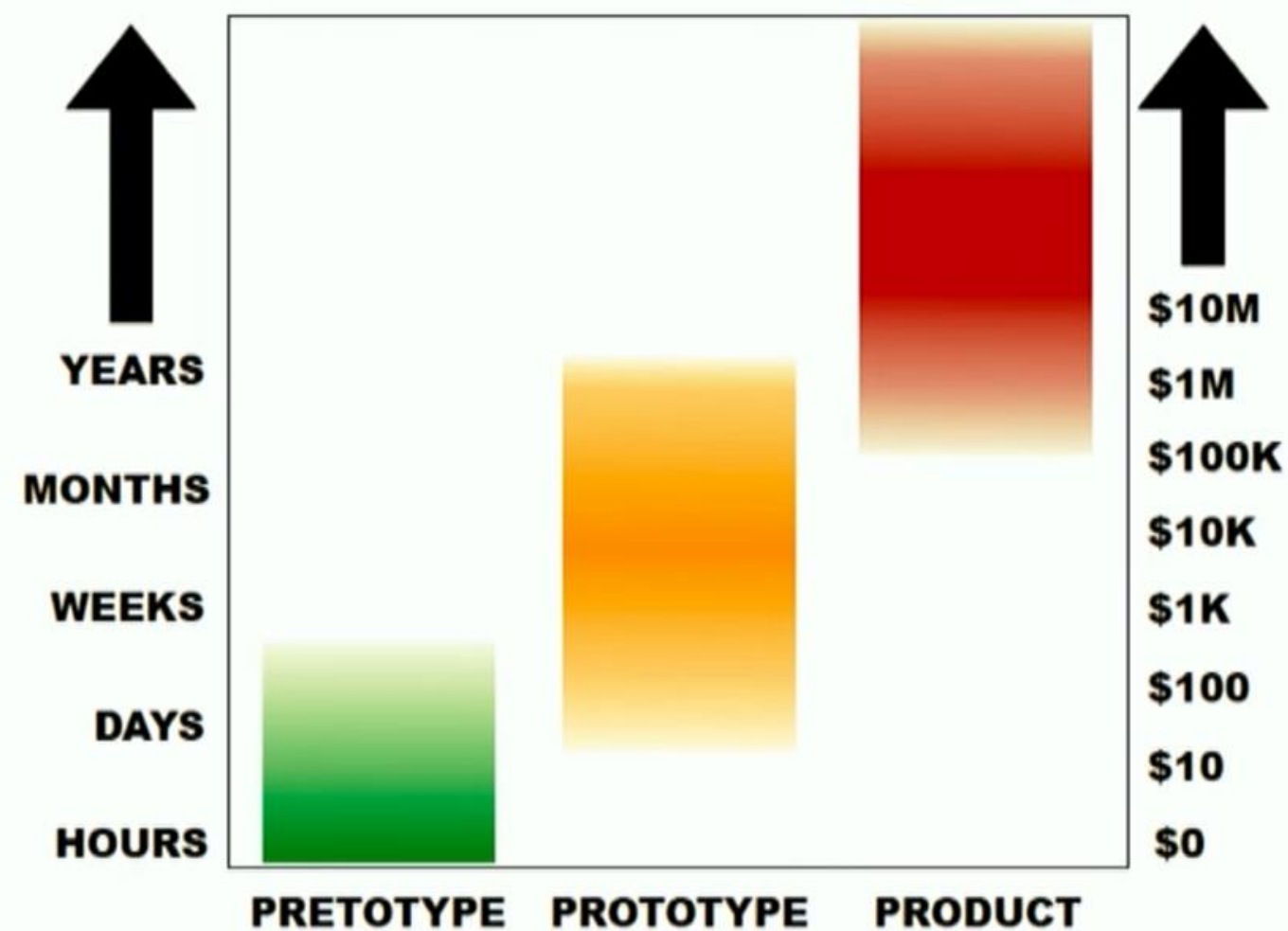
ALBERTO SAVOIA

Stanford Entrepreneurial Thought Leaders Series, March 6, 2019

THE RIGHT IT

PRETOTYPE:

An artifact or technique
that can be used
to collect YODA
**very quickly and
very inexpensively.**



Stanford

Pretotyping Manifesto

Innovation beats Ideation

**Pretotypes beat Products
(initially)**

Data beats Opinions

Doing beats Talking

Simple beats Complex

Now beats Later

**Commitment beats
Committees**

Adapted from work by Alberto Savoia; see this for template for 'Right It'

Pretotyping Manifesto

Pretotyping Manifesto

Innovation beats Ideation

Pretotypes beat Products (initially)

Data beats Opinions

Doing beats Talking

Simple beats Complex

Now beats Later

Commitment beats Committees

Conducting Interviews

(Steve Blank, Kauffman Foundation, 5:36 m)



Video Source: <https://vimeo.com/groups/204136/videos/75535337>

Asking the right questions

(Steve Blank, Kauffman Foundation, 2:45 m)

Pretotyping

(Alberto Savoia, 9:11 m)



<https://www.youtube.com/watch?v=LfbLRT4QtWs&t=179s>

Resources

1.Video: Pre-Planning Pt1 (CD30):

<https://vimeo.com/groups/204136/videos/75308828>

2.Video: Customer Discovery Interview (Part2; CD34):

<https://vimeo.com/groups/204136/videos/75536337>

3. For the full catalogue of videos, visit: <https://vimeo.com/groups/204136>

Founder/Expert Talk : Prof. Nagarjuna Sadineni

Announcement



**COHORT 9
IS HERE!**

**TURN YOUR IDEA
INTO IMPACT.**

Scan the QR code to be part of Cohort 9 and join a community of student innovators building real-world solutions.



For Tomorrow's Innovators

CIE Ignite Season 02: A curriculum-integrated innovation program delivering to 4,500+ engineering and non-engineering students at your most motivated moment.

From classroom theory to real-world impact!

Quick Updates

Mentor Mapping Done

You now have an industry/alumni mentor assigned. Check your email for details.

Google Group Live

Join for active discussions, quick help, and mentor connect.

2 Mentor Sessions

Come prepared with your blockers and questions. Make them count.

Quick Updates



March 08 – Deck Deadline

Final Phase 2 submission. No extensions.

March 11 – Pitch Day @ PESU 52

Your moment to shine in front of faculty, mentors & partners.

Team Leads: Submit Tonight

Google Form closes tonight. Don't miss out.

Stuck? Tell Us Early

We can help – but only if you reach out in time.



The CIE Ignite Journey

A three-phase innovation pipeline that transforms classroom learning into validated, market-ready solutions



Phase 1: Department Hackathons

January 2026

600+ teams compete across 21 departments. Top 20% advance with validated concepts and initial prototypes.

Phase 3: Grand Finale

April 2026

15 finalists pitch refined solutions to investors, partners, and faculty. Winners announced with prizes and opportunities.

1

2

3

4

Phase 2: Mentor Sprint

February 2026

100+ selected teams receive structured guidance from industry & alumni experts through two intensive mentorship sessions.

Beyond: Sustained Innovation

April 2026+

Top projects can continue as capstone work, enter pre-incubation programs, or launch as startups with ongoing support.

Innovation starts here—and it doesn't stop.

Our Website cie.pes.edu

Reach out to
tarunrama@pes.edu

Essentials of Innovation & Entrepreneurship (EIE) - II

Module (M6): Innovation Matrix, Disruptive Innovation

EIE2-Icebreaker:

I&E Roundup this Week (Feb 25)

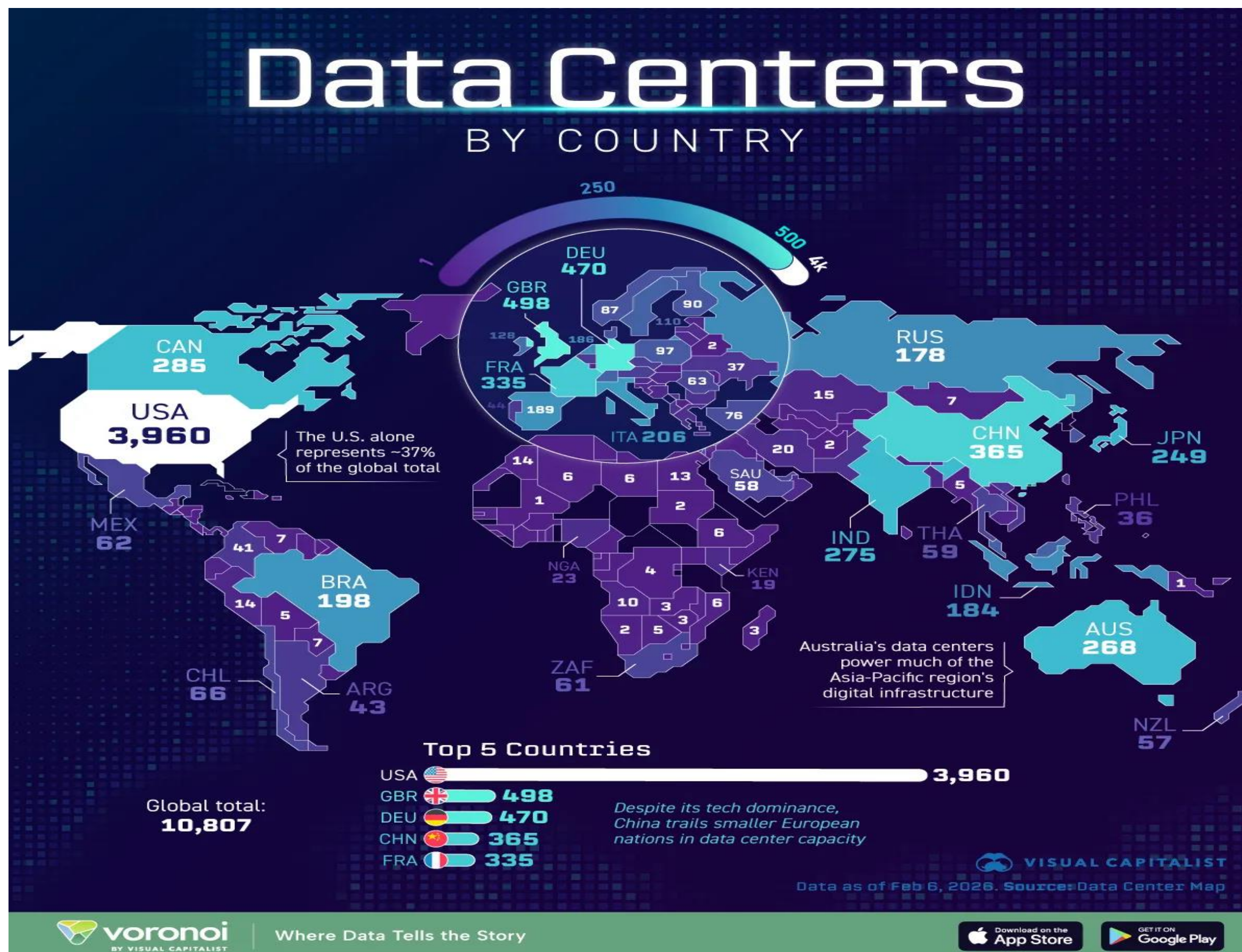


COBOL Comes Alive...Again!

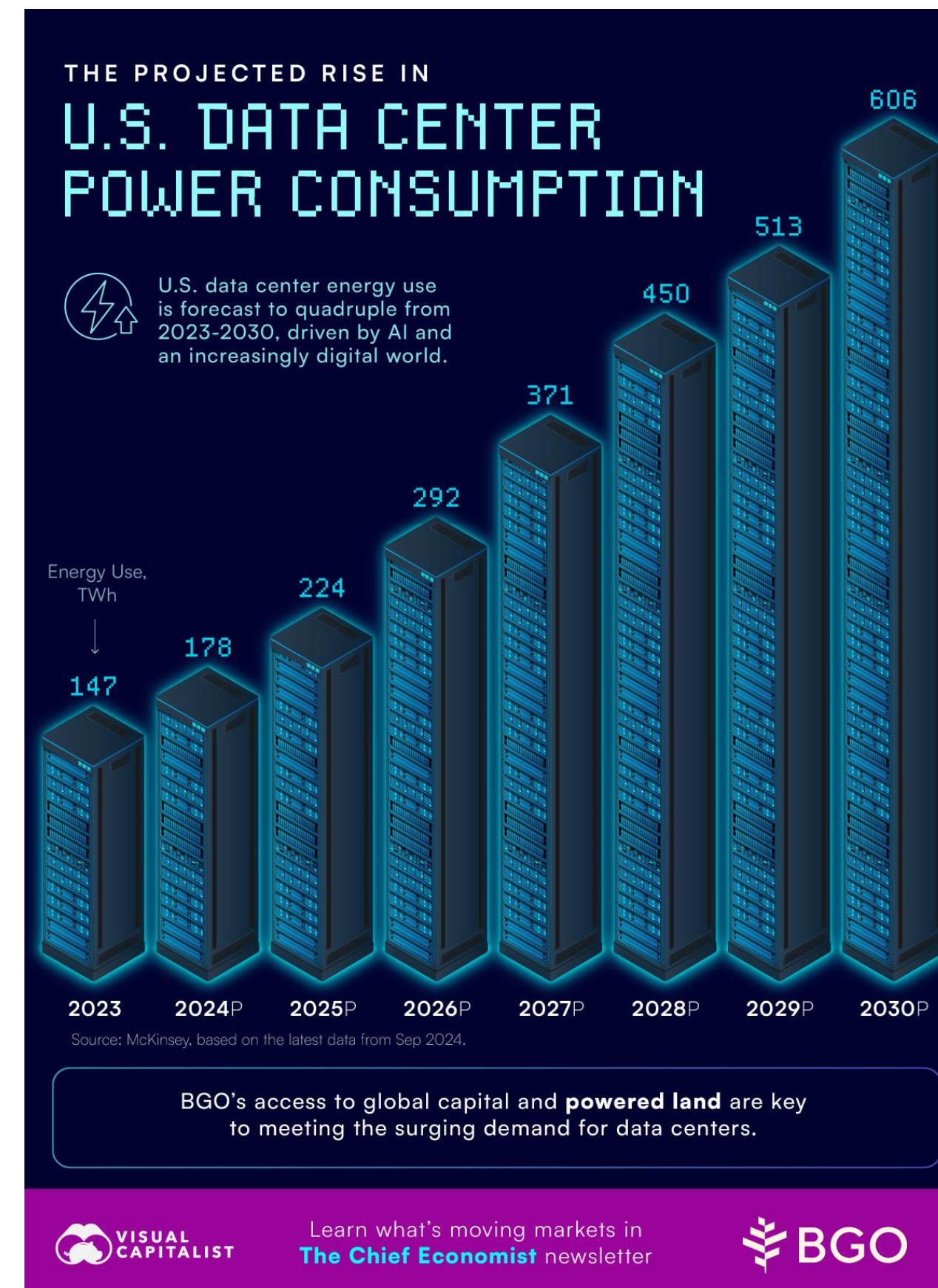
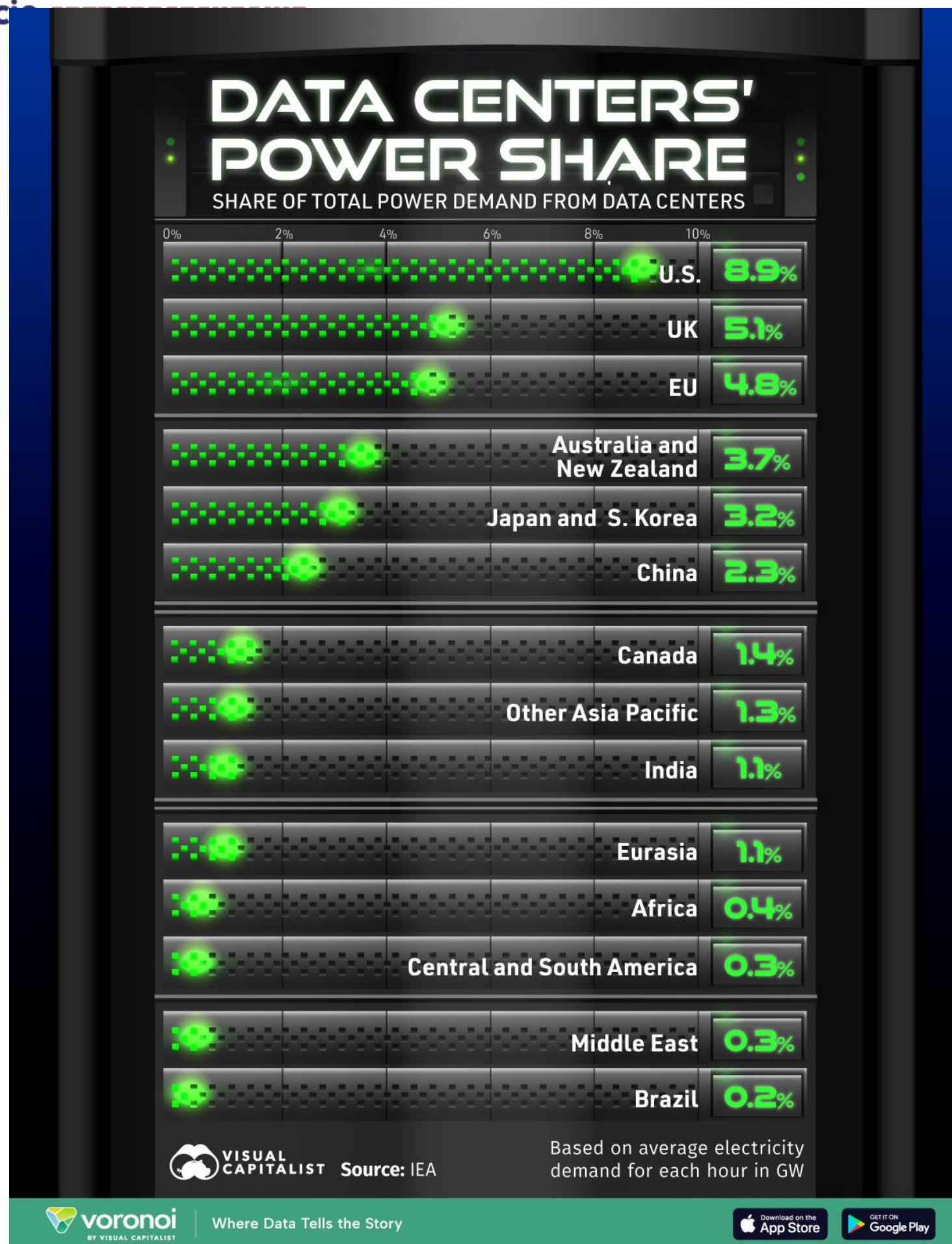


“IBM is the latest AI casualty. Shares tank 13% on Anthropic programming language threat”
“Anthropic's new AI tool can write 67-year-old COBOL code, sending 115-year-old IBM's stock tumbling by 13%
IBM stock has worst day in 26 years, down 25% MoM and counting”

Data Centers – The Happening IT!



Data Centers – Energy Monsters?



Source: VisualCapitalist

Cisco ThingQbator

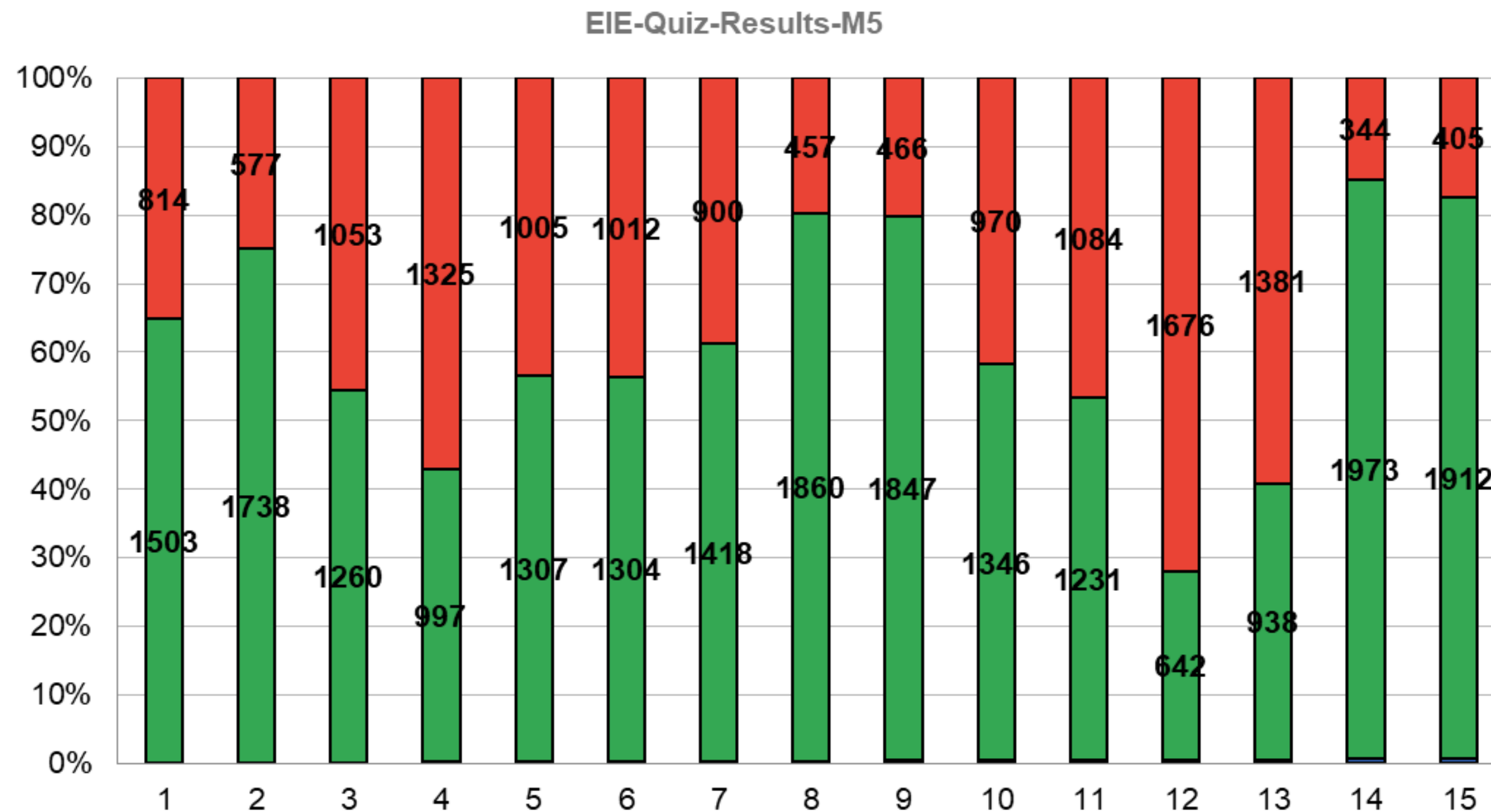


**COHORT 9
IS HERE!**

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INTO IMPACT.**

Scan the QR code to be part of Cohort 9 and join a community of student innovators building real-world solutions.

M5 Session End Quiz



M5 Session End Quiz

Qstn #	StudentScore (Correct Answer)	StudentScore (Wrong Answer)	Total Students	% Correct	Options-wise StudentScore			
					Option A	Option B	Option C	Option D
1	1503	814	2317	65%	1503	375	184	255
2	1738	577	2315	75%	172	1738	174	231
3	1260	1053	2313	54%	212	1260	693	148
4	997	1325	2322	43%	385	301	639	997
5	1307	1005	2312	57%	449	347	1307	209
6	1304	1012	2316	56%	1304	318	174	520
7	1418	900	2318	61%	106	1418	155	639
8	1860	457	2317	80%	113	228	1860	116
9	1847	466	2313	80%	1847	178	167	121
10	1346	970	2316	58%	263	552	1346	155
11	1231	1084	2315	53%	560	160	364	1231
12	642	1676	2318	28%	148	1403	642	125
13	938	1381	2319	40%	1051	938	147	183
14	1973	344	2317	85%	115	118	111	1973
15	1912	405	2317	83%	124	150	1912	131

M5 Session End Quiz

4. When users try a product but “simply don’t care,” it is mainly a failure of:

1. Technology
2. Operations
3. Design

4. Premise - 997 students got it right!

12. A startup mocks up a user interface and shows it to potential customers.

This is:

1. Move fast and break things
2. Validate assumptions before building

3. Data beats opinion - 642 students got it right!

4. Focus on scalability first

13. Spending weeks building a prototype before checking user interest mainly increases:

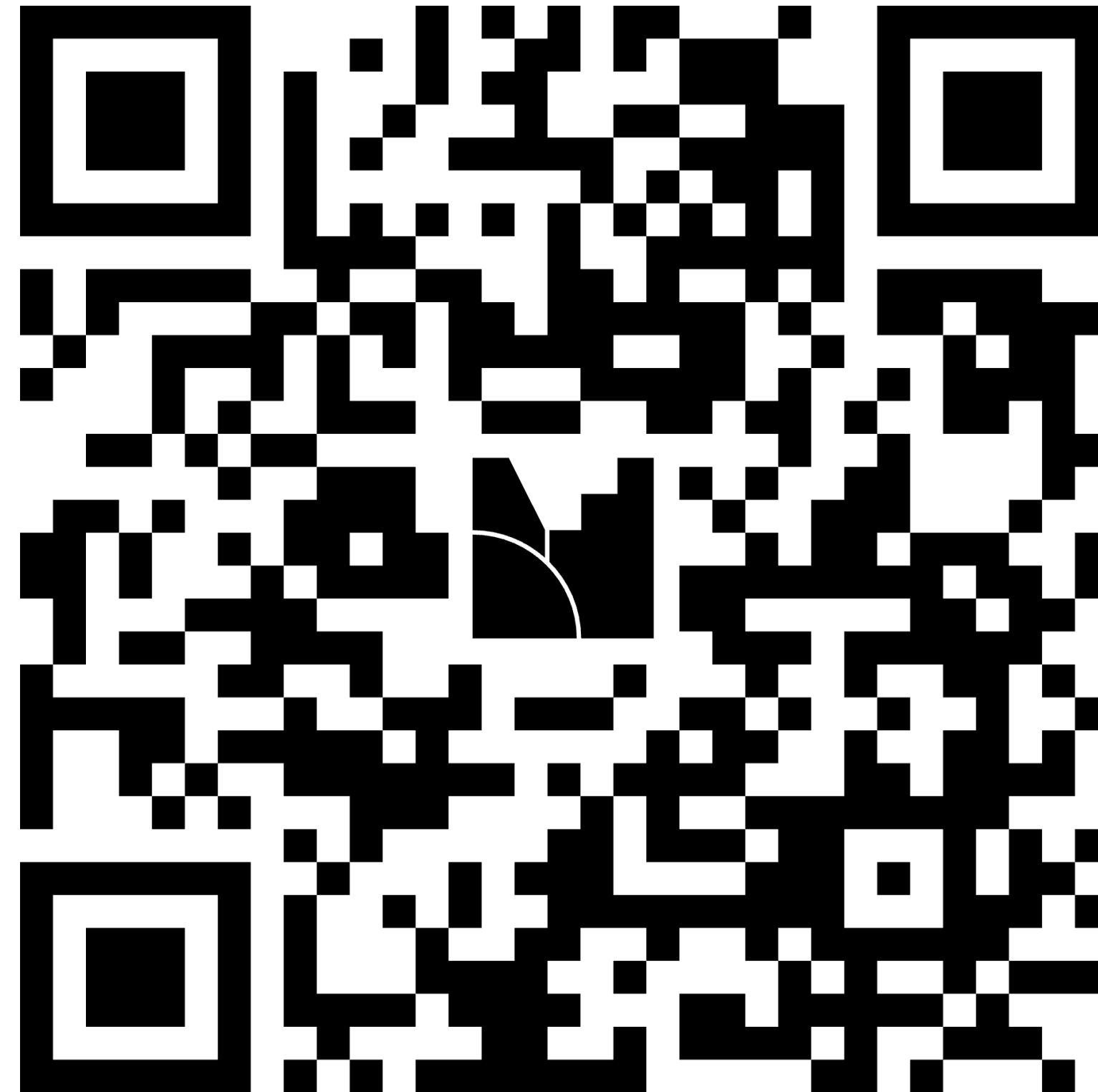
1. Market uncertainty

2. Premise risk - 938 students got it right!

3. Legal exposure
4. Technical debt

M5 Recap – Menti Quiz

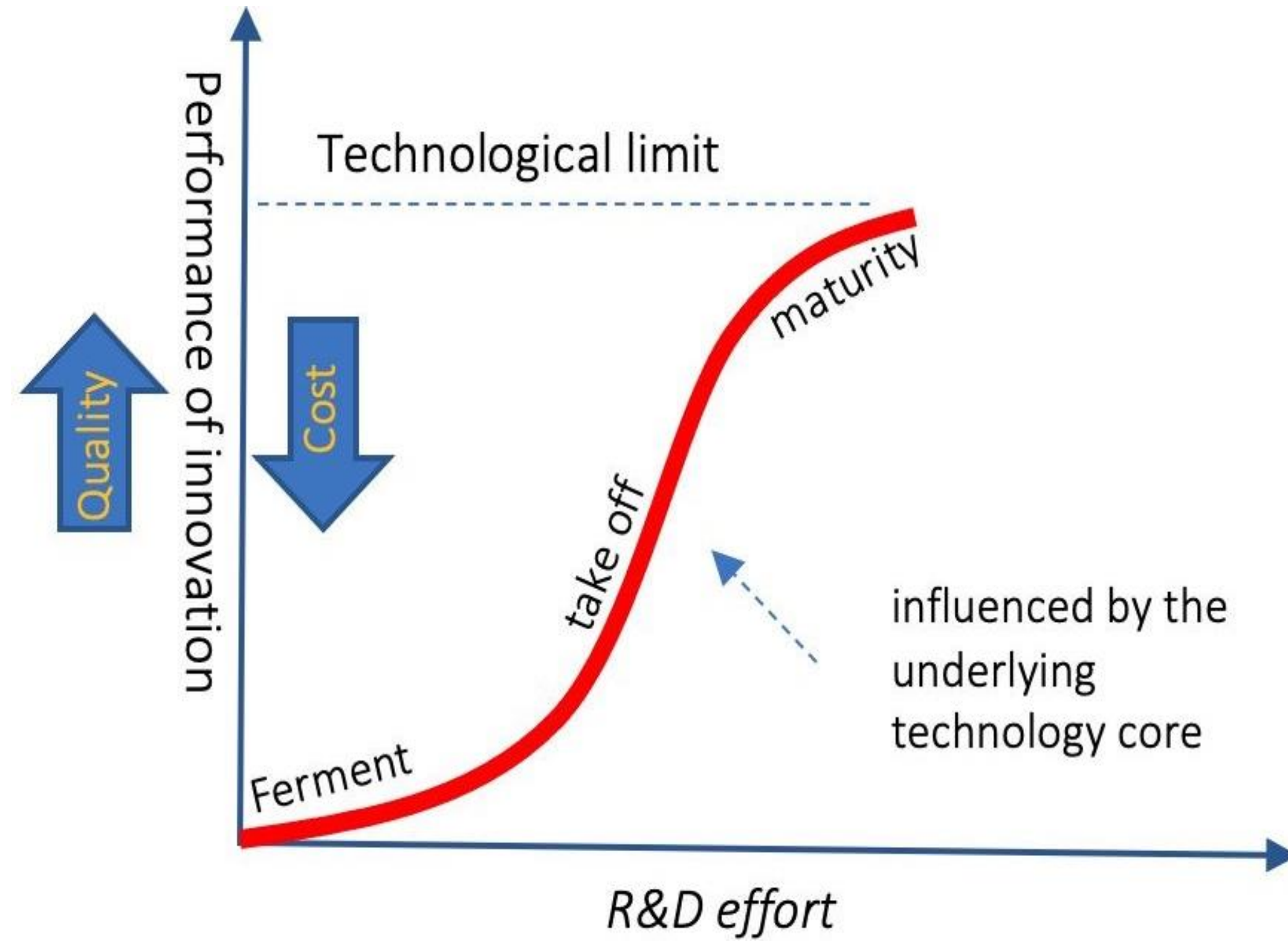
Menti Code: 1752 4998



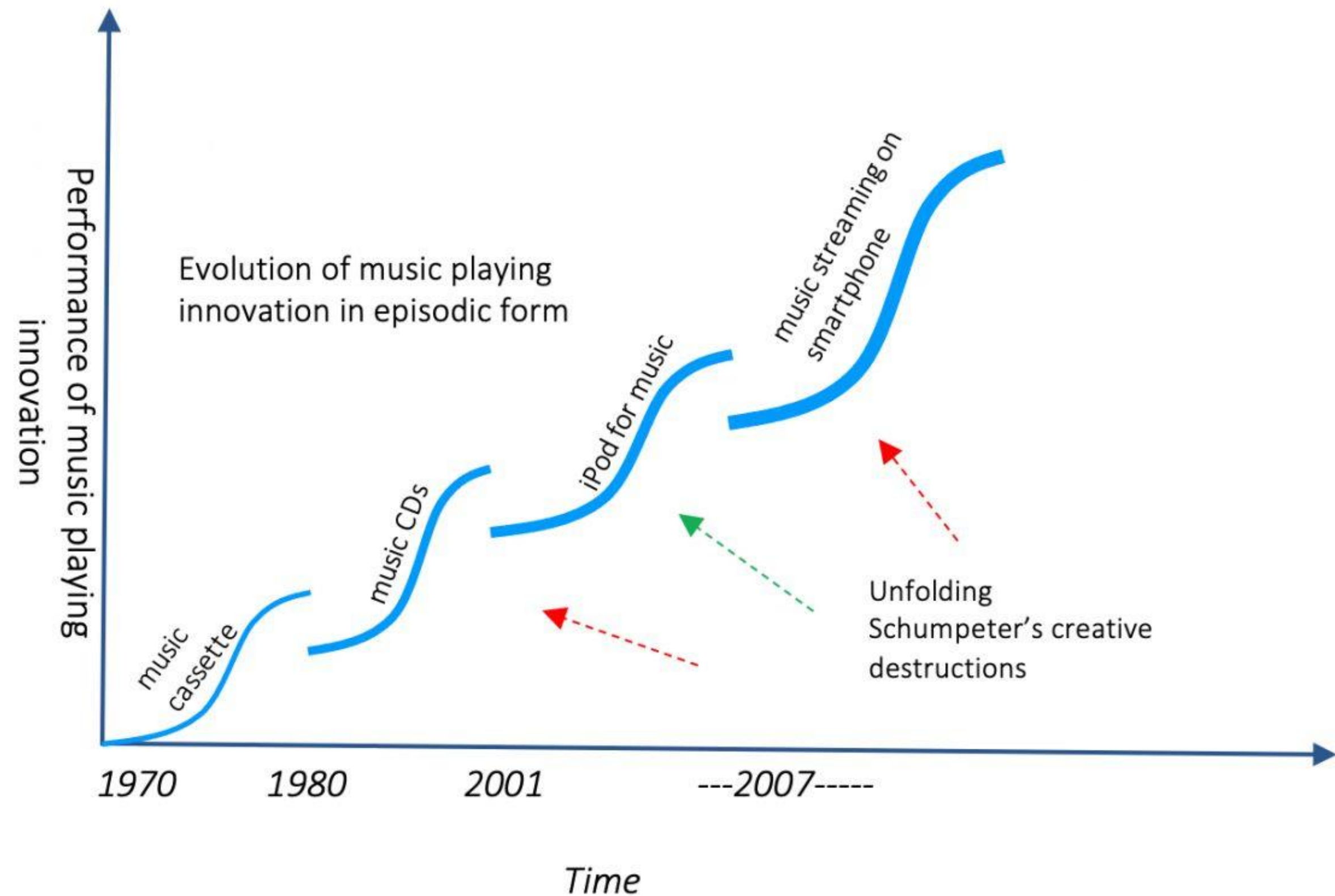
M5: The Right It, Pretotyping Recap

- **75%** of Venture-Backed Startups fail – Harvard Business School
- Harvard Professor Clayton Christensen said more than 30,000 new consumer products are launched every year and **85%** of them fail.
- “Most new products will **fail** to return the investment in the market, even if competently executed”.
- Risks exist beyond market – Technology, Product, Execution risks.
- Find the “**Right It**” for your Product/Service.
- Before Prototyping – Think of a Pretotype – It tells you “Should we build it”.
- Keep the Pretotyping Manifesto handy.
- Be Curious, Be Open.

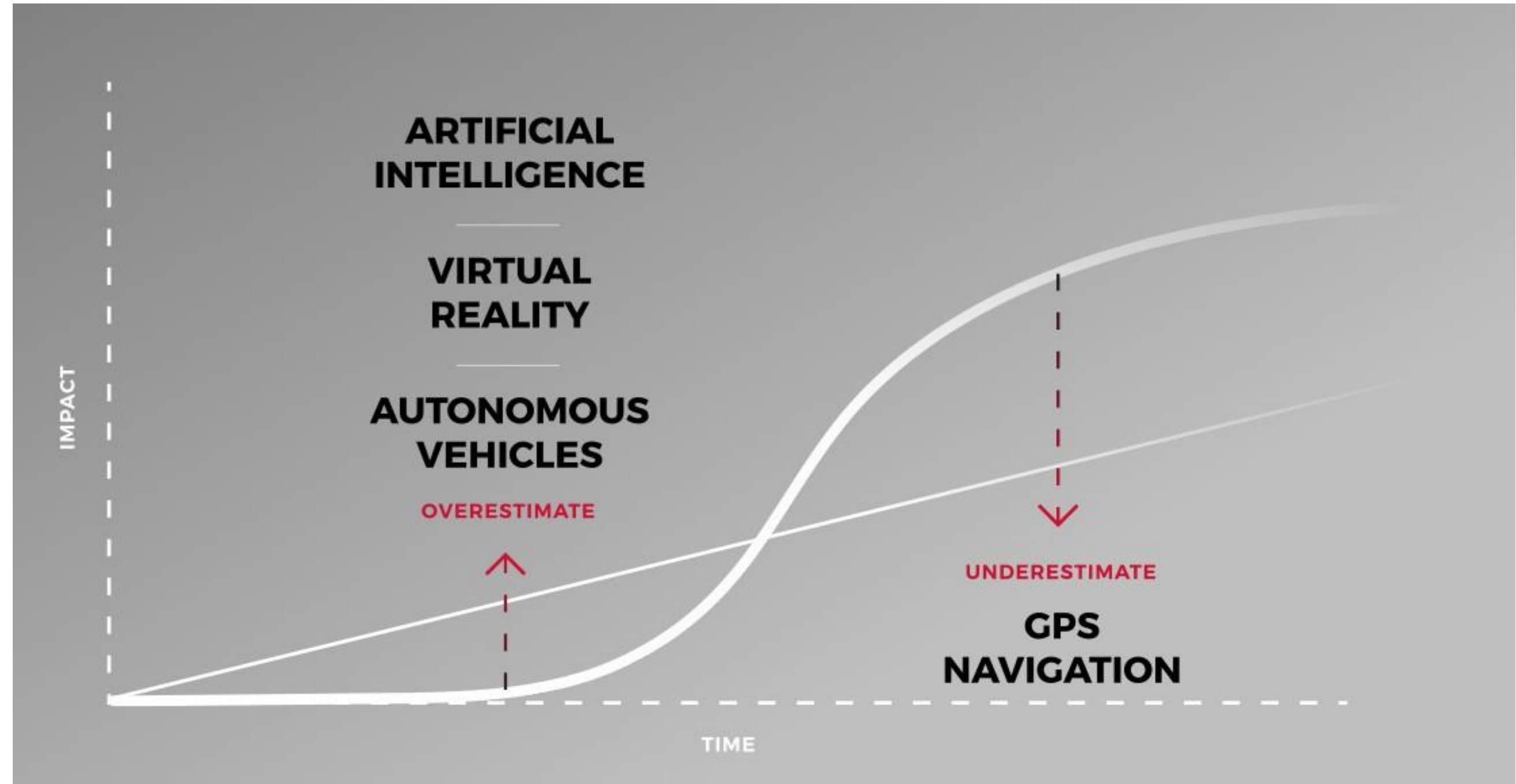
'S-Curve' of Technology-Driven Innovation



'S-Curve' of Technology-Driven Innovation (Example)



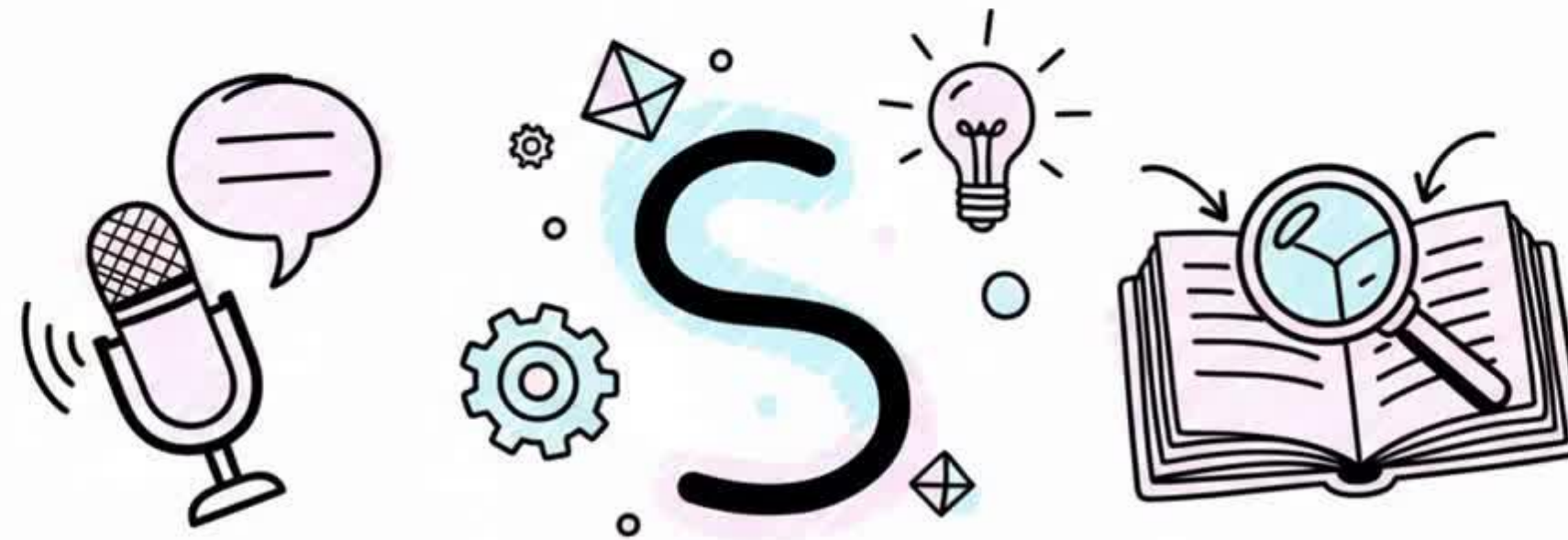
'Amara's Law'



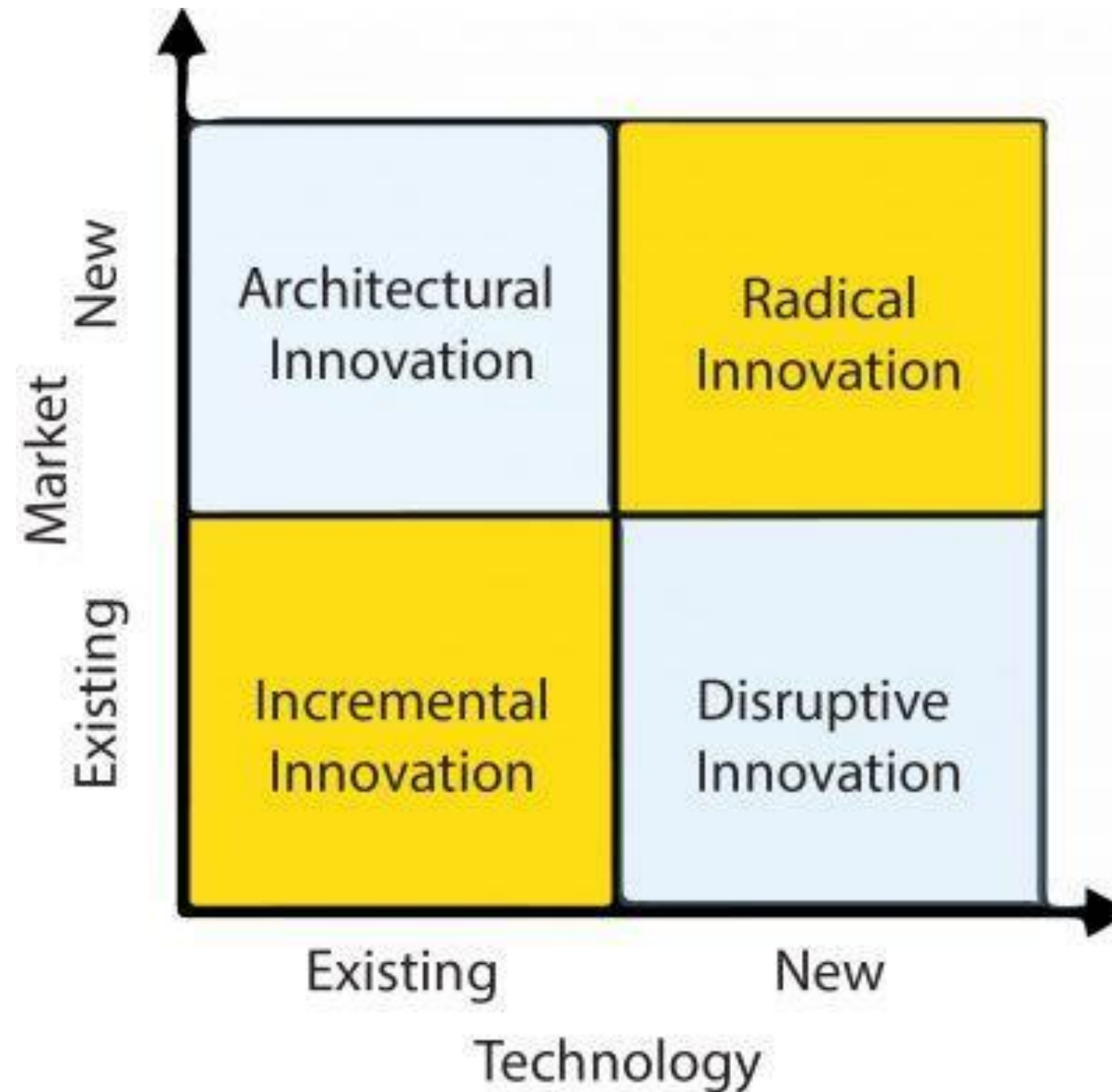
“We tend to overestimate the short-term impact of new technologies while underestimating their long-term effects”

Pattern of Innovation

The Pattern of Innovation



Innovation Type: Technology vs. Market



Type: Technology vs. Market (Examples)

Incremental Innovation

(Small improvements to existing products or processes)

Example: Apple's iPhone updates – Each new iPhone model brings incremental improvements, such as better cameras, faster processors, and enhanced battery life, without fundamentally changing the smartphone industry.

Architectural Innovation

(Reconfiguring existing tech for a new market or purpose)

Example: Sony's Walkman – Sony took existing components (cassette players, compact speakers, and headphones) and reconfigured them into a portable music player, creating a new market for personal audio devices.

Radical Innovation

(Breakthroughs that fundamentally change industries)

Example: The first personal computer (PC) – The introduction of PCs by companies like IBM and Apple revolutionized computing by making it accessible to individuals and businesses, fundamentally changing the way people work and communicate.

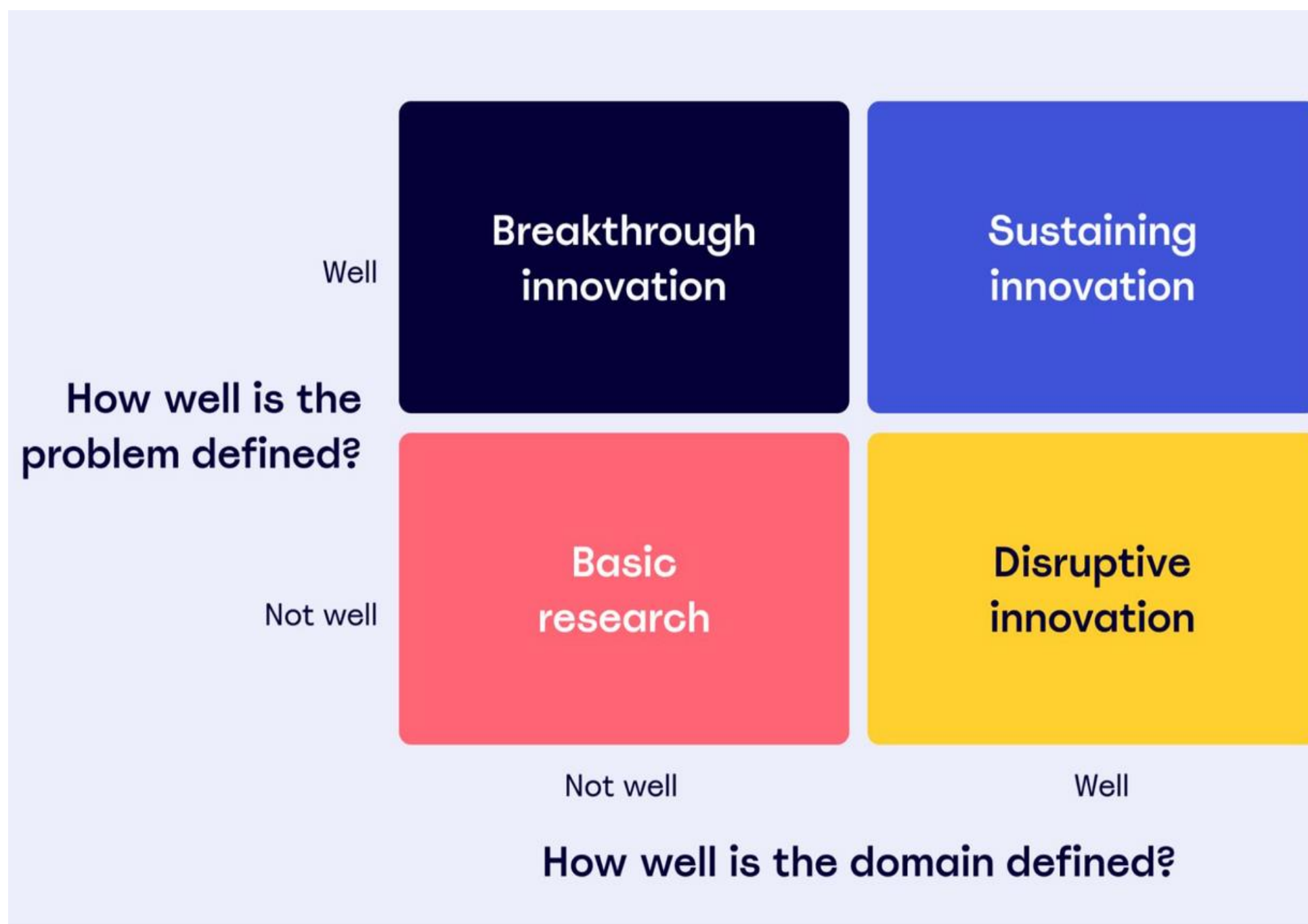
Disruptive Innovation

(Starts as a low-end/niche product; eventually upends an entire industry)

Example: Netflix's streaming service – Initially, Netflix offered DVD rentals by mail, targeting a niche market. Eventually, it shifted to streaming, disrupting traditional cable TV and movie rental businesses like Blockbuster.

Innovation Type:

Problem vs. Domain Maturity



Basic Research:

fundamentalscientific exploration without immediate commercial application

Sustaining Innovation:

Incremental improvementsstoexisting products or services to maintain competitiveness

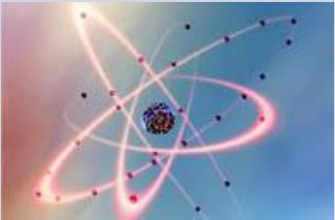
Breakthrough Innovation:

Asignificant technological advancement that creates new markets or changes existing industries

Disruptive Innovation:

A new tech/businessmodel that starts at the low end of the market & eventually displaces established players

Innovation Type: Problem vs. Domain Maturity (Examples)

Basic Research		Discovery of quantum mechanics in the early 20th century
Sustaining Innovation		Apple's release of newer iPhone models with better cameras, faster processors, and improved battery life
Breakthrough Innovation		The development of CRISPR gene-editing technology, which revolutionized genetics and opened possibilities for treating previously incurable diseases
Disruptive Innovation		Netflix disrupting Blockbuster. Initially, Netflix's DVD-by-mail service appealed to niche customers, but as streaming technology improved, it overtook traditional video rental businesses entirely

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[illegible]

Ten Types of Innovation (Deloitte)

THE TEN TYPES OF INNOVATION AN OVERVIEW

The Ten Types framework is simple and intuitive. It is a useful tool you can use both to diagnose and enrich an innovation you're working on, or to analyze existing competition. It makes it especially easy to spot errors of omission — missing dimensions that will make a concept stronger.

The Ten Types framework is structured into three color-coded categories. The types on the left side of the framework are the most internally focused and distant from customers; as you move toward the right side, the types become increasingly apparent and obvious to end users. To use a theatrical metaphor, the left of the framework is backstage; the right is onstage.



STRENGTH IN NUMBERS

Simple innovations use one or two types of innovation and every company needs to pursue them. Failure to consistently, relentlessly improve the known is one of the surest routes to failure. Unfortunately, too many firms do only simple innovation. In today's contested markets, simple innovations alone are never enough for long-term success. They can build a lead on competitors, but they don't create the firms, brands, or platforms that thrill us.

When a market grows up and gets complicated, it demands more **sophisticated innovation**, which uses many types of innovation combined elegantly and orchestrated with care. Under the covers, inside your firm, these require working across internal boundaries and silos — challenges that bring additional complexity. You can cut through this with multi-disciplinary teams to bring in the necessary talent and knowledge, and with systems in place to tell everyone how they can tackle tough challenges with curiosity, confidence, and courage.

Naturally, sophisticated innovations are more difficult to pull off, not least because they have longer development horizons than simple innovations. But consider the flip side: once you launch them, they are likely both to delight customers and confound competitors. Often you will be able to succeed with them for years before challengers can catch up. Almost all of the enterprises that we identify as leading innovators routinely use multiple types of innovation — and handily outperform the average firms that innovate more naively.

SIX PRINCIPLES FOR USING THE TEN TYPES EFFECTIVELY

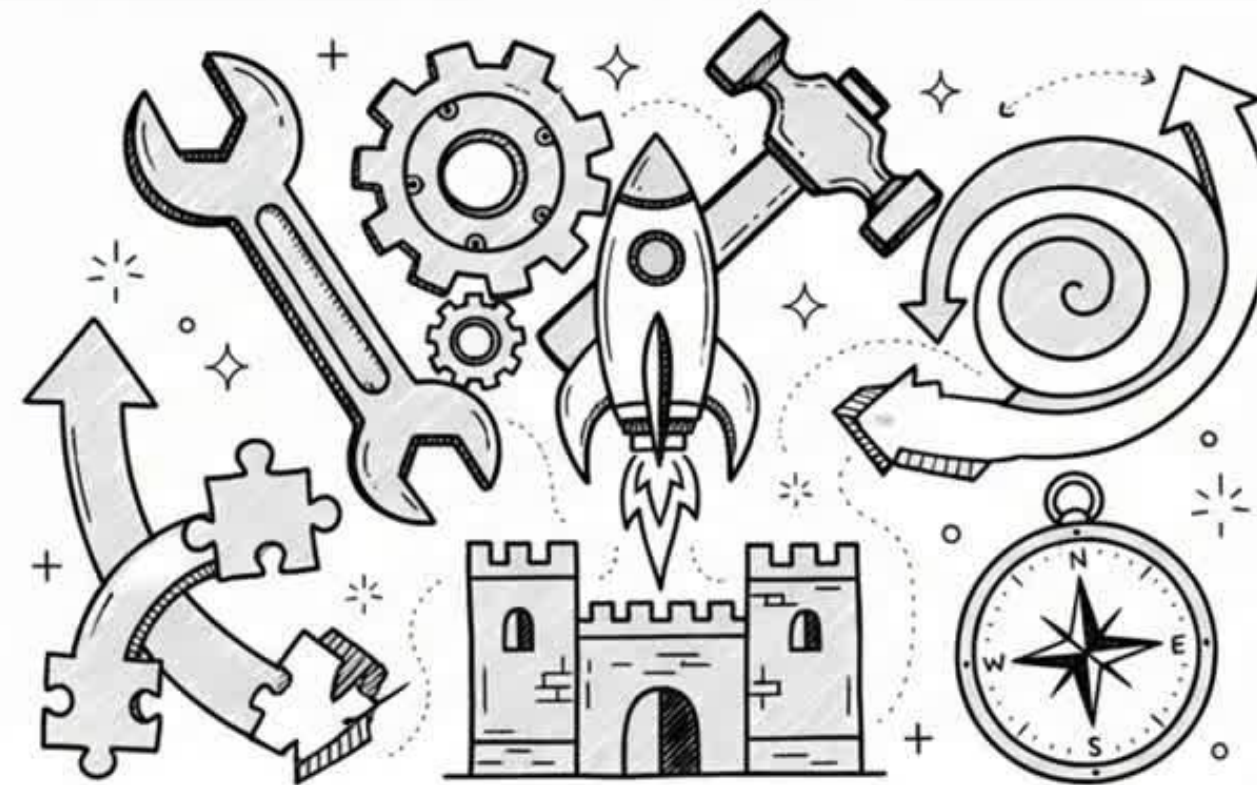
- 1 UNDERSTAND ALL TEN TYPES**
Virtually all projects can improve just by knowing and deeply understanding the value and subtleties of each of the types.
- 2 DE-EMPHASIZE RELIANCE ON PRODUCTS AND TECHNOLOGY**
These are the easiest capabilities for competitors to copy.
- 3 THINK ABOUT CATEGORIES AS WELL AS TYPES**
Consciously try to imagine new ways to configure assets, build platforms, and foster fresh experiences.
- 4 USE THE TYPES THAT MATTER MOST**
Use diagnostics to understand which types you and others in your industry tend to overlook.
- 5 UNDERSTAND WHAT YOUR USERS REALLY NEED**
User research can help you know what is relevant to customers and what surprises other types might help to deliver.
- 6 USE ENOUGH OF THE TYPES TO MAKE A SPLASH**
Using five or more types, integrated with care, is nearly always enough to reinvent a category and become newsworthy.

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Credit: [Deloitte](#)

Business Growth Frameworks

Business Growth Frameworks



Essentials of Innovation & Entrepreneurship (EIE) - II

Module (M6): Disruptive Innovation

Disruptive Innovation Explained

Disruptive Innovation



Disruptive Innovation Explained

THE EXPLAINER: **DISRUPTIVE INNOVATION**

Harvard
Business
Review



(Harvard Business Review, 2:08 m)

<https://www.youtube.com/watch?v=aMsvWmvTmxw>

Disruptive Innovation Explained ([link](#))



<https://www.youtube.com/watch?v=qDrMAzCHFUU>

Essentials of Innovation & Entrepreneurship (EIE) - II

Module (M6):

Jobs to be Done: A deeper understanding of Customer Needs/Behaviour

Jobs to be Done in Brief

- **'Jobs-To-Be-Done' or 'Jobs' was proposed by Clayton Christensen.**
- **Focuses on the User's needs**
- **Target audience is based on the needs and not category or background**
- **Iterative flow of process**

'Jobs to be Done' - Example



“People don't want a quarter-inch drill bit; they want a quarter-inch hole.” - Theodore Levitt

What's the Opposite of 'Jobs-to-be-Done'?

Feature	'Jobs-to-be-Done' (Customer-Centric)	Traditional Business/Sales Perspective
Primary Focus	Customer Oriented: Built on what is necessary for the user to achieve a goal.	Business Oriented: Driven by sales targets and internal business goals.
Source of Truth	Customer Insights: Focuses on the specific needs and context of the user.	Idea Generation: Focuses on brainstorming new concepts from within the company.
Validation	Input from Real People: Relies on direct feedback and observation of users.	Ideal Person Character: Uses a "persona" or hypothetical character to guide the product.
Methodology	Iterative Process: A continuous loop of learning, building, and refining.	Single Flow: A more linear, straightforward path toward a final product.

Job-To-Be-Done Theory ([link](#))



Job-To-Be-Done Theory

Jobs to be Done: Milkshake



Summary (For Startup Founders)

- Target underserved markets ignored by incumbents to gain an early foothold.
- Use disruptive innovation to enter niche markets overlooked by incumbents & then scale
- Identifying the “job” your product is hired to do is crucial for achieving product-market fit
- Leverage incremental improvements for competitive advantage in the shorter term
- Build solutions that solve genuine customer pain points, ensuring relevance and demand

Refer: Jobs to be Done in some detail

During	Customers	Companies can innovate by	Example
Define	Determine their goals and plan resources	Simplifying planning	Weight Watchers streamlines diet planning by offering a system that doesn't require calorie counting.
Locate	Gather items and information needed to do the job	Making required inputs easier to gather and ensuring they're available when and where needed	U-Haul provides customers with prepackaged moving kits containing the number and types of boxes required for a move.
Prepare	Set up the environment to do the job	Making set-up less difficult and creating guides to ensure proper set-up of the work area	Bosch added adjustable levers to its circular saw to accommodate common bevel angles used by roofers to cut wood.
Confirm	Verify that they're ready to preform the job	Giving customers information they need to confirm readiness	Oracle's ProfitLogic merchandising optimisation software confirms optimal timing and level of a store's markdowns for each product.
Execute	Carry out the job	Preventing problems or delays	Kimberly-Clark's Patient Warning System automatically circulates heated water through thermal pads placed on surgery patients to maintain their normal body temperature during surgery.
Monitor	Assess whether the job is being successfully executed	Linking monitoring with improved execution	Nike makes a running shoe containing a sensor that communicates audio feedback about time, distance, pace, and calories burned to an iPhone or iPod worn by the runner.
Modify	Make alternations to improve execution	Reducing the need to make alterations and the number of alterations needed	By automatically downloading and installing updates, Microsoft's operating systems remove hassle for computer users. People don't have to determine which updates are necessary, find the updates, or ensure the updates compatible with their operating system.
Conclude	Finish the job or prepare to repeat it	Designing products that simplify the process of concluding the job.	3M makes a wound dressing that stretches and adheres only to itself-not to patients' skin or sutures. It thus offers a convenient way for medical personnel to secure dressing at the conclusion of treatment and then remove them after a wound has healed.

Expert Talk Series: Dr. Shailashree Haridas (Dean, FOMC, PESU)



Dr. Shailashree Haridas

Dean of FOMC at PES University

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You now have an industry/alumni mentor assigned. Check your email for details.

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Join for active discussions, quick help, and mentor connect.

2 Mentor Sessions

Come prepared with your blockers and questions. Make them count.

Quick Updates



March 08 – Deck Deadline

Final Phase 2 submission. No extensions.



March 11 – Pitch Day @ PESU 52

Your moment to shine in front of faculty, mentors & partners.



Team Leads: Submit Tonight

Google Form closes tonight. Don't miss out.



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A three-phase innovation pipeline that transforms classroom learning into validated, market-ready solutions



Phase 1: Department Hackathons

January 2026

600+ teams compete across 21 departments. Top 20% advance with validated concepts and initial prototypes.

Phase 3: Grand Finale

April 2026

15 finalists pitch refined solutions to investors, partners, and faculty. Winners announced with prizes and opportunities.

1

2

3

4

Phase 2: Mentor Sprint

February 2026

100+ selected teams receive structured guidance from industry & alumni experts through two intensive mentorship sessions.

Beyond: Sustained Innovation

April 2026+

Top projects can continue as capstone work, enter pre-incubation programs, or launch as startups with ongoing support.

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Reach out to
tarunrama@pes.edu